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Optical waveguide structure with partially overlapping loops in direction dependent material

Abstract

An optical waveguide structure comprises a nonlinear optical waveguide, a set of tuning optical waveguides, a set of wavelength selective couplers that couples light between the nonlinear optical waveguide and one or more tuning optical waveguides in the set of tuning optical waveguide based on a wavelength of light, and a set of phase shifters located along one or more tuning optical waveguides in the set of tuning optical waveguides.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3842289	12/1973	Yariv et al.	N/A	N/A
5002349	12/1990	Cheung et al.	N/A	N/A
5206868	12/1992	Deacon	N/A	N/A
5295218	12/1993	Agostinelli et al.	N/A	N/A
5875272	12/1998	Kewitsch et al.	N/A	N/A
5999548	12/1998	Mori	N/A	N/A
6078717	12/1999	Nashimoto et al.	N/A	N/A

6081632	12/1999	Yoshimura et al.	N/A	N/A
6269205	12/2000	Peral et al.	N/A	N/A
6438291	12/2001	Duck et al.	N/A	N/A
6445848	12/2001	Islam et al.	N/A	N/A
6483953	12/2001	McBrien et al.	N/A	N/A
6721481	12/2003	Terahara et al.	N/A	N/A
6888667	12/2004	Nicolaescu	N/A	N/A
7046714	12/2005	Nicolaescu et al.	N/A	N/A
7262902	12/2006	Burns et al.	N/A	N/A
7266258	12/2006	Liu et al.	N/A	N/A
7489439	12/2008	Kuo et al.	N/A	N/A
7495823	12/2008	Kanner et al.	N/A	N/A
8173982	12/2011	Edamatsu et al.	N/A	N/A
9291837	12/2015	Yap	N/A	N/A
9798219	12/2016	Pant	N/A	G02F 1/39
10372014	12/2018	Vidrighin	N/A	G02F 1/3536
10451951	12/2018	Yap et al.	N/A	N/A
10698292	12/2019	Nagano	N/A	N/A
11003046	12/2020	Liscidini et al.	N/A	N/A
11092875	12/2020	Xu et al.	N/A	N/A
11221540	12/2021	Srinivasan et al.	N/A	N/A
11226538	12/2021	Marandi et al.	N/A	N/A
11307484	12/2021	Zhang et al.	N/A	N/A
11550201	12/2022	Yap et al.	N/A	N/A
11561454	12/2022	Yap et al.	N/A	N/A
2003/0048527	12/2002	Kimerling et al.	N/A	N/A
2005/0047702	12/2004	Parker et al.	N/A	N/A
2006/0132901	12/2005	Miller	N/A	N/A
2012/0093459	12/2011	Mathai et al.	N/A	N/A
2014/0193155	12/2013	Popovic	29/428	H04J 14/0307
2015/0016767	12/2014	Akiyama	385/3	G02F 1/225
2017/0199443	12/2016	Vermeulen et al.	N/A	N/A
2018/0031949	12/2017	Mookherjea et al.	N/A	N/A
2019/0361315	12/2018	Zhou	N/A	N/A
2020/0256722	12/2019	Najafi et al.	N/A	N/A
2021/0026222	12/2020	Nagano	N/A	N/A
2021/0157177	12/2020	Kharel et al.	N/A	N/A
2022/0107548	12/2021	Yap	N/A	N/A
2022/0236621	12/2021	Kashiwazaki et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110333637	12/2018	CN	N/A
H01134309	12/1988	JP	N/A
H01134310	12/1988	JP	N/A
H05107577	12/1992	JP	N/A
H05216079	12/1992	JP	N/A
H10319444	12/1997	JP	N/A
2014222331	12/2013	JP	N/A
2017078786	12/2016	JP	G02F 1/313
7160194	12/2021	JP	N/A
WO2019208582	12/2020	WO	N/A

OTHER PUBLICATIONS

English Translation of JP-2017078786-A, 8 pages. (Year: 2017). cited by examiner

Runde et al., “Mode-selective coupler for wavelength multiplexing using LiNbO₃:Ti optical waveguides”, Cent. Eur. J. Phys. 6(3), 2008, pp. 588-592, found at DOI: 10.2478/311534-008-0078-1 (Year: 2008). cited by examiner

Office Action, dated Jun. 11, 2024, regarding U.S. Appl. No. 17/655,930, 33 pages. cited by applicant

Final Office Action dated Feb. 21, 2025, regarding U.S. Appl. No. 18/059,624, 31 pages. cited by applicant

Final Office Action dated Oct. 10, 2024, regarding U.S. Appl. No. 17/655,930, 22 pages. cited by applicant

Lumerical, “Lithium Niobate Nonlinear Thermal Waveguide,” ANSYS, available at <https://optics.ansys.com/hc/en-us/articles/360047509134-Lithium-Niobate-Nonlinear-Thermal-Waveguide>, last accessed on Dec. 5, 2022, 19 pages. cited by applicant

Notice of Allowance, dated Nov. 23, 2022, regarding U.S. Appl. No. 17/450,046, 10 pages. cited by applicant

PCT International Search Report and Written Opinion, dated Mar. 28, 2023, regarding Application No. PCT/US2021/071721, 8 pages. cited by applicant

PCT International Search Report and Written Opinion, dated Mar. 28, 2023, regarding Application No. PCT/US2021/071723, 9 pages. cited by applicant

Bosshard, “Cascading of second-order nonlinearities in polar materials,” Advanced Materials, vol. 8, No. 5, May 1, 1996, pp. 385-397. cited by applicant

Biaggio, “Coupling-length phase matching for nonlinear optical frequency conversion in parallel waveguides,” Physical Review A, vol. 90, Oct. 13, 2014, pp. 043816, 11 pages. cited by applicant

Boyd, "Nonlinear Optics," Third Edition, Academic Press, Mar. 28, 2008, pp. 69-133. cited by applicant

Cai et al., "Integrated optics on single-crystal lithium niobate thin film: some recent progress," 18th International Conference on Transparent Optical Networks ICTON, Jul. 2016, paper Tu.D5.5, pp. 1-4. cited by applicant

Chen et al., "Modal phase matched lithium niobate nanocircuits for integrated nonlinear photonics," OSA Continuum, vol. 1, No. 1, Sep. 2018, pp. 229-242. cited by applicant

Clark et al., "Depositing Light in a Photonic Stop Gap by Use of Kerr Nonlinear Microresonators," Optics Letters, vol. 28, No. 10, 2003, pp. 1966-1968. cited by applicant

Doerr, "Planar Lightwave Devices for WDM," Optical Fiber Telecommunications, vol. IVA, Jan. 1, 2002, pp. 405-476. cited by applicant

Dong et al., "Nonlinear frequency conversion in waveguide directional couplers," Phys. Rev. Lett., vol. 93, No. 13, Sep. 20, 2004, pp. 133901, 4 pages. cited by applicant

Dong et al., "Observation of continuous-wave second-harmonic generation in semiconductor waveguide directional couplers," Optics Express, vol. 14, No. 6, Mar. 20, 2006, pp. 2256-2262 (2006). cited by applicant

Fan et al., "290 Hz intrinsic linewidth from an integrated optical chip-based widely tunable InP—Si₃N₄ hybrid laser," Digest 2017 Conference on Lasers and Electro-Optics CLEO, May 2017, paper JTh5C.9, pp. 1-2. cited by applicant

Fejer et al., "Quasi-phase-matched second harmonic generation tuning and tolerances," IEEE Journal of Quantum Electronics, vol. 28, No. 11, Nov. 1992, pp. 2631-2654. cited by applicant

Gad et al., "Compound ring resonator circuit for integrated optics applications," Journal Optical Society America A, vol. 26, No. 9, Sep. 9, 2009, pp. 2023-2032. cited by applicant

Halir et al., "Compact High-Performance Multimode Interference Couplers in Silicon-on-Insulator," IEEE Photonics Technology Letters, vol. 21, No. 21, Nov. 1, 2009, pp. 1600-1602. cited by applicant

Helt et al., "How does it scale Comparing quantum and classical nonlinear optical processes in integrated devices," Journal of the Optical Society of America B, vol. 29, No. 8, Aug. 1, 2012, pp. 2199-2212. cited by applicant

Huang et al., "A novel quasi-phase-matching frequency doubling technique," Optics Communications, vol. 150, May 1, 1998, pp. 235-238. cited by applicant

Lin et al., "Broadband Quasi-Phase-Matched Harmonic Generation in an On-Chip Monocrystalline Lithium Niobate Microdisk Resonator," Phys. Rev. Lett., vol. 122, No. 17, May 3, 2019, pp. 173903, 5 pages. cited by applicant

Lin et al., "Characterization of hybrid InP-TriPlex photonic integrate tunable lasers based on silicon nitride (Si₃N₄/SiO₂) microring resonators for optical coherent systems," IEEE Photonics Journal, vol. 10, No. 3, Jun. 2018, pp. 1400108, 9 pages. cited by applicant

Luo et al., "On-chip second-harmonic generation and broadband parametric down-conversion in a lithium niobate microresonator," Optics Express, vol. 25, No. 20, Oct. 2, 2017, pp. 24531-24539. cited by applicant

Luo et al., "Semi-nonlinear nanophotonic waveguides for highly efficient second-harmonic generation," Sep. 17, 2018, 8 pages, accessed Mar. 30, 2022, available at <https://arxiv.org/abs/1809.06476>. cited by applicant

Lu et al., "Chip-integrated visible-telecom entangled photon pair source for quantum communication," Nature Physics, vol. 15, Jan. 21, 2019, pp. 373-381, available at <https://doi.org/10.1038/s41567-018-0394-3>. cited by applicant

Matsushita et al., "Quasi-phase-matched parametric fluorescence in a periodically inverted GaP waveguide," Applied Physics Express, vol. 2, No. 6, May 22, 2009, pp. 061101. cited by applicant

May et al., "Second-harmonic generation in AlGaAs-on-insulator waveguides," Optics Letters, vol. 44, No. 6, Mar. 15, 2019, pp. 1339-1342. cited by applicant

Pasiskevicius et al., "Quasi-phase matched nonlinear media: Progress towards nonlinear optical engineering," Optical Materials, vol. 34, No. 3, Jul. 22, 2011, pp. 513-523. cited by applicant

Pernice et al., "Second harmonic generation in phase matched aluminum nitride waveguides and micro-ring resonators," Applied Physics Letters, vol. 100, No. 22, May 30, 2012, pp. 223501. cited by applicant

Rao et al., "Second-harmonic generation in periodically-poled thin film lithium niobate wafer-bonded on silicon," Optics Express, vol. 24, No. 26, Dec. 26, 2016, pp. 29941-29947. cited by applicant

Rao et al., "Second-harmonic generation in single-mode integrated waveguides based on mode-shape modulation," Applied Physics Letters, vol. 110, No. 11, Jan. 19, 2017, pp. 111109. cited by applicant

Schunemann et al., "Optical parametric oscillation in quasi-phase-matched GaP," Proc SPIE, Nonlinear Frequency Generation and Conversion: Materials, Devices, and Applications XIV, vol. 9347, Mar. 24, 2015, pp. 93470J. cited by applicant

Spencer et al., "Low kappa, narrow bandwidth Si₃N₄ Bragg gratings," Optics Express, vol. 23, No. 23, Nov. 16, 2015, pp. 30329-30336. cited by applicant

Thomson et al., "Low Loss MMI Couplers for High Performance MZI Modulators," IEEE Photonics Technology Letters, vol. 22, No. 20, Oct. 15, 2010, pp. 1485-1487. cited by applicant

Tison et al., "Path to increasing the coincidence efficiency of integrated resonant photon sources," Optics Express, vol. 25, No. 26, Dec. 25, 2017, pp. 33088-33096. cited by applicant

Wang et al., "Ultrahigh-efficiency wavelength conversion in nanophotonic periodically poled lithium niobate waveguides," Optica, vol. 5, No. 11, Nov. 7, 2018, pp. 1438-1441. cited by applicant

Wang et al., "Second harmonic generation in nano-structured thin-film lithium niobate waveguides," Optics Express, vol. 25, No. 6, Mar. 2017, pp. 6963-6973. cited by applicant

Yang et al., "Realization of controllable photonic molecule based on three ultrahigh-Q microtoroid cavities", Laser Photonics Reviews, vol. 11, No. 2, Jan. 16, 2017, paper 1600178, 8 pages. cited by applicant

Yoo et al., "Wavelength conversion by difference frequency generation in AlGaAs waveguides with periodic domain Inversion achieved by wafer bonding," Applied Physics Letters, vol. 68, No. 19, May 6, 1996, pp. 2609-2611. cited by applicant

Yu et al., "Efficient continuous wave second harmonic generation pumped at 1.55 m in quasi-phasesmatched AlGaAs waveguides," Optics Express, vol. 13, No. 26, Dec. 26, 2005, pp. 10742-10748. cited by applicant

International Search Report dated Jan. 11, 2022, regarding application No. PCT/US2021/071721, 4 pages. cited by applicant

Written Opinion of the International Searching Authority dated Jan. 11, 2022, regarding application No. PCT/US2021/071721, 7 pages. cited by applicant

International Search Report dated Jan. 14, 2022, regarding application No. PCT/US2021/071722, 5 pages. cited by applicant

Written Opinion of the International Searching Authority dated Jan. 14, 2022, regarding application No. PCT/US2021/071722, 6 pages. cited by applicant

International Search Report dated Feb. 8, 2022, regarding application No. PCT/US2021/071723, 5 pages. cited by applicant
Written Opinion of the International Searching Authority dated Feb. 8, 2022, regarding application No. PCT/US2021/071723, 8 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority dated Feb. 9, 2022, regarding application No. PCT/US2021/071725, 24 pages. 021/071725, 24 pages. cited by applicant

Non-Final Office Action dated Jul. 8, 2022, regarding U.S. Appl. No. 17/450,046, 23 pages. cited by applicant

Non-Final Office Action dated Aug. 10, 2022, regarding U.S. Appl. No. 17/450,031, 29 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation in part application of U.S. patent application Ser. No. 17/450,038 filed Oct. 5, 2021, and entitled “Optical Waveguide Structure With Partially Overlapping Loops In Direction Dependent Material,” which is related to and claims the benefit of priority of provisional U.S. Patent Application Ser. No. 63/088,220, entitled “Directional Phase Matching (DPM) Optical Waveguide”, filed on Oct. 6, 2020; provisional U.S. Patent Application Ser. No. 63/201,661, entitled “Directional Phase Matching Optical Waveguide”, filed on May 7, 2021; and provisional U.S. Patent Application Ser. No. 63/201,664, entitled “Nonlinear Optical Waveguide Structures for Light Generation and Conversion”, filed on May 7, 2021, all of which are hereby incorporated by reference. (2) This application is related to U.S. patent application Ser. No. 17/450,031 filed on Oct. 5, 2021 and entitled “Optical Waveguide Structure With Triple Partially Overlapping Loops”, and U.S. patent application Ser. No. 18/059,624 filed on Nov. 29, 2022, entitled “Optical Waveguide Structure With Partially Overlapping Loops In Direction Dependent Material,” assigned to the same assignee, and incorporated herein by reference in their entirety.

BACKGROUND INFORMATION

1. Field

(1) The present disclosure relates generally to optical waveguide structures and, in particular, to phase matching optical waveguide structures with partially overlapping loops to generate light using non-linear optical processes.

2. Background

(2) Optical waveguides are physical structures that guide electromagnetic waves in an optical spectrum. Optical waveguides can be used as components in integrated optical circuits. With respect to quantum communications and processing, nonlinear optical material structures can be used to create photon transmitters, repeaters, and other quantum devices for communications. Nonlinear optical structures can be used to change the light passing through them depending on factors such as orientation, temperature, wavelength of light, polarization of light, and other factors. For example, a waveguide with light of a blue wavelength passing through the waveguide can generate one or more photons of light that has a longer wavelength, such as green or red, and a correspondingly lower photon energy. This type of conversion can be performed using waveguides that incorporate a material having a second order nonlinear optical susceptibility or a third order nonlinear optical susceptibility.

(3) Current waveguides and structures that implement second order nonlinear optical processes are not as efficient as desired. Therefore, it would be desirable to have a method and apparatus that take into account at least some of the issues discussed above, as well as other possible issues. For example, it would be desirable to have a method and apparatus that overcome a technical problem with increasing efficiency in generating light in nonlinear optical waveguide structures.

SUMMARY

(4) In one illustrative embodiment, an optical waveguide structure comprises a nonlinear optical waveguide, a set of tuning optical waveguides, a set of wavelength selective couplers that couples light between the nonlinear optical waveguide and a tuning optical waveguide based on a wavelength of light, and a set of phase shifters located along one or more tuning optical waveguides in the set of tuning optical waveguides.

(5) In another illustrative embodiment, an optical waveguide structure comprises a nonlinear optical waveguide, a tuning optical waveguide, a set of wavelength selective couplers that couples light between the nonlinear optical waveguide and the tuning optical waveguide based on a wavelength of light, and a set of phase shifters located along the set of tuning optical waveguide.

(6) In yet another illustrative embodiment, a method facilitates a nonlinear optical interaction process. A wavelength selective coupler couples a first wavelength light from a first segment in a nonlinear optical waveguide into a second segment in the nonlinear optical waveguide. The wavelength selective coupler couples a second wavelength light from the first segment in the nonlinear optical waveguide into a tuning optical waveguide. A phase shifter applies an activation to the tuning optical waveguide to change a phase shift for the second wavelength light in the tuning optical waveguide.

(7) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is an illustration of a high level block diagram of an optical waveguide structure in accordance with an illustrative embodiment;

(3) FIG. 2 is another illustration of an optical waveguide structure in accordance with an illustrative embodiment;

(4) FIG. 3 is an illustration of a block diagram of optical waveguides in accordance with an illustrative example;

(5) FIG. 4 is an illustration of loops in optical waveguides in accordance with an illustrative embodiment;

- (6) FIG. 5 is an illustration of a block diagram of a configuration for nonlinear optical waveguides in accordance with an illustrative embodiment;
- (7) FIG. 6 is an illustration of phase shifters used to obtain at least one of resonance matching or roundtrip phase matching in accordance with an illustrative embodiment;
- (8) FIG. 7 is an illustration of a cross-section of an optical waveguide in accordance with an illustrative embodiment;
- (9) FIG. 8 is an illustration of light coupling by a wavelength-selective coupler in accordance with an illustrative embodiment;
- (10) FIG. 9 is an illustration of light coupling by a wavelength-selective coupler in accordance with an illustrative embodiment;
- (11) FIG. 10 is an illustration of light coupling by a wavelength-selective coupler in accordance with an illustrative embodiment;
- (12) FIG. 11 is an illustration of simulation results of light coupling by a wavelength-selective coupler in accordance with an illustrative embodiment;
- (13) FIG. 12 is an illustration of simulation results of light coupling by a wavelength-selective coupler in accordance with an illustrative embodiment;
- (14) FIG. 13 is an illustration of an optical waveguide structure with five optical waveguides in accordance with an illustrative embodiment;
- (15) FIG. 14 is an illustration of an optical waveguide structure with five optical waveguides in accordance with an illustrative embodiment;
- (16) FIG. 15 is an illustration of an optical waveguide structure with five optical waveguides in accordance with an illustrative embodiment;
- (17) FIG. 16 is an illustration of an optical waveguide structure with five optical waveguides in accordance with an illustrative embodiment;
- (18) FIG. 17 is an illustration of an optical waveguide structure with five optical waveguides in accordance with an illustrative embodiment;
- (19) FIG. 18 is an illustration of an optical waveguide structure with ten optical waveguides in accordance with an illustrative embodiment;
- (20) FIG. 19 is an illustration of a flowchart of a process for a non-linear optical process in accordance with an illustrative embodiment;
- (21) FIG. 20 is an illustration of a flowchart of additional operations for a process for a non-linear optical process in accordance with an illustrative embodiment;
- (22) FIG. 21 is an illustration of a flowchart of additional operations for a process for a non-linear optical process in accordance with an illustrative embodiment;
- (23) FIG. 22 is an illustration of a flowchart of an additional operation for a process for a non-linear optical process in accordance with an illustrative embodiment;
- (24) FIG. 23 is an illustration of a flowchart of additional operation for a process for a non-linear optical process in accordance with an illustrative embodiment;
- (25) FIG. 24 is an illustration of a flowchart of an additional operation for a process for a non-linear optical process in accordance with an illustrative embodiment;
- (26) FIG. 25 is an illustration of a flowchart of an additional operation for a process for a non-linear optical process in accordance with an illustrative embodiment;
- (27) FIG. 26 is an illustration of a flowchart of an additional operation for a process for a non-linear optical process in accordance with an illustrative embodiment;
- (28) FIG. 27 is an illustration of a block diagram of a product management system in accordance with an illustrative embodiment;
- (29) FIG. 28 is an illustration of a block diagram of an optical waveguide structure in accordance with an illustrative embodiment;
- (30) FIG. 29 is an illustration of an optical waveguide structure in accordance with an illustrative embodiment;
- (31) FIGS. 30A and 30B are illustrations of an optical waveguide structure in accordance with an illustrative embodiment;
- (32) FIGS. 31A and 31B are illustrations of an optical waveguide structure in accordance with an illustrative embodiment;
- (33) FIGS. 32A and 32B are illustrations of a block diagram of routes for light traveling through an optical waveguide structure in accordance with an illustrative embodiment;
- (34) FIG. 33 is an illustration of graphs of the effect of waveguide cross-sectional dimensions on the phase walk-off associated with imperfect wave vector matching in accordance with an illustrative embodiment;
- (35) FIG. 34 is an illustration of an optical waveguide structure with phase shifters for tuning light in separate tuning optical waveguides for signal light and for idler light in accordance with an illustrative example;
- (36) FIG. 35 is an illustration of an optical waveguide structure with phase shifters for tuning light in accordance with an illustrative example;
- (37) FIG. 36 is an illustration of a graph of light generation in accordance with an illustrative embodiment;
- (38) FIGS. 37A-37G are illustrations of cross-sections for nonlinear optical waveguide structures in accordance with an illustrative embodiment;
- (39) FIG. 38 is an illustration of an optical waveguide structure in accordance with an illustrative embodiment;
- (40) FIG. 39 is an illustration of an optical waveguide structure with offset tuning optical waveguides in accordance with an illustrative embodiment;
- (41) FIG. 40 is an illustration of an optical waveguide structure formed on an xy plane in accordance with an illustrative embodiment;
- (42) FIGS. 41A and 41B are illustrations of phase shifter cross sections in accordance with an illustrative embodiment;
- (43) FIG. 42 is an illustration of an optical waveguide structure in accordance with an illustrative embodiment;
- (44) FIG. 43 is an illustration of an optical waveguide structure with separate tuning optical waveguides for signal light and for idler light in accordance with an illustrative embodiment; and
- (45) FIG. 44 is an illustration of an optical waveguide structure with two groups of segments in which one group of the segments is associated with tuning optical waveguides in accordance with an illustrative embodiment.

DETAILED DESCRIPTION

- (46) The illustrative embodiments recognize and take into account one or more different considerations. For example, the illustrative embodiments recognize and take into account that a nonlinear optical structure can function as a resonator such that light of a particular wavelength in resonance with the resonator can travel over a longer distance in a nonlinear optical waveguide of the nonlinear optical structure as compared to light of other wavelengths that are not in resonance with the resonator.
- (47) The illustrative embodiments recognize and take into account that the loss of light from a resonator occurs when some of the light exits the resonator instead of continuing to travel in the resonator. The resonator selects certain wavelengths of light to continue traveling in the resonator. The illustrative embodiments recognize and take into account that different resonators can have different resonances that match to different wavelengths of light. The illustrative embodiments recognize and take into account that the same resonator can have multiple resonances, with different resonances matching to different wavelengths of the light.
- (48) The illustrative embodiments recognize and take into account that currently used nonlinear optical waveguide structures employ a

resonator that implements three-wave mixing and four-wave mixing processes to generate light of one wavelength from light of a different wavelength. In other words, the process changes the wavelength of the light. The illustrative embodiments recognize and take into account that spontaneous parametric down conversion (SPDC) is an example of a three-wave mixing process for generating certain wavelengths of light, such as a signal light and an idler light, in response to the introduction of source light of a different wavelength, such as a pump light, into the nonlinear optical waveguide structure. The illustrative embodiments recognize and take into account that spontaneous parametric down conversion can generate a pair of photons, such as a signal photon and an idler photon, from a pump photon.

(49) The illustrative embodiments recognize and take into account that the nonlinear optical waveguide structure, in forming a ring-shaped route for the travel by the light, can employ a nonlinear optical waveguide in which light of three wavelengths involved in spontaneous parametric down conversion and spontaneous four-wave mixing propagates within the nonlinear optical waveguide structure. The illustrative embodiments recognize and take into account that the ring-shaped route formed from the nonlinear optical waveguide structure can be a closed path of a ring resonator. The illustrative embodiments recognize and take into account that for current nonlinear optical waveguides structures, different wavelengths of the light must match resonances of the same resonator. The illustrative embodiments recognize and take into account that this requirement results in severe limitation on allowable wavelengths for the signal light and the idler light that are generated.

(50) The illustrative embodiments recognize and take into account that current optical waveguide structures can have optical structures to input and output light from the ring resonator. The illustrative embodiments recognize and take into account that the addition of these input and output optical structures is unhelpful for achieving the resonance match because the three wavelengths for the pump light, the signal light, and the idler light propagate through the ring resonator and are constrained to match the modes of the same ring resonator.

(51) The illustrative embodiments recognize and take into account that current nonlinear optical waveguide structures can employ two coupled ring resonators having different values for their circumferences. The illustrative embodiments recognize and take into account that these different values can result in different sets of resonance modes for the two resonators. The illustrative embodiments recognize and take into account that a first resonator can have all three wavelengths for the pump light, the signal light, and the idler light matched to the modes for the first resonator. The illustrative embodiments recognize and take into account that the second resonator can have modes matched to the wavelengths of the signal light and the idler light. The illustrative embodiments recognize and take into account that these two coupled resonators still have the same limitations on resonance matching as a single ring resonator since wavelengths of the signal light and of the idler light must match with resonances of both resonators. The illustrative embodiments recognize and take into account that the use of three coupled ring resonators may provide some improvement, but still have limitations because at least some of the light from all of the three wavelengths travels through all three rings in the current nonlinear optical waveguide structures.

(52) The illustrative embodiments recognize and take into account that current nonlinear optical waveguide structures employ multiple resonators that are coupled together directly through common wavelengths and not through a nonlinear optical process. The illustrative embodiments recognize and take into account that at least some light for all of the wavelengths travel through all of these multiple resonators. In other words, the illustrative embodiments recognize and take into account that the light with different wavelengths and traveling through all of the resonators is resonant with each of the individual resonators that are coupled together.

(53) With currently used spontaneous parametric down conversion or spontaneous four-wave mixing, all three wavelengths involved in the nonlinear optical process are adjusted to match resonances of the same ring resonator or to match common resonances of multiple coupled rings. However, this type of adjustment of the wavelengths may not be possible if an entangled photon pair, such as entangled pair of idler and signal photons, is used in a quantum photonic circuit that also contains other sources of such photon pairs. The need in quantum photonics to perform optical interference functions involving photons produced by different sources of entangled photons may require those photons to have the same wavelength, so that photons can be indistinguishable.

(54) As a result, adjusting the wavelengths associated with a first ring resonator whose output photons are involved in an optical interference function can cause a need to also adjust the wavelengths associated with a second ring resonator whose output photons are interfered with the photons from the first ring resonator. However, if those two ring resonators are not identical, such adjustment may be beyond what is permitted by the spectral width of the resonances of the two ring resonators.

(55) For example, a departure of a dimension of the fabricated waveguide, such as the waveguide width, by only 1-2 nm would shift the resonance wavelength beyond the spectral width associated with a quality factor (Q) of 10.^{sup.3}. Resonators with a higher Q have resonances with narrower spectral width, thereby making them impractical for use in quantum photonic circuits. Thus, if multiple currently available ring resonators are used in a quantum photonic circuit, those resonators would need to have a low Q.

(56) As a result, the nonlinear optical interaction distance for producing the entangled photon pairs by spontaneous parametric down conversion or spontaneous four-wave mixing would be much shorter and the photon-pair generation rates would be much lower.

(57) The optical waveguide structure in the illustrative examples provides design flexibility to enable three loops through the waveguides to have resonances that correspond to three pre-specified wavelengths. Also, if multiple optical waveguide structures are used together in a quantum photonic circuit, these optical waveguide structures can be adjusted to make the resonances of the optical waveguide structures correspond to specified wavelengths. This type of adjustment is in contrast to having all of the wavelengths adjusted to correspond to one resonator. Thus, the loops in the optical waveguide structures in a quantum photonic circuit can have a higher Q, enabling those optical waveguide structures to generate photon pairs at higher generation rates.

(58) In an illustrative example, the optical waveguide structure can be a triple partially overlapping loops for entanglement (TriPOLE) optical waveguide structure that is used in illustrative examples to produce entangled photon pairs by nonlinear optical (NLO) processes. These nonlinear optical processes can be, for example, spontaneous parametric down conversion and spontaneous four-wave mixing. The two entangled photons produced by spontaneous parametric down conversion can be entangled when those photons are produced from the same pump photon. In a similar fashion, the two entangled photons produced by spontaneous four-wave mixing can be entangled when those photons are produced from the same two degenerate pump photons.

(59) In this illustrative example, nonlinear optical waveguides in the form of ring resonators can be used to increase the generation rate of these entangled photon pairs, comprising a signal photon and an idler photon. In a high-Q ring resonator, light can travel many times around the circumference of the ring resonator. Thus, the interaction length of a ring resonator can be many times greater than its physical size. In implementing spontaneous parametric down conversion or spontaneous four-wave mixing with three partially overlapping ring resonators as in this example, all three wavelengths of light involved in the nonlinear optical process correspond to resonances of their individual resonators.

(60) In an illustrative example, the optical waveguide structure is configured such that light of a particular wavelength can travel on a particular loop through the optical waveguide structure in which the loop is present for that particular wavelength of the light. In the illustrative examples, the loops are partially overlapping such that light of two different wavelengths are not required to travel along the

same exact loop.

(61) In one illustrative example, an optical waveguide structure comprises a main nonlinear optical waveguide; an extension optical waveguide; a secondary optical waveguide; a first wavelength-selective coupler; and a second wavelength-selective coupler. The first wavelength-selective coupler optically couples a first main location in the main nonlinear optical waveguide and a primary location in the extension optical waveguide to each other. The second wavelength-selective coupler optically couples a second main location in the main nonlinear optical waveguide and a secondary location in the extension optical waveguide to each other. The first wavelength-selective coupler also optically couples a first main location in the main nonlinear optical waveguide and a first location in the secondary optical waveguide to each other. The second wavelength-selective coupler also optically couples a second main location in the main nonlinear optical waveguide and a second location in the secondary optical waveguide to each other.

(62) With this example, light of different wavelengths travels on different loops in the optical waveguide structure. A route is a path in which the light travels. In this illustrative example, a loop is a closed route. For example, a first loop can be present in which light of a first wavelength (a first-wavelength light) travels on a first loop having a first length. This first loop can extend through the main nonlinear optical waveguide and a portion of an extension optical waveguide. A second loop can extend through a portion of the main nonlinear optical waveguide and a portion of a secondary optical waveguide. Light of a second wavelength (a second-wavelength light) can travel in the second loop having a second length. The second length can be different from the first length.

(63) In this example, the first wavelength-selective coupler and the second wavelength-selective coupler can be selected to cause light of a particular wavelength to travel from one optical waveguide to another optical waveguide. For example, the first wavelength-selective coupler can cause the second-wavelength light to be coupled from the main nonlinear optical waveguide to the secondary optical waveguide. The second wavelength-selective coupler can cause the second-wavelength light to be coupled from the secondary optical waveguide back to the main nonlinear optical waveguide. The second length is determined by the first-main and second-main locations and by the first-secondary and second-secondary locations as well as by the length of the secondary optical waveguide portion (or portions) between these first-secondary and second-secondary locations. The length of the secondary optical waveguide portion (or portions) between the first-secondary and second-secondary locations can be selected to obtain a desired value for the second length.

(64) The length of the portions of secondary optical waveguide are selected to achieve a desired value for the second length. This desired value can be selected to achieve a resonance condition for a particular wavelength of light.

(65) In this example, the first wavelength-selective coupler also can cause the first-wavelength light to be coupled from the main nonlinear optical waveguide to the extension optical waveguide. The second wavelength-selective coupler can cause the first-wavelength light to be coupled from the extension optical waveguide back to the main nonlinear optical waveguide. The first length is determined by first main location and the second main location in the main nonlinear optical waveguide, the primary-extension location and secondary-extension location in the extension waveguide as well as by the length of the primary optical waveguide portion between these primary-extension and secondary-extension locations. The length of the primary optical waveguide portion between these primary-extension and secondary-extension locations can be selected to obtain a desired value for the first length.

(66) In the illustrative example, with this optical waveguide structure, the loops for the different light of different wavelengths in the optical waveguides can have lengths that can be selected such that at least one of resonance or round-trip phase matching is present for the different light of different wavelengths traveling on the different routes.

(67) As used herein, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

(68) For example, without limitation, “at least one of item A, item B, or item C” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

(69) In other words, the length can be selected for a loop such that resonance is achieved for the light traveling in a loop. This type of selection of the length can be made for each loop independently of the lengths for other loops in the optical waveguide structure. In the illustrative example, resonance occurs for each wavelength of the light separately from the other wavelengths of the light.

(70) Round-trip phase matching can be achieved for the combination of three loops in which three wavelengths of light travel. Round-trip phase matching involves all three wavelengths of the light. In the illustrative example, the lengths of all three loops are selected jointly such that round-trip phase matching is achieved for the nonlinear optical interaction between the three wavelengths of the light.

(71) In some illustrative examples, an optical waveguide structure can be a resonator-enhanced structure for nonlinear optical (NLO) three-wave mixing processes. These nonlinear optical three-wave mixing processes can include difference frequency generation, sum frequency generation, and spontaneous parametric down conversion (SPDC). In other illustrative examples, an optical structure can be a resonator-enhanced structure for degenerately pumped or degenerate output nonlinear optical (NLO) four-wave mixing processes. These degenerate output nonlinear optical four-wave mixing processes can be, for example, difference frequency generation, sum frequency generation, and spontaneous four-wave mixing (SFWM). In these illustrative examples, degenerate means at least two of the waves participating in the nonlinear optical process have the same wavelength. Further, a degenerate three-wave mixing process, such as second harmonic generation, can be used. With second harmonic generation, the two input waves have the same wavelength and produce an output wave of a different wavelength.

(72) In the illustrative examples, the nonlinear optical processes can involve three distinct wavelengths of light, a first wavelength, a second wavelength, and a third wavelength. The nonlinear optical waveguide structure in the different illustrative examples comprises triple partially overlapping loops for entanglement (TriPOLE). This optical waveguide structure comprises a main nonlinear optical waveguide, a first extension optical waveguide, a second extension optical waveguide, and a third extension optical waveguide in which light of different wavelengths travels in loops that extend through different combinations of these different optical waveguides. A first loop extends through the main nonlinear optical waveguide and a first extension optical waveguide. This first loop is overlapped by parts of two other loops, which are a second loop and a third loop. A second loop extends through the main nonlinear optical waveguide and a second extension optical waveguide. A third loop extends through the main nonlinear optical waveguide and a third extension optical waveguide. These loops can be closed routes that define optical resonators having resonances at specific sets of wavelengths.

(73) The parts of the first loop, the second loop and the third loop that are in common with or that extend through the main nonlinear optical waveguide in the optical waveguide structure are the portions of the optical waveguide structure in which the nonlinear optical three-wave mixing or four-wave mixing processes can occur. In the illustrative examples, the first extension optical waveguide is physically separate

from the main nonlinear optical waveguide and is connected to the main nonlinear optical waveguide by a first wavelength-selective coupler that selectively couples only the first-wavelength light into that first extension optical waveguide, but does not couple the second-wavelength light or the third-wavelength light into that first extension optical waveguide. In other words, the first wavelength-selective coupler optically connects the first extension optical waveguide to the main nonlinear optical waveguide only for the first-wavelength light. A second wavelength-selective coupler can couple the first-wavelength light from the first extension optical waveguide back into the main nonlinear optical waveguide.

(74) In this illustrative example, the second extension optical waveguide and the third extension optical waveguide are connected to the main nonlinear optical waveguide through a segment of a secondary optical waveguide. In this example, the first wavelength-selective coupler couples the second-wavelength light and the third-wavelength light into a first segment of the secondary optical waveguide. A third wavelength-selective coupler selectively couples the second-wavelength light into the second extension optical waveguide, but the third wavelength-selective coupler does not couple the third-wavelength light into that second extension optical waveguide.

(75) In other words, the third wavelength-selective coupler optically connects the second extension optical waveguide to the secondary optical waveguide. The third wavelength-selective coupler also selectively couples the third-wavelength light into the third extension optical waveguide, but the third wavelength-selective coupler does not couple the second-wavelength light into that third extension optical waveguide. In other words, this third wavelength-selective coupler optically connects the third extension optical waveguide to the secondary optical waveguide. As a result, the selection is between the second wavelength and the third wavelength. The first wavelength is assumed to not be present in the secondary optical waveguide in this example.

(76) In an illustrative example, the first wavelength-selective coupler couples the second-wavelength light from the main nonlinear optical waveguide to travel in the second extension optical waveguide of the second loop, via a third wavelength-selective coupler, but does not couple light of the first wavelength from the main nonlinear optical waveguide to travel in the second extension optical waveguide. The first wavelength-selective coupler also couples the third-wavelength light from the main nonlinear optical waveguide to travel in the third extension optical waveguide of the third loop, via the third wavelength-selective coupler, but does not couple light of the first wavelength from the main nonlinear optical waveguide to travel in the third extension optical waveguide.

(77) The third wavelength-selective coupler couples the light of the second wavelength from the main nonlinear optical waveguide, via the first waveguide-selective coupler, to the second extension optical waveguide of the second loop but does not couple light of the first or third wavelengths into the second extension optical waveguide. The third wavelength-selective coupler also couples the light of the third wavelength from the main nonlinear optical waveguide, via the first wavelength-selective coupler, to travel in the third extension optical waveguide of the third loop but does not couple the light of the first or second wavelengths into the third extension optical waveguide. Thus, only the second-wavelength light travels a second length through the entire second loop. Also, only the third-wavelength light travels a third length through the entire third loop. The first-wavelength light travels only a first length through the first loop that includes the main nonlinear optical waveguide and the first extension optical waveguide, but does not include the second extension optical waveguide or the third extension optical waveguide.

(78) The main nonlinear optical waveguide is common to all three loops. The first-wavelength light travels in a first loop that includes the main nonlinear optical waveguide and the first extension optical waveguide. In this example, the first loop also can include the first wavelength-selective coupler and the second wavelength-selective coupler. The second-wavelength light travels in a loop that includes the main nonlinear optical waveguide and the second extension optical waveguide. The third-wavelength light travels in a third loop that includes the main nonlinear optical waveguide and the third extension optical waveguide. Each of the three loops has a length that is designed to be resonant for the light that travels in the loop. The three loops can have different lengths.

(79) The length of the first loop for the light of the first wavelength can be selected such that the first-wavelength light is at a resonance of a first resonator comprising the main nonlinear optical waveguide and the first extension optical waveguide.

(80) The length of the second loop for the light of the second wavelength can be selected such that the second-wavelength light is at a resonance of a second resonator comprising the main nonlinear optical waveguide and the second extension optical waveguide. The length of the third loop for the light of the third wavelength can be selected so that the third-wavelength light is at a resonance of a third resonator formed by the main nonlinear optical waveguide and the third extension nonlinear optical waveguide.

(81) In the illustrative example, a loop may traverse one or more of these optical waveguides. The loops through these optical waveguides can partially overlap with each other. In other words, the different loops are not identical to each other but may have overlaps within the optical waveguide structures.

(82) Thus, although the propagation constants or wave vectors for the three wavelengths may be different from each other, the light at the three different wavelengths can still be at resonances when propagating in their respective loops in the optical waveguide structure. The propagation can occur such that the light of the three wavelengths can propagate constructively over many cycles through loops within the optical waveguide structure. This type of propagation can occur because the three loops have different lengths. Furthermore, the relative lengths of the three loops can be selected to meet the phase-matching requirement to sustain the nonlinear optical process over an interaction distance that is greater than the length of the main nonlinear optical waveguide in the optical waveguide structure.

(83) The phase matching can be a feature distinct from the resonance that occurs for a resonator in the optical waveguide structure. Thus, five constraints may be applied to the nonlinear optical interaction that occurs in the optical waveguide structure. One constraint is on “energy conservation” which constrains the relationship between the three wavelengths. The other four constraints relate to the propagation constants or wave vectors of the light of the three different wavelengths.

(84) The phase-matching condition for the nonlinear optical process occurring in the main nonlinear optical waveguide can be described by a phase walk-off and by a constructive interaction distance. The constructive interaction distance is the distance at which a phase walk-off for the nonlinear optical interaction between the light of the three wavelengths equals 180 degrees or π radians.

(85) When the phase walk-off has a value between 0 and π radians, the nonlinear optical interaction is “constructive” and transfers power from the pump into the signal and idler. This transfer of power increases the generation of signal and idler light. However, when the phase walk-off has a value between π and 2π radians, the nonlinear optical interaction is “destructive” and transfers power from the signal and idler back to the pump, thereby reducing the generation of signal and idler light.

(86) Constructive generation of signal and idler light occurs for values of the phase walk-off between 0 and π , between 2π and 3π , between 4π and 5π , etc. Destructive generation of signal and idler occurs for values of the phase walk-off between π and 2π , between 3π and 4π , between 5π and 6π , etc.

(87) Whether the nonlinear optical generation is constructive or destructive can also depend on the sign of the nonlinear optical coefficient of the nonlinear optical material involved in that nonlinear optical process. For the same value of the phase walk-off, if the sign of the nonlinear optical coefficient changes, the generation can change from being constructive to being destructive, and vice versa.

(88) In some illustrative examples, the length of the main nonlinear optical waveguide, in which all three wavelengths of light travel, can be set to be no greater than the constructive interaction distance. This length of the main nonlinear optical waveguide can be the length of multiple separate segments.

(89) The length of the first extension optical waveguide, the length of the second extension optical waveguide, and the length of the third extension optical waveguide (when present) can be set such that the roundtrip phase walk-off for the nonlinear optical interaction between the light of the three wavelengths is a specified value. This round-trip phase walk-off can be set equal to zero or as close to being zero as possible, or can be set as close as possible to being a multiple of 2π radians or 360 degrees.

(90) In some examples, tuning electrodes can be located at optical waveguides. For example, the first extension optical waveguide can have a set of tuning electrodes that operates to adjust the roundtrip phase of the light of the first wavelength. The second extension optical waveguide can have a set of tuning electrodes that operate to adjust the roundtrip phase of the light of the second wavelength. The third extension optical waveguide can have a set of tuning electrodes that operate to adjust the roundtrip phase of the light of the third wavelength. The main nonlinear optical waveguide can have a set of phase shifters, such as a set of tuning electrodes, that operate to adjust the roundtrip phase of the light of all three wavelengths, and in particular of the first wavelength. Thus, these tuning electrodes can enable adjusting the resonance conditions to compensate for changes in at least one of the wavelengths of the light, the cross-sectional dimensions of the optical waveguides, and environmental conditions, such as temperature, or other factors. These tuning electrodes can also allow the optical waveguide structure to adjust the phase walk-off for the nonlinear optical interaction occurring in the main nonlinear optical waveguide.

(91) For example, a structure for spontaneous parametric down conversion can have the light such as, the pump light, supplied to the main nonlinear optical waveguide through an input optical coupler and travel in the first loop. The optical coupler can be connected to an input optical waveguide that receives the pump light. The signal light and the idler light generated by the spontaneous parametric down conversion process would travel in the second loop and the third loop, respectively.

(92) A nonlinear optical generation process such as spontaneous parametric down conversion can result in generation of lower intensity light from higher intensity light. A nonlinear optical generation process also can result in the generation of a higher intensity light from a lower intensity light. However, since the efficiency of a nonlinear optical generation process depends on the intensity of the input or source light for that process, which typically is the pump light, a nonlinear optical process typically results in generation of additional lower intensity light from the higher intensity light. Typically, the pump light has an intensity that is at least twice the intensity of the signal light and at least twice the intensity of the idler light. In some examples, such as many examples as spontaneous parametric down conversion, the intensity of the pump light is at least ten times greater than the intensity of the signal light or of the idler light. Thus, even when a phase-matched condition is present, if the pump light is absent from an optical waveguide comprising nonlinear optical material, and only signal and idler light are present, there is much less generation of pump light from that weaker signal and idler light.

(93) In the illustrative examples, an optical waveguide structure can comprise a first nonlinear optical waveguide segment, a second nonlinear optical waveguide segment, an extension optical waveguide, a first wavelength-selective coupler, and a second wavelength-selective coupler. A first-wavelength light and a second-wavelength light travel in the first nonlinear optical waveguide segment. A second-order nonlinear optical process such as spontaneous parametric down conversion can occur in the first and second nonlinear optical waveguide segments. The first nonlinear optical waveguide segment has a nonlinear optical coefficient of a first sign. The second nonlinear optical waveguide segment has a nonlinear optical coefficient of a second sign, which is opposite from the first sign. In this illustrative example, this second nonlinear optical segment is part of the second extension waveguide or the third extension waveguide. It is desirable to divert the pump light away from these extension segments for the signal and idler light. In this example these extension segments comprise electro-optic material to enable them to provide voltage-controlled phase shifts.

(94) The first wavelength-selective coupler can optically couple a first location in the first nonlinear optical waveguide segment and a primary extension location in the extension optical waveguide to each other such that the first-wavelength light is coupled from the first nonlinear optical waveguide at the first location to the extension optical waveguide at the primary extension location. The second wavelength-selective coupler can optically couple a second location in the first nonlinear optical waveguide segment and a secondary extension location in the extension optical waveguide to each other such that the first-wavelength light is coupled from the extension optical waveguide at the secondary extension location to the main nonlinear optical waveguide at a location in the first nonlinear optical waveguide segment. Thus, the first-wavelength light bypasses the second nonlinear optical waveguide segment that has a nonlinear optical coefficient of a second sign, which is opposite from the first sign. Instead, the first-wavelength light travels only through the first nonlinear optical waveguide segment that has a nonlinear optical coefficient of the first sign.

(95) In the illustrative examples, the wavelength-selective couplers enable selective coupling of light in a manner that directs light of different wavelengths to either travel through or to bypass two different nonlinear optical waveguide segments that have nonlinear optical coefficients of opposite sign.

(96) Some examples of the optical waveguide structures can avoid undesired effects of the sign reversal in the nonlinear optical coefficient by removing the pump light or by having an absence of a non-linear optical material in part of the loop traversed by the pump light. Other examples of the optical waveguide structures can take advantage of a sign reversal in the nonlinear optical coefficient by adjusting the phase walk-off to compensate for the sign reversal in the nonlinear optical coefficient for two different segments of nonlinear optical waveguide.

(97) With reference now to the figures and, in particular, with reference to FIG. 1, an illustration of a high level block diagram of an optical waveguide structure is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **100** comprises optical waveguides **102** in which at least one of optical waveguides **102** is comprised of nonlinear optical material **104**. One or more other optical waveguides in optical waveguides **102** can be comprised of at least one of nonlinear optical material **104** or non-nonlinear optical material **105**. In this example, nonlinear optical material **104** can have first sign **107** and second sign **109** for nonlinear optical coefficient **112** for a nonlinear optical interaction of light with nonlinear optical material **104**. As depicted, first sign **107** is opposite of second sign **109**. Nonlinear optical coefficient **112** is a coefficient that quantifies the strength of the nonlinear optical interaction.

Nonlinear optical coefficient **112** can be a second order nonlinear optical coefficient or a third order nonlinear optical coefficient depending on the type of optical process implemented in optical waveguide structure **100**. One or more optical waveguides in optical waveguides **102** also can be comprised of an electro-optic material **103**. The refractive index of an electro-optic material **103** can be changed by applying a DC or low-frequency (as compared to the optical frequency) electric field to the material. In some cases, a material can be both a nonlinear optical material **104** as well as an electro-optic material **103**.

(98) In this illustrative example, optical waveguide structure **100** can also include at least one of input optical waveguides **164** or output optical waveguides **166**. In this illustrative example, input optical waveguides **164** and output optical waveguides **166** are connected to one or more of optical waveguides **102** using optical couplers **130**.

(99) For example, a set of input optical waveguides **164** can input input-light **168** into one or more of optical waveguides **102**. As another example, a set of output optical waveguides **166** can output output-light **170** from one or more of optical waveguides **102**. The input of input light **168** and output of output light **170** can be facilitated by a set of optical couplers **130** that connect the set of input optical waveguides or the set of output optical waveguides to one or more of optical waveguides **102**.

(100) As used herein, a “set of” when used with reference items means one or more items. For example, a set of input optical waveguides **164** is one or more of input optical waveguides **164**.

(101) In this illustrative example, light generation can be improved for optical waveguide structure **100** using optical waveguides **102** arranged as loops **116** through optical waveguides **102**. In the illustrative example, loops **116** are defined as the course of travel of light **118** within one or more of optical waveguides **102**. In other words, loops **116** are defined as where light **118** travels within optical waveguides **102**.

(102) The manner in which optical waveguides **102** are coupled to each other is through mechanisms such as wavelength-selective couplers **114**, which can be used to define loops **116** along which light **118** can travel within optical waveguides **102**. In the illustrative example, loops **116** can use different portions of optical waveguides **102** and wavelength-selective couplers **114** in optical waveguide structure **100**.

(103) As depicted, optical waveguide structure **100** also includes wavelength-selective couplers **114** that can be used to define routes **115** in the form of loops **116** for light **118** traveling within optical waveguide structure **100**. These wavelength-selective couplers can selectively direct light **118** from one optical waveguide to another optical waveguide in optical waveguides **102**.

(104) Wavelength-selective couplers **114** can take a number of different forms. For example, wavelength-selective couplers **114** can be selected from at least one of a two-waveguide coupler, a multi-mode interference coupler, a pulley coupler, a Mach-Zehnder interferometer, a 4-port micro-ring resonator coupler, or some other suitable wavelength-selective coupler that can couple light and determine which wavelengths of light are directed through coupling from one optical waveguide to another optical waveguide.

(105) As used herein, a “number of” when used with reference items means one or more items. For example, a number of different forms is one or more different forms.

(106) In this illustrative example, optical waveguides **102** in optical waveguide structure **100** can support the propagation of light **118** through routes **115** in the form of loops **116**, which are closed routes. Light **118** travels within optical waveguides **102** along routes **115**. In the illustrative example, a closed route is a route for which a starting point and ending point are common or for which no distinct starting point that is separate from an ending point is present. The closed route is also referred to as a loop.

(107) In this illustrative example, loops **116** can traverse multiple optical waveguides **102** in optical waveguide structure **100**. Loops **116** also can traverse one or more of wavelength-selective couplers **114** in optical waveguide structure **100**. Loops **116** can comprise multiple loops that overlap each other in portions of some of optical waveguides **102** in optical waveguide structure **100** but do not overlap each other for other optical waveguides **102** traversed by a loop of loops **116**. Different wavelengths of light **118** can travel through different loops. In other words, overlap is present between portions of loops **116** for the different wavelengths of light **118** traveling through optical waveguides **102**.

(108) As depicted, wavelength-selective couplers **114** can operate to define different loops in loops **116** for the different wavelengths of light **118**, with these different loops having different lengths.

(109) As depicted in this illustrative example, nonlinear optical material **104** has nonlinear optical coefficient **112**. In the illustrative example, nonlinear optical coefficient **112** can be a second order nonlinear optical coefficient or a third order nonlinear optical coefficient depending on the type of optical process implemented in optical waveguide structure **100**.

(110) Nonlinear polarization can occur in nonlinear optical material **104** in which the material polarization no longer varies linearly with the electric field amplitude. This nonlinear relationship can be expressed as follows:

$$P = \chi_{\text{sup}}(1)E + \chi_{\text{sup}}(2)EE + \chi_{\text{sup}}(3)EEE + \dots$$

(111) where E is the electric field, $\chi_{\text{sup}}(1)$ is the linear optical susceptibility, $\chi_{\text{sup}}(2)$ is the second order nonlinear optical susceptibility, etc. The nonlinear susceptibilities, such as $\chi_{\text{sup}}(2)$ and $\chi_{\text{sup}}(3)$, represent the nonlinear parts of the material dipolar characteristics.

(112) In this example, the electric field amplitude is the electric field amplitude of the light wave, which is an electromagnetic field. An electromagnetic field has a traveling (or propagating) electric field and a traveling (or propagating) magnetic field.

(113) In this illustrative example, nonlinear optical process **140** can be nonlinear optical mixing processes that can occur within optical waveguide structure **100**. These nonlinear optical mixing processes can be used to generate light **118**. For example, the propagation of first-wavelength light **132** can result in the generation of at least one of second-wavelength light **134** or third-wavelength light **136** using one or more nonlinear optical waveguides employing nonlinear optical mixing processes in optical waveguides **102**.

(114) In the illustrative example, nonlinear optical mixing processes can include nonlinear optical three-wave mixing processes and nonlinear optical four-wave mixing processes. In this illustrative example, the nonlinear optical three-wave mixing processes and the nonlinear optical four-wave mixing processes can include difference frequency generation (DFG) and sum frequency generation (SFG). The nonlinear optical three-wave mixing processes can also include spontaneous parametric down conversion (SPDC). The nonlinear optical four-wave mixing can also include spontaneous four-wave mixing (SFWM).

(115) In this illustrative example, nonlinear optical wave-mixing processes can include three types of light with three distinct wavelengths such as first-wavelength light **132**, second-wavelength light **134**, and third-wavelength light **136**.

(116) For example, nonlinear optical process **140** such as spontaneous three-wave mixing is a second-order nonlinear optical process that can occur in an optical waveguide having nonlinear optical material **104** in optical waveguides **102**. In this process, pair of generated photons **142** are generated from source photons **144** in optical waveguides **102** that have nonlinear optical material **104**. Generated photons **142** of a pair can have different wavelengths from each other, such as of second-wavelength light **134** and third-wavelength light **136** and have wavelengths different from the wavelength, such as first-wavelength light **132**, of source photons **144**.

(117) In this illustrative example, “resonance matching” means a given wavelength is matched to a resonance of a resonator. A resonator can have many resonances. Also, a resonator can be designed such that different lengths can still produce resonance matching for a particular wavelength of light. Resonance is achieved every time the round-trip phase is a multiple of 2π . In this illustrative example, lengths for loops **116** can be selected such that at least one of resonance matching or roundtrip phase matching is present for different wavelengths of light **118**.

(118) The lengths for loops **116** can be selected based on the locations where wavelength-selective couplers **114** connect to optical waveguides **102**.

(119) Thus, optical waveguide structure **100** can have multiple optical waveguides in optical waveguides **102** that are configured or constructed to enable the propagation of light **118** of different wavelengths to travel within optical waveguide structure **100** in a constructive manner. In one illustrative example, the light **118** of the different wavelengths can travel on loops **116** in which each loop is selected to

enable light **118** of a particular wavelength to travel in a particular manner. For example, a loop in loops **116** can traverse through both a main nonlinear optical waveguide and extension optical waveguides in optical waveguides **102** that extend the length of the loop in loops **116** for different wavelengths of light beyond that provided by the main nonlinear optical waveguide.

(120) Additionally, some loops in loops **116** can extend through both the main nonlinear optical waveguide and one or more parts of a secondary waveguide in addition to or in place of the extension optical waveguides. As a result, a loop in loops **116** for a light of a particular wavelength can traverse one or more of optical waveguides **102**.

(121) Thus, although the propagation constants or wave vectors for the light of three wavelengths may be different from each other, the light at the three different wavelengths can still be at resonances when propagating on their respective loops in optical waveguides **102**. The propagation can occur such that light **118** of the three wavelengths can propagate constructively over many cycles through loops **116** within the optical waveguide structure **100**. This type of propagation can occur because loops **116** have different lengths that are selected to be constructive for light of a particular wavelength.

(122) Turning next to FIG. 2, another illustration of an optical waveguide structure is depicted in accordance with an illustrative embodiment. In the illustrative examples, the same reference numeral may be used in more than one figure. This reuse of a reference numeral in different figures represents the same element in the different figures.

(123) As depicted in this illustrative example, optical waveguide structure **100** comprises optical waveguides **102**. As depicted, optical waveguides **102** include main nonlinear optical waveguide **106**, first extension optical waveguide **108**, secondary optical waveguide **113**, and first wavelength-selective coupler **120**, and second wavelength-selective coupler **122**. In this example, main nonlinear optical waveguide **106** comprises a nonlinear optical material **104**. Main nonlinear optical waveguide **106** also can comprise an electro-optic material **103**. First extension optical waveguide **108** and secondary optical waveguide **113** can comprise a nonlinear optical material **104**, a non-nonlinear optical material **105**, or a combination of a nonlinear optical material and one or more non-nonlinear optical materials. Main nonlinear optical waveguide **106** can comprise a single optical waveguide segment or can comprise multiple optical waveguide segments that are physically separate from each other. Secondary optical waveguide **113** likewise can comprise a single optical waveguide segment or can comprise multiple optical waveguide segments that are physically separate from each other.

(124) In this example, first-wavelength light **512** of a first wavelength and second-wavelength light **518** of a second wavelength travel in the main nonlinear optical waveguide **106**. As an example, first-wavelength light **512** can be a pump light with second-wavelength light **518** being at least one of a signal light or an idler light.

(125) In this illustrative example, first wavelength-selective coupler **120** optically couples first main location **520** in main nonlinear optical waveguide **106** and primary extension location **522** in first extension optical waveguide **108** to each other. First wavelength-selective coupler **120** optically couples these two optical waveguides such that first-wavelength light **512** is coupled from main nonlinear optical waveguide **106** at first main location **520** to first extension optical waveguide **108** at primary extension location **522**.

(126) Second wavelength-selective coupler **122** optically couples second main location **524** in main nonlinear optical waveguide **106** and secondary extension location **526** in first extension optical waveguide **108** to each other. In this example, second wavelength-selective coupler **122** optically couples these two optical waveguides such that first-wavelength light **512** is coupled from first extension optical waveguide **108** at secondary extension location **526** to main nonlinear optical waveguide **106** at second main location **524**.

(127) In this example, first-wavelength light **512** travels in first loop **528** that traverses through portions of main nonlinear optical waveguide **106**, portions of first extension optical waveguide **108**, first wavelength-selective coupler **120** and second wavelength-selective coupler **122**. In this example, first loop **528** has first length **530**.

(128) In this illustrative example, first wavelength-selective coupler **120** also optically couples first main location **520** in main nonlinear optical waveguide **106** and first secondary location **511** in secondary optical waveguide **113** to each other. First wavelength-selective coupler **120** optically couples these two optical waveguides such that second-wavelength light **518** is coupled from main nonlinear optical waveguide **106** at first main location **520** to secondary optical waveguide **113** at first secondary location **511**.

(129) In this example, second wavelength-selective coupler **122** also optically couples second main location **524** in main nonlinear optical waveguide **106** and second secondary location **513** in secondary optical waveguide **113** to each other. In this example, second wavelength-selective coupler **122** optically couples these two optical waveguides such that second-wavelength light **518** is coupled from secondary optical waveguide **113** at second secondary location **513** to main nonlinear optical waveguide **106** at second main location **524**.

(130) In this illustrative example, second-wavelength light **518** travels in main nonlinear optical waveguide **106** and is coupled from main nonlinear optical waveguide **106** at first main location **520** to secondary optical waveguide **113** at first secondary location **511** and travels in secondary optical waveguide **113** to second secondary location **513**. Second-wavelength light **518** is coupled from secondary optical waveguide **113** at second secondary location **513** to main nonlinear optical waveguide **106** at second main location **524** by second wavelength-selective coupler **122** such that second-wavelength light **518** travels in second loop **534** having second length **536** for second-wavelength light **518**.

(131) Second loop **534** includes portions of main nonlinear optical waveguide **106**, portions of secondary optical waveguide **113**, first wavelength-selective coupler **120** and second wavelength-selective coupler **122**.

(132) With reference next to FIG. 3, an illustration of a block diagram of optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguides **102** comprise main nonlinear optical waveguide **106**, secondary optical waveguide **113**, first extension optical waveguide **108**, second extension optical waveguide **110**, and third extension optical waveguide **119**. Each of these waveguides, main nonlinear optical waveguide **106**, secondary optical waveguide **113**, first extension optical waveguide **108**, second extension optical waveguide **110**, and third extension optical waveguide **119** can comprise one or more segments.

(133) As depicted in this example, main nonlinear optical waveguide **106** is an optical waveguide in the set of optical waveguides **102** and is comprised of nonlinear optical material **104**. Additionally, secondary optical waveguide **113** is another optical waveguide in the set of optical waveguides **102** and can be comprised of nonlinear optical material **104** or non-nonlinear optical material **105**. Secondary optical waveguide **113** can comprise a single optical waveguide segment or can comprise multiple optical waveguide segments that are physically separate from each other. First extension optical waveguide **108** is an example of first extension optical waveguide **108** depicted in FIG. 2.

(134) In this example, light **118** of three different wavelengths can travel through main nonlinear optical waveguide **106**. For example, first-wavelength light **132**, second-wavelength light **134**, and third-wavelength light **136** can travel in main nonlinear optical waveguide **106**.

(135) Light of two different wavelengths can travel through secondary optical waveguide **113**. For example, second-wavelength light **134** and third-wavelength light **136** can travel in secondary optical waveguide **113**.

(136) As depicted in this example, light **118** traveling through optical waveguides **102** can comprise at least one of first-wavelength light **132**, second-wavelength light **134**, or third-wavelength light **136**. In this particular example, first-wavelength light **132**, second-wavelength light **134**, and third-wavelength light **136** can refer to a pump light **161**, a signal light **163**, and an idler light **165**, but not necessarily in any

particular order.

(137) For example, first-wavelength light **132** can also be the signal light **163**, second-wavelength light **134** can be the pump light **161**, and third-wavelength light **136** can be idler light **165**. As another example, first-wavelength light **132** can also be pump light **161**, second-wavelength light **134** can be signal light **163**, and third-wavelength light **136** can be idler light **165**.

(138) Typically, pump light **161** has an intensity that is at least twice the intensity of signal light **163** and at least twice the intensity of idler light **165**. In some examples, the intensity of pump light **161** is at least ten times greater than the intensity of signal light **163** or of idler light **165**. Typically, pump light **161** is supplied as an input to optical waveguide structure **100**. In some cases, either of signal light **163** and idler light **165** also can be supplied as a second input to optical waveguide structure **100**. Either or both of signal light **163** and idler light **165** can be generated through nonlinear optical process **140** that occurs in portions of optical waveguide structure **100** that comprise a nonlinear optical material **104**.

(139) First extension optical waveguide **108** can be comprised of one at least one of nonlinear optical material **104** or a non-nonlinear optical material **105**. In this example, a light such as a pump light **161** can travel through first extension optical waveguide **108**.

(140) Second extension optical waveguide **110** can be comprised of at least one of nonlinear optical material **104** or non-nonlinear optical material **105**. A light such as signal light **163** can travel through second extension optical waveguide **110**.

(141) Third extension optical waveguide **119** can also be comprised of one of nonlinear optical material **104** and a non-nonlinear optical material **105**. In this example, a light such as idler light **165** can travel through third extension optical waveguide **119**, which can be a nonlinear optical waveguide.

(142) In one illustrative example, first extension optical waveguide **108**, second extension optical waveguide **110**, and third extension optical waveguide **119** are not constructed using nonlinear optical material **104**. In another illustrative example, at least one of first extension optical waveguide **108**, second extension optical waveguide **110** and third extension optical waveguide **119** can be constructed using nonlinear optical material **104**. In yet another illustrative example, at least one of first extension optical waveguide **108**, second extension optical waveguide **110** and third extension optical waveguide **119** can be constructed using electro-optic material **103**. Main nonlinear optical waveguide **106** also can be constructed using electro-optic material **103**.

(143) In this illustrative example, wavelength-selective couplers **114** include first wavelength-selective coupler **120**, second wavelength-selective coupler **122**, third wavelength-selective coupler **133**, and fourth wavelength-selective coupler **131**. Wavelength-selective couplers **114** can couple light **118** of different wavelengths to different optical waveguides based on the wavelengths in light **118**. For example, wavelength-selective couplers **114** can be configured to couple first-wavelength light **132**, second-wavelength light **134** and third-wavelength light **136** to selected different routes for travel of light **118** through optical waveguides in optical waveguides **102** based on the wavelengths of the light. For another example, wavelength-selective couplers **114** can be configured to couple at least one of second-wavelength light **134** or third-wavelength light **136** to different selected optical waveguides in optical waveguides **102** based on the wavelengths of the light.

(144) For example, first wavelength-selective coupler **120** optically couples first main location **146** in main nonlinear optical waveguide **106** and primary first extension location **148** in first extension optical waveguide **108** to each other such that first-wavelength light **132** is coupled from main nonlinear optical waveguide **106** at the first main location **146** to first extension optical waveguide **108** at primary first extension location **148**.

(145) First-wavelength light **132** can travel from primary first extension location **148** to secondary first extension location **150** through first extension segment **141**. In this illustrative example, locations at which first wavelength-selective coupler **120** and second wavelength-selective coupler **122** connect to main nonlinear optical waveguide **106** define the extent of main segment **143** of main nonlinear optical waveguide **106**. Further, main nonlinear optical waveguide **106** also can include additional segments. These additional segments can be defined by additional locations in main nonlinear optical waveguide **106** at which those segments are coupled to wavelength-selective couplers.

(146) In this example, second wavelength-selective coupler **122** optically couples second main location **152** in main nonlinear optical waveguide **106** and secondary first extension location **150** in first extension optical waveguide **108** to each other such that first-wavelength light **132** is coupled from first extension optical waveguide **108** at secondary first extension location **150** to main nonlinear optical waveguide **106** at second main location **152**.

(147) First-wavelength light **132** can travel from second main location **152** to first main location **146** through main segment **143** in main nonlinear optical waveguide **106**.

(148) In this illustrative example, first main location **146** and second main location **152** define main segment **143**, which is the portion of main nonlinear optical waveguide **106** through which first-wavelength light **132**, second-wavelength light **134**, and third-wavelength light **136** can travel. In this example, main segment **143** is comprised of a nonlinear optical material **104** and nonlinear optical processes can occur within main segment **143**.

(149) In this example, third wavelength-selective coupler **133** optically couples third secondary location **123** in secondary optical waveguide **113** and primary second extension location **156** in second extension optical waveguide **110** to each other such that second-wavelength light **134** is coupled from secondary optical waveguide **113** at third secondary location **123** to second extension optical waveguide **110** at primary second extension location **156**.

(150) In this example, second-wavelength light **134** can travel from primary second extension location **156** to secondary second extension location **158** through second extension segment **145** in second extension optical waveguide **110**.

(151) Illustration of waveguide configurations for optical waveguides **102** in FIG. 1, FIG. 2 and FIG. 3 are presented as illustrations of some configurations for optical waveguides **102**. These illustrations are not meant to limit the manner in which other illustrative examples can be implemented. For example, one or more waveguide segments can be present in addition to or in place of main segment **143**. As yet another example, additional ones of wavelength-selective couplers **114** can be connected to additional segments of main nonlinear optical waveguide **106**, additional segments of secondary optical waveguide **113** and additional extension optical waveguides in optical waveguide **102**. In other illustrative examples, optical waveguide **102** can omit at least one of second extension optical waveguide **110** or third extension optical waveguide **119**.

(152) Turning to FIG. 4, fourth wavelength-selective coupler **131** optically couples fourth secondary location **125** in the secondary optical waveguide **113** and secondary second extension location **158** in second extension optical waveguide **110** to each other such that second-wavelength light **134** is coupled from second extension optical waveguide **110** at secondary second extension location **158** to secondary optical waveguide **113** at fourth secondary location **125**. Second-wavelength light **134** can travel from primary second extension location **156** to secondary second extension location **158** through second extension segment **145** in second extension optical waveguide **110**.

(153) Second-wavelength light **134** can travel from first secondary location **124** to third secondary location **123** through first secondary

segment **147** (in FIG. 4). Second-wavelength light **134** can travel from fourth secondary location **125** to second secondary location **129** through second secondary segment **127** (in FIG. 4). Similarly, third-wavelength light **136** can travel from first secondary location **124** to third secondary location **123** through second secondary segment **127** (in FIG. 4). Third-wavelength light **136** can travel from fourth secondary location **125** to second secondary location **129** through second secondary segment **127** (in FIG. 4). In this illustrative example, first-wavelength light **132** can be pump light **161**, second-wavelength light **134** can be one of signal light **163** and idler light **165**. (154) Additionally, third wavelength-selective coupler **133** can optically couple third secondary location **123** in secondary optical waveguide **113** and primary third extension location **171** in third extension optical waveguide **119** to each other such that third-wavelength light **136** is coupled from secondary optical waveguide **113** at third secondary location **123** to third extension optical waveguide **119** at primary third extension location **171**. (155) Furthermore, fourth wavelength-selective coupler **131** can optically couple fourth secondary location **125** in secondary optical waveguide **113** and secondary third extension location **173** in the third extension optical waveguide **119** to each other such that third-wavelength light **136** is coupled from third extension optical waveguide **119** at secondary third extension location **173** to secondary optical waveguide **113** at fourth secondary location **125**. Third-wavelength light **136** can travel from primary third extension location **171** to secondary third extension location **173** through third extension segment **175** in third extension optical waveguide **119**. Third-wavelength light **136** can travel from third secondary location **123** to fourth secondary location **125** through second secondary segment **127** (in FIG. 4). (156) When second extension optical waveguide **110** and third extension optical waveguide **119** are present and coupled to secondary optical waveguide **113**, both second-wavelength light **134** and third-wavelength light **136** can travel through secondary optical waveguide **113**. In this example, first-wavelength light **132** can be pump light **161**, second-wavelength light **134** can be signal light **163**, and third-wavelength light **136** can be idler light **165**. (157) With reference now to FIG. 4, an illustration of loops in optical waveguides is depicted in accordance with an illustrative embodiment. In this example, first loop **200**, second loop **202**, and third loop **204** are examples of loops **116** in FIG. 1. (158) In this illustrative example, first-wavelength light **132** travels in first loop **200** through main segment **143** between first main location **146** and second main location **152** within the main nonlinear optical waveguide **106** and first extension segment **141** between primary first extension location **148** and secondary first extension location **150** in the first extension optical waveguide **108**. In this example, first loop **200** has first length **191**. (159) Second-wavelength light **134** travels in second loop **202** through first secondary segment **147** between first secondary location **124** and third secondary location **123** in secondary optical waveguide **113**, second extension segment **145** between primary second extension location **156** and secondary second extension location **158** in second extension optical waveguide **110**, second secondary segment **127** between third secondary location **123** and second secondary location **129** in secondary optical waveguide **113**, and main segment **143** in main nonlinear optical waveguide **106**. In this illustrative example, second loop **202** has second length **193** for second-wavelength light **134**. (160) Third-wavelength light **136** travels in third loop **204** through first secondary segment **147** between first secondary location **124** and third secondary location **123** in secondary optical waveguide **113**, third extension segment **175** between primary third extension location **171** and secondary third extension location **173** in third extension optical waveguide **119**, second secondary segment **127** between fourth secondary location **125** and second secondary location **129** in secondary optical waveguide **113**, and main segment **143** in main nonlinear optical waveguide **106**. In this example, third loop **204** as third length **195**. (161) As depicted, first-wavelength light **132** travels within main segment **143** in main nonlinear optical waveguide **106** and first extension segment **141** in first extension optical waveguide **108** in first loop **200**. In this example, first loop **200** has first length **191**. (162) As depicted, first length **191** can also comprise the length of first wavelength-selective coupler **120** and the length of second wavelength-selective coupler **122**. Second length **193** can also comprise the lengths of third wavelength-selective coupler **133** and fourth wavelength-selective coupler **131** as well as the lengths of first wavelength-selective coupler **120** and second wavelength-selective coupler **122**. Third length **195** of third loop **204** can also comprise the lengths of third wavelength-selective coupler **133** and the length of fourth wavelength-selective coupler **131** as well as the lengths of first wavelength-selective coupler **120** and second wavelength-selective coupler **122**. (163) The lengths of first loop **200**, second loop **202**, and third loop **204** can be selected based on the locations where wavelength-selective couplers **114** connect optical waveguides **102** to each other. First length **191** for first loop **200**, second length **193** for second loop **202**, and third length **195** for third loop **204** can have different lengths from each other. (164) For example, first length **191** of first loop **200** can be selected based on a selection of first main location **146** and primary first extension location **148** for first wavelength-selective coupler **120** connecting main nonlinear optical waveguide **106** to first extension optical waveguide **108** and based on a selection of secondary second extension location **158** and second main location **152** for second wavelength-selective coupler **122** connecting first extension optical waveguide **108** to main nonlinear optical waveguide **106**. (165) As another example, second length **193** of second loop **202** can be selected based on a selection of first secondary location **124** in secondary optical waveguide **113**, and second secondary location **129** and primary second extension location **156** for third wavelength-selective coupler **133** connecting secondary optical waveguide **113** to second extension optical waveguide **110**; and based on a selection of secondary second extension location **158** and third secondary location **123** for fourth wavelength-selective coupler **131** connecting second extension optical waveguide **110** to secondary optical waveguide **113**, and fourth secondary location **125** in secondary optical waveguide **113**. (166) As yet another example, third length **195** of third loop **204** can be selected based on a selection of first secondary location **124** in secondary optical waveguide **113**, and second secondary location **129** and primary third extension location **171** for third wavelength-selective coupler **133** connecting secondary optical waveguide **113** to third extension optical waveguide **119** and based on a selection of secondary third extension location **173** and third secondary location **123** for fourth wavelength-selective coupler **131** connecting third extension optical waveguide **119** to secondary optical waveguide **113**, and fourth secondary location **125** in secondary optical waveguide **113**. (167) With reference next to FIG. 5, an illustration of a block diagram of a configuration for nonlinear optical waveguides is depicted in accordance with an illustrative embodiment. In illustrative example, at least one of resonance matching **300** or roundtrip phase matching **302** for optical waveguides **102** can be achieved through the selection of dimensions **304** for optical waveguides **102**. This selection of dimensions **304** can be made in addition to the selection of lengths, such as first length **191**, second length **193**, and second length **193** for loops **116** optical waveguides **102** to achieve at least one of resonance matching **300** or roundtrip phase matching **302** for optical waveguides **102**. (168) For example, main nonlinear optical waveguide **106** can have main cross-section **308** with a set of dimensions **310** in dimensions **304** selected to achieve resonance condition **306** for first-wavelength light **132** traveling in main nonlinear optical waveguide **106**. In this

example, secondary optical waveguide **113** can have second cross-section **301** with secondary dimensions **303** selected to achieve resonance condition **306** for one of first-wavelength light **132** and second-wavelength light **134** traveling in secondary optical waveguide **113**.

(169) As another example, first extension optical waveguide **108** can have first cross-section **312** with first dimensions **314** selected to achieve resonance condition **306** for first-wavelength light **132** traveling in first extension optical waveguide **108**. Further, second extension optical waveguide **110** can have second cross-section **316** with a set of second dimensions **318** selected to achieve resonance condition **306** for second-wavelength light **134** traveling in second extension optical waveguide **110**. Also, third extension optical waveguide **119** can have third cross-section **317** with a set of third dimensions **319** selected to achieve resonance condition **306** for third-wavelength light **136** traveling in second extension optical waveguide **110**.

(170) With reference now to FIG. **6**, an illustration of phase shifters used to obtain at least one of resonance matching or roundtrip phase matching is depicted in accordance with an illustrative embodiment. At least one of manufacturing deviations from specifications, environmental factors, or other influences can affect whether a resonance condition is present during the operation of optical waveguide structure **100**.

(171) When roundtrip phase matching **302** in FIG. **5** is not present during operation of optical waveguide structure **100**, a set of phase shifters **400** can be used to adjust a set of phases **402** for light **118** propagating within optical waveguides **102**. In one illustrative example, the set of phase shifters **400** can be structures that are located adjacent to one or more of optical waveguides **102**; connected to one or more of optical waveguides **102**; include part of one or more of optical waveguides **102**; or a combination thereof.

(172) The set of phase shifters **400** can operate to ensure a desired level of roundtrip phase matching **302** is achieved for light **118** that is generated within optical waveguides **102** in optical waveguide structure **100**. As depicted, light **118** can be generated in an optical waveguide in optical waveguides **102** that is comprised of nonlinear optical material **104**. In the illustrative example, main nonlinear optical waveguide **106** is comprised of nonlinear optical material **104**. Optionally, at least one of first extension optical waveguide **108**, second extension optical waveguide **110** or third extension optical waveguide **119** can be comprised of nonlinear optical material **104**. In an illustrative example, at least one of first extension optical waveguide **108**, second extension optical waveguide **110** or third extension optical waveguide **119** can be comprised of electro-optic material **103**.

(173) In one illustrative example, a set of phase shifters **400** can be connected to a set of optical waveguides **102** comprising at least one of main nonlinear optical waveguide **106**, first extension optical waveguide **108** second extension optical waveguide **110**, or third extension optical waveguide **119**. The set of phase shifters **400** can apply a set of activations **404** to achieve a change or shift in the phase of at least one of first-wavelength light **132**, second-wavelength light **134**, or third-wavelength light **136** in light **118** traveling in the set of optical waveguides **102** to which the set of activations **404** is applied.

(174) In one illustrative example, the set of phase shifters **400** comprises a set of elements that can be located adjacent to a waveguide. The set of phase shifters **400** can take a number of different forms. For example, the set of phase shifters **400** can be selected from at least one of a tuning electrode, a thermal element, shape memory alloy element, piezo electric element, or some other element that can change the phase of light of a particular wavelength propagating through the optical waveguide. These elements for the set of phase shifters **400** can be at least one of adjacent to part of an optical waveguide, connected to part of an optical waveguide, or include part of an optical waveguide.

(175) The set of activations **404** can take a number of different forms. For example, the set of activations **404** can be selected from at least one of a voltage, a current, a thermal energy, an electrically induced strain, or some other type of energy that can be applied to an optical waveguide to affect the manner in which light propagates through the optical waveguide. In particular, the energy can be used to affect the phase of light of a particular wavelength propagating through the optical waveguide.

(176) In other words, the set of phase shifters **400** can selectively apply the set of activations **404** to adjust the phase for a particular wavelength of light **118** traveling within loops **116** in optical waveguides **102**. This adjustment can be made by applying the activations **404** using a particular phase shifter located adjacent to an optical waveguide in the set of optical waveguides **102** in a loop in loops **116** for a particular wavelength of light to maintain or reach resonance matching **300** for that particular wavelength of light.

(177) For example, a phase shifter, such as main phase shifter **406**, can be located adjacent to a portion of main nonlinear optical waveguide **106**. Main phase shifter **406** can apply an activation in activations **404** such that a phase shifts in first-wavelength light **132** to achieve resonant condition **306** for first-wavelength light **132** for light traveling in first loop **200**.

(178) Another phase shifter, such as secondary phase shifter **408** can be located adjacent to a portion of secondary optical waveguide **113**. Secondary phase shifter **408** can apply an activation in activations **404** such that a phase shifts in one or both of second-wavelength light **134** and third-wavelength light **136** to achieve a roundtrip phase matching **302** for the nonlinear optical process.

(179) A phase shifter, such as first phase shifter **410**, can be located adjacent to a portion of first extension optical waveguide **108**. First phase shifter **410** can apply an activation in activations **404** such that a phase shifts in first-wavelength light **132** to achieve a resonance condition **306** for first-wavelength light **132** in first loop **200**. First phase shifter **410** also can apply an activation in activations **404** such that a phase shifts in first-wavelength light **132** to achieve a roundtrip phase matching **302** for the nonlinear optical process.

(180) In another illustrative example, a phase shifter, such as second phase shifter **412**, can be located adjacent to a portion of second extension optical waveguide **110**. Second phase shifter **412** can apply an activation in activations **404** such that a phase shifts in second-wavelength light **134** to achieve a resonance condition **306** for second-wavelength light **134** in second loop **202**.

(181) As another illustrative example, a phase shifter, such as third phase shifter **414**, can be located adjacent to a portion of third extension optical waveguide **119**. Third phase shifter **414** can an activation in activations **404** such that a phase shifts in third-wavelength light **136** to achieve resonance condition **306** for third-wavelength light **136** in third loop **204**.

(182) In one illustrative example, the set of phase shifters **400** can be a set of tuning electrodes that apply a set of activations **404** as a set of voltages **418**. With this type of phase shifters in the form of tuning electrodes that apply activations **404** in the form of voltages **418**, the optical waveguides associated with the tuning electrodes can be comprised of an electro-optic material **103**. One example of an electro-optical material **103** is lithium niobate. This material does not have to be use throughout the entire optical waveguide. Lithium niobate can be used in the sections that are associated with or adjacent to the tuning electrodes.

(183) Lithium niobate is an electro-optic material for which the material refractive index can be changed by applying an electric field to the lithium niobate material. Lithium niobate has a second order nonlinear optical coefficient that is large enough to result in undesired light generation. As a result, in some illustrative examples the regions in a nonlinear optical waveguide containing the lithium niobate containing regions used for electro-optic tuning from the lithium niobate can be separated from regions containing lithium niobate used for the nonlinear optical generation of signal photons and idler photons.

(184) With this example, main phase shifter **406** in the set of phase shifters **400** can be main tuning electrode **420** located adjacent to a portion of main nonlinear optical waveguide **106**. Secondary phase shifter **408** in the set of phase shifters **400** can be secondary tuning

electrode **422** located adjacent to a portion of secondary optical waveguide **113**.

(185) In this illustrative example, first phase shifter **410** can be first tuning electrode **424** located adjacent to a portion of first extension optical waveguide **108**. Second phase shifter **412** can be second tuning electrode **426** located adjacent to a portion of second extension optical waveguide **110**, and third phase shifter **414** in the set of phase shifters **400** can be third tuning electrode **428** located adjacent to a portion of third extension optical waveguide **119**.

(186) First tuning electrode **424**, second tuning electrode **426**, and third tuning electrode **428** can apply the set of activations **404** in the form of a set of voltages **418** to adjust the set of phases **402** in at least one of first-wavelength light **132**, second-wavelength light **134**, or third-wavelength light **136** traveling in a set of loops **116** through optical waveguides **102**. This shift in the set of phases **402** can be made to maintain or reach resonance condition **306** for one or more of the wavelengths of light **118**. These wavelengths of light can be for example, at least one of first-wavelength light **132**, second-wavelength light **134**, or third-wavelength light **136**. This shift in the set of phases **402** also can be made to achieve or maintain roundtrip phase matching **302**.

(187) In the illustrative example, when an optical waveguide in the set of optical waveguides **102** comprises an electro-optic material **103**, the activation can take the form of a voltage. When the optical waveguide does not comprise an electro-optic material, other forms of energy such as, for example, thermal energy, such as heat, or strain can be used as the set of activations **404**. In this illustrative example, heat can be generated by applying electrical current to a resistor that forms a phase shifter in the set of phase shifters **400** such that heat is generated. As another example, a voltage can be applied to a piezo electric element for phase shifter in the set of phase shifters **400** to change the dimensions of the tuning electrode to cause strain in the portion of the optical waveguide adjacent to the phase shifter in the set of phase shifters **400**.

(188) The illustration of optical waveguide structure **100** and the different components in FIGS. **1-6** is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

(189) For example, additional extension waveguides can be present in optical waveguide structure **100**. For example, another extension waveguide can be optically coupled to second extension optical waveguide **110**. This coupling can be performed using another pair of wavelength-selective couplers to form a third extension segment for third-wavelength light.

(190) In another illustrative example, fewer components can be present than depicted in optical waveguide structure **100** in FIGS. **1-6**. In another illustrative example, third extension optical waveguide **119** can be omitted from optical waveguides **111**. In other illustrative examples, phase shifters **400** may be used with some but not all of optical waveguides **102**. In one example, only main phase shifter **406** may be present.

(191) With reference now to FIG. **7**, an illustration of a cross-section of an optical waveguide is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide **700** is shown in a cross-sectional view. This cross-section can be used in the optical waveguides in optical waveguide structure **100** in FIGS. **1-6**.

(192) As depicted, optical waveguide **700** comprises core region **702** and cladding region **704**. Core region **702** can be comprised of a material such as silicon nitride (Si.sub.3N.sub.4), silicon (Si) or silicon carbide (SiC) for optical processes based on four wave mixing. Core region **702** can be comprised of a material such as lithium niobate (LiNbO.sub.3), gallium phosphide (GaP), aluminum nitride (AlN), aluminum gallium arsenide (AlGaAs), or silicon carbide (SiC) for optical processes based on three-wave mixing. Cladding region **704** can be comprised of silicon dioxide (SiO.sub.2) or other material whose refractive index is lower than the refractive index of the material comprising core region **702**. The particular material used in optical waveguide **700** can vary in other illustrative examples depending on the optical process used.

(193) In this illustrative example, core region **702** has width w **706** and height t_w **708**. Cladding region **704** has height t_{ox} **710**. Cladding region **704** can cover any combination of the top, the two sides and the bottom of core region **702**.

(194) Optical waveguide **700** can be adjusted to achieve values for the effective refractive indices (n_{eff}) of the wavelengths of light **118** traveling through optical waveguide **700**. The effective refractive indices can be adjusted through the selection of the material refractive index at a specific wavelength and varying the waveguide dimensions such as width w **706**, height t_w **708**, and top oxide thickness, height t_{ox} **710**.

(195) The selection of at least one of the material and dimensions for optical waveguide **700** can be based on the conditions for momentum conservation and phase matching. In the illustrative example, momentum conservation is an automatic consequence of the nonlinear optical interaction. Whether the phase matching associated with the particular waveguide structure is consistent with momentum conservation determines the degree of phase walk-off that results as the light travels in the waveguide over some distance.

(196) For example, an effective refractive index can be a function of the height and width of core region **702**. The constructive nonlinear generation length is the propagation length at which the phase walk-off equals π radians. The constructive nonlinear generation length is inversely proportional to the phase mismatch. In an illustrative example, the length of the main nonlinear optical waveguide should be no larger than the constructive nonlinear generation length that can be achieved for the main nonlinear optical waveguide. In illustrative examples, the nonlinear optical interaction occurs in all three loops.

(197) Additionally, the cross-section shown for optical waveguide **700** is provided as an example and is not meant to limit the manner in which other illustrative examples can implement cross-sections for waveguides. For example, optical waveguide **700** is shown with side **720** and side **722** that are angled for core region **702**. In other illustrative examples, these two sides can be parallel to each other rather than angled. As another example, other components may be present in this cross-section such as side regions that may be located adjacent to side **720** and side **722**. In yet another illustrative example, the cross-section of optical waveguide **700** may also include a phase shifter such as a tuning electrode. As another example, optical waveguide **700** can include a second core region in addition to core region **702** when optical waveguide **700** is used to implement a two-waveguide optical coupler.

(198) Turning to FIG. **8**, an illustration of light coupling by a wavelength-selective coupler is depicted in accordance with an illustrative embodiment. In this illustrative example, pump light **802**, signal light **804**, and idler light **806** travel through optical waveguide **808** and are input into wavelength-selective coupler **810**. Signal light **804** and idler light **806** also travel through optical waveguide **808** and are input into wavelength-selective coupler **810**. As depicted, at the output of wavelength-selective coupler **810**, pump light **802** continues through to optical waveguide **828**. In this example, signal light **804** and idler light **806** cross over from optical waveguide **808** to optical waveguide **822** at the output of wavelength-selective coupler **810**. Signal light **804** and idler light **806** also cross over from optical waveguide **812** at the input of wavelength-selective coupler **810** to optical waveguide **828** at the output of wavelength-selective coupler **810**. Wavelength-selective coupler **810** is an illustration of an implementation for first wavelength-selective coupler **1220** and second wavelength-selective

coupler **1222** in optical waveguide structure **1200** in FIG. **16** and for first wavelength-selective coupler **1320** and second wavelength-selective coupler **1322** in optical waveguide structure **1300** in FIG. **17**.

(199) With reference next to FIG. **9**, an illustration of light coupling by a wavelength-selective coupler is depicted in accordance with an illustrative embodiment. In this illustrative example, pump light **902**, signal light **904**, and idler light **906** travel through optical waveguide **908** into wavelength-selective coupler **910**. Pump light **912** also travels through optical waveguide **914** and is input into wavelength-selective coupler **910**.

(200) As depicted, at the output of wavelength-selective coupler **910**, signal light **904** and idler light **906** continues through into optical waveguide **928**. Pump light **902** traveling into wavelength-selective coupler **910** from optical waveguide **908** crosses over to optical waveguide **924** at the output of wavelength-selective coupler **910**. In a similar fashion, pump light **912** traveling through optical waveguide **914** into wavelength-selective coupler **910** crosses over to optical waveguide **928** at the output of wavelength-selective coupler **910**. Wavelength-selective coupler **910** is illustrative of first wavelength-selective coupler **1120** and second wavelength-selective coupler **1122** in optical waveguide structure **1100** in FIG. **13**, first wavelength-selective coupler **1020** and second wavelength-selective coupler **1022** in optical waveguide structure **1000** in FIG. **14**, first wavelength-selective coupler **1420**, second wavelength-selective coupler **1422** in optical waveguide structure **1400** in FIG. **15**, and first wavelength-selective coupler **1580**, second wavelength-selective coupler **1586**, third wavelength-selective coupler **1584** and fourth wavelength-selective coupler **1582** in optical waveguide structure **1500** in FIG. **18**, described below.

(201) With reference to FIG. **10**, an illustration of light coupling by a wavelength-selective coupler is depicted in accordance with an illustrative embodiment. In this illustrative example, signal light **4502** and idler light **4504** travel through optical waveguide **4506** and are input into wavelength-selective coupler **4508**. In this depicted example, light is not input into optical waveguide **4510** which is connected to wavelength-selective coupler **4508**. As depicted, at the output of wavelength-selective coupler **4508**, idler light **4504** continues through into optical waveguide **4526** and signal light **4502** crosses over into optical waveguide **4520**.

(202) This crossover of signal light **4502** is caused by the design of wavelength-selective coupler **4508**. In illustrative examples, wavelength-selective coupler **4508** can be used for a signal wavelength-selective coupler to selectively couple signal light from a secondary optical waveguide to a signal extension optical waveguide. Wavelength-selective coupler **4508** can also be used to selectively couple signal light from a signal extension optical waveguide to the secondary optical waveguide. Wavelength-selective coupler **4508** is illustrative of wavelength-selective couplers used in optical waveguide structure **1500** in FIG. **18**.

(203) In FIG. **11**, an illustration of simulation results of light coupling by a wavelength-selective coupler is depicted in accordance with an illustrative embodiment. As depicted, simulation results **4600** comprises plots and. Simulation results **4600** comprise signal extraction plot **4602** for a signal extraction result and idler retention plot **4604** for an idler retention result. These plots are of the optical-field distributions for a signal light and an idler light having different wavelengths from each other.

(204) Simulation results **4600** are generated using a wavelength-selective coupler such as wavelength-selective coupler **4508** in FIG. **10**. This wavelength-selective coupler can be implemented as a two-waveguide optical coupler. In this illustrative example, simulation results **4600** are for a case in which signal light **4502**, that is coupled and exits from the “cross” output of wavelength-selective coupler **4508**, has a larger guided-mode effective index of refraction $n_{\text{sub.eff}}$ and is confined more strongly than the idler light **4504**, that exits from the “through” output of wavelength-selective coupler **4508**.

(205) As depicted, signal extraction plot **4602** depicts the electric-field magnitude of the signal light. Plot **4602** has x-axis **4606** that represents the longitudinal direction of the two-guide wavelength-selective coupler structure and y-axis **4608** that represents the transverse direction of the two-guide wavelength-selective coupler structure. Signal extraction plot **4602** in simulation results **4600** shows that signal light is coupled from the lower left waveguide to the upper right waveguide and is illustrative of the cross-state of a coupler.

(206) In this illustrative example, idler retention plot **4604** depicts the electric-field magnitude of the idler light. Idler retention plot **4604** has x-axis **4610** that represents the longitudinal direction of the two-guide wavelength-selective coupler structure and y-axis **4612** that represents the transverse direction of the two-guide wavelength-selective coupler structure. As depicted, idler retention plot **4604** shows that the idler light couples from the lower waveguide to the upper waveguide in a few portions of the coupling region but eventually remains in the lower waveguide away from that coupling section and exits from the lower right waveguide, illustrative of the thru-state of a coupler.

(207) In this example, these simulation results can be obtained using a wavelength-selective coupler that comprises two curved waveguides that are coupled by a section of a straight waveguide of a length and a gap for wavelength-selective coupler that are selected to result in the coupling of the signal light from a first optical waveguide to a second optical waveguide when passing through the wavelength-selective coupler.

(208) Thus, if light of both signal light **4502** and idler light **4504** are supplied to wavelength-selective coupler **4508** through optical waveguide **4506**, signal light **4502** exits wavelength-selective coupler **4508** via optical waveguide **4520** and idler light **4504** exits wavelength-selective coupler **4508** via optical waveguide **4526**.

(209) For this example, an example length $d_{\text{sub.s}}$ for the coupling section for wavelength-selective coupler **4508** can be described by the following relation: $\kappa_{\text{sub.s}}(\lambda_{\text{sub.S}}) \cdot d_{\text{sub.s}} = \pi$, where $\kappa_{\text{sub.s}}$ is the coupling coefficient. To achieve the desired wavelength selectivity, wavelength-selective coupler **4508** can also be constrained by another relation:

$\kappa_{\text{sub.s}}(\lambda_{\text{sub.I}}) \cdot d_{\text{sub.s}} = 2 \cdot \pi \cdot X$, where $\lambda_{\text{sub.I}}$ is the longer wavelength and X is an integer. In the illustrative example, the value of X is 2, such that the photons of signal light wavelength $\lambda_{\text{sub.S}}$ have approximately 100% coupling between the two waveguides being coupled, while the photons of idler light wavelength $\lambda_{\text{sub.I}}$ are coupled back again to the starting waveguide.

(210) With reference now to FIG. **12**, an illustration of simulation results of light coupling by a wavelength-selective coupler is depicted in accordance with an illustrative embodiment. Simulation results **4700** comprise plots that illustrate light coupling using a wavelength-selective coupler such as a two-waveguide optical coupler.

(211) As depicted, simulation results **4700** are for pump light in pump plot **4718**, signal light in signal plot **4710**, and idler light in idler plot **4714**. These simulation results are plots of the electric field magnitude distributions of light at the pump, signal, and idler wavelengths. Pump plot **4718** is a plot for field magnitude distribution in linear scale. As depicted, pump plot **4718** has x-axis **4708** that represents the longitudinal direction of the two-guide wavelength-selective coupler structure and y-axis **4702** that represents the transverse direction of the two-guide wavelength-selective coupler structure.

(212) In this illustrative example, signal plot **4710** and idler plot **4714** are plots for the signal and idler field magnitude distributions in a logarithmic scale. As depicted, signal plot **4710** has x-axis **4712** that represents the longitudinal direction of the two-guide wavelength-selective coupler structure and y-axis **4705** that represents the transverse direction of the two-guide wavelength-selective coupler structure. Idler plot **4714** has x-axis **4716** that represents the longitudinal direction of the two-guide wavelength-selective coupler structure and y-axis **4707** that represents the transverse direction of the two-guide wavelength-selective coupler structure.

(213) In this depicted example, the optical waveguide at the lower portion of the plots for simulation results **4700** has a smaller radius of curvature than the optical waveguide at the upper portion of those plots. The light travels from left to right in these plots for simulation results **4700**. Pump light enters in the upper guide from the upper left of pump plot **4718**. Signal light and idler light enter in the lower, curved guide from the lower left of signal plot **4704** and idler plot **4706**.

(214) In this example, the pump light experiences primarily the “cross” state of this coupler and is coupled into the curved, lower guide and exits from the lower right of the plot. The signal and idler light experience the “through” state of this coupler and remain in the curved guide to also exit from the lower right of the plots. For this example, the pump light is carried by a higher-order transverse mode of the lower, curved guide. Thus, the field magnitude distribution of the pump light in that curved guide has several brighter regions. The signal and idler light, however, are carried by the fundamental transverse modes at those wavelengths. Thus, the intensity distributions for the signal and idler light have just one bright region that is brighter near the center of the guide. In this illustrative example, the pump light is carried in the upper guide by the fundamental transverse mode at the pump wavelength. Thus, the intensity distribution for the pump light in the upper waveguide has just one bright region that is brighter near the center of that upper guide. The simulation results **4700** can be examples of the performance of some implementations of wavelength-selective coupler **910** illustrated in FIG. **9**.

(215) The examples of FIGS. **13-18** illustrate different aspects of optical waveguide structure **100** as shown in FIGS. **1-6**. These illustrations are intended to be inclusive rather than exclusive. Thus, although only some features are illustrated in one example and other features are illustrated in another example, this difference in features in different figures is used only for the purpose of clarity and to simplify the description of features in the illustrative examples.

(216) With reference to FIG. **13**, an illustration of an optical waveguide structure with five optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **1100** is an example of an implementation for optical waveguide structure **100** as shown in FIGS. **1-6**. More specifically, FIG. **13** is an implementation for optical waveguides **102** as depicted in FIG. **3**.

(217) In this illustrative example, optical waveguide structure **1100** can be designed to achieve the concurrent requirements that the three wavelengths are at resonances of their respective resonator loops and also that the phase match condition is met for sustaining the nonlinear optical process over many cycles of travel through the loops.

(218) In this illustrative example, optical waveguide structure **1100** comprises optical waveguides in the form of main nonlinear optical waveguide **1110**, segment **1107** in secondary optical waveguide **1108**, segment **1109** in secondary optical waveguide **1108**, pump loop extension **1102**, signal loop extension **1104**, and idler loop extension **1106**. Main nonlinear optical waveguide **1110** is an example of main nonlinear optical waveguide **106** in FIG. **3** and main nonlinear optical waveguide **106** in FIG. **2**. Pump loop extension **1102** is an example of an implementation for first extension optical waveguide **108** in FIG. **3** and first extension optical waveguide **108** in FIG. **2**. Signal loop extension **1104** and idler loop extension **1106** are optical waveguides that can be coupled to segments of secondary optical waveguide **113** in FIG. **3** or secondary optical waveguide **113** in FIG. **2**. Signal loop extension **1104** and idler loop extension **1106** are examples of second extension optical waveguide **110** and third extension optical waveguide **119**, respectively, in FIGS. **3-6**.

(219) In these illustrative examples, the individual optical waveguides can be portions or segments from which loops can be established through the use of wavelength selective optical couplers to connect those segments or portions to other segments or portions.

(220) In this illustrative example, main nonlinear optical waveguide **1110** of optical waveguide structure **1100** is comprised of a nonlinear optical material **104**. For some second-order nonlinear optical materials, such as x-cut lithium niobate, the nonlinear optical coefficient is much larger for light whose electric-field vector is aligned parallel to one crystallographic axis than for light whose electric-field vector is aligned perpendicular to that crystallographic axis. Thus, for x-cut lithium niobate, a larger second-order nonlinear optical coefficient applies for a nonlinear optical waveguide aligned parallel to the material Y-axis, with the electric-field vector of the propagating transverse-electric (TE) polarized light aligned parallel to the material Z-axis. In this illustrative example, main nonlinear optical waveguide **1110** has a linear shape and is aligned parallel to the lithium niobate material Y-axis. Thus, the propagation direction would be in the ty direction or the -y direction of the lithium niobate crystalline material.

(221) In this illustrative example, pump loop extension **1102** is comprised of a non-nonlinear optical material **105**. As depicted, idler loop extension **1106** is comprised of an electro-optic material **103**. As depicted, signal loop extension **1104** is comprised of a nonlinear optical material **104** as well as an electro-optic material **103**. An electro-optic material is a material with a large electro-optic coefficient. Examples of electro-optic materials that can be used are lithium niobate, gallium arsenide, gallium phosphide and silicon carbide.

(222) In an illustrative example, the use of an electro-optic material can provide desired propagation properties for light. Electro-optical materials often also are nonlinear optical materials having nonlinear optical coefficient.

(223) As depicted, optical waveguide structure **1100** also includes pump input optical waveguide **1132** that inputs pump light **1112**. Optical waveguide structure **1100** also includes signal output optical waveguide **1134** and idler output optical waveguide **1136**. Signal output optical waveguide **1134** can output signal light **1114**. Idler output optical waveguide **1136** can output idler light **1116**.

(224) As depicted, first wavelength-selective coupler **1120** and second wavelength-selective coupler **1122** connect pump loop extension **1102** to main nonlinear optical waveguide **1110**. In this illustrative example, third wavelength-selective coupler **1124** and fourth wavelength-selective coupler **1126** connect signal loop extension **1104** and idler loop extension **1106** to segment **1107** and segment **1109** of secondary optical waveguide **1108**.

(225) In this illustrative example, pump optical coupler **1131** couples pump input optical waveguide **1132** to pump loop extension **1102**. Signal optical coupler **1135** couples signal output optical waveguide **1134** to signal loop extension **1104**. Idler optical coupler **1137** couples idler output optical waveguide **1136** to idler loop extension **1106**.

(226) In this illustrative example, pump light **1112** travels in pump loop **1152** which extends through main nonlinear optical waveguide **1110** and pump loop extension **1102**. Signal light **1114** travels in signal loop **1154** which extends through main nonlinear optical waveguide **1110**, secondary optical waveguide **1108** and signal loop extension **1104**. Idler light **1116** travels in idler loop **1156** which extends through main nonlinear optical waveguide **1110**, secondary optical waveguide **1108** and idler loop extension **1106**.

(227) As depicted, optical waveguide structure **1100** also includes phase shifters in the form of tuning electrodes. In this illustrative example, tuning electrode **1160** is located adjacent to a portion of main nonlinear optical waveguide **1110**. Tuning electrode **1164** is located adjacent to a portion of signal loop extension **1104**. Tuning electrode **1166** is located adjacent to a portion of idler loop extension **1106**.

(228) In this illustrative example, each wavelength-selective coupler in optical waveguide structure **1100** produces a phase shift for each given wavelength of light at its “thru” state output and a possibly different phase shift for each given wavelength of light at its “cross” state output. For example, first wavelength-selective coupler **1120** extracts pump light **1112** from main nonlinear optical waveguide **1110** into pump loop extension **1102**. First wavelength-selective coupler **1120** also extracts signal light **1114** and idler light **1116** from main nonlinear linear optical waveguide **1110** into segment **1107** of secondary optical waveguide **1108**.

(229) In this illustrative example, first wavelength-selective coupler **1120** produces a phase shift of $\phi_{\text{sub.M1p}}$ for the pump light **1112** coupled from main nonlinear optical waveguide **1110** to pump loop extension **1102** via a “cross” state output of first wavelength-selective coupler **1120**. First wavelength-selective coupler **1120** produces a phase shift of $\phi_{\text{sub.1s}}$ for signal light **1114** that is coupled from main nonlinear optical waveguide **1110** into segment **1107** of secondary optical waveguide **1108**, and a phase shift of $\phi_{\text{sub.1i}}$ for idler light **1116** that is coupled from main nonlinear optical waveguide **1110** into segment **1107** of secondary optical waveguide **1108** via a “thru” state output of first wavelength-selective coupler **1120**.

(230) Furthermore, second wavelength-selective coupler **1122** causes a phase shift of $\phi_{\text{sub.1Mp}}$ for pump light **1112** coupled from pump loop extension **1102** back to main nonlinear optical waveguide **1110**. Second wavelength-selective coupler **1122** produces a phase shift of $\phi_{\text{sub.1s}}$ for signal light **1114** that is coupled from segment **1109** of secondary optical waveguide **1108** into main nonlinear optical waveguide **1110**, and produces a phase shift of $\phi_{\text{sub.1i}}$ for idler light **1116** that is coupled from segment **1109** of secondary optical waveguide **1108** into main nonlinear optical waveguide **1110**.

(231) In this illustrative example, third wavelength-selective coupler **1124** and fourth wavelength-selective coupler **1126** between the secondary optical waveguide **1108** and idler loop extension **1106** produce phase shifts of $\phi_{\text{sub.2i}}$ and $\phi_{\text{sub.2i}}$ for idler light **1116** coupled in their “cross” state output. Third wavelength-selective coupler **1124** and fourth wavelength-selective coupler **1126** between the secondary optical waveguide **1108** and signal loop extension **1104** produce phase shifts of $\phi_{\text{sub.2s}}$ and $\phi_{\text{sub.2s}}$ for signal light **1114** that exits from their “thru” state outputs.

(232) The light propagating in a waveguide can experience a phase shift associated with the length of the waveguide and with the effective refractive index of the wave-guided mode. For transverse-electric (TE) polarized light in x-cut lithium niobate, the material index depends on the direction of propagation. Thus, the phase shift can be estimated by performing a numerical simulation. The phase shifters, such as tuning electrodes, can contribute an additional phase shift that can either advance the phase or retard the phase, depending on the sign of the applied voltage, for an electro-optic phase shifter.

(233) For the example in optical waveguide structure **1100** in FIG. 13, tuning electrode **1164** for signal loop extension **1104** in signal loop **1154** and tuning electrode **1166** for idler loop extension **1106** in idler loop **1156** can contribute additional phase shifts of $\Delta\phi_{\text{sub.Es}}$ and $\Delta\phi_{\text{sub.Ei}}$, respectively. These phase shifts can have a positive or negative value.

(234) In this illustrative example, tuning electrode **1160** for main nonlinear optical waveguide **1110** affects pump light **1112**, signal light **1114**, and idler light **1116** and can produce additional phase shifts of $\Delta\phi_{\text{sub.MEp}}$, $\Delta\phi_{\text{sub.MEs}}$, and $\Delta\phi_{\text{sub.MEi}}$ to the pump light **1112**, signal light **1114**, and idler light **1116**, respectively.

(235) The resonator for pump light **1112** is comprised of components of optical waveguide structure **1100** in pump loop **1152**. This pump loop comprises main nonlinear optical waveguide **1110**, the cross-state of first wavelength-selective coupler **1120**, the cross-state of second wavelength-selective coupler **1122**, and pump loop extension **1102**. The round-trip phase shift $\phi_{\text{sub.RTp}}$ for pump light **1112** at the pump wavelength should be equal to a multiple of 2π for pump light **1112** to remain circulating for many round-trips through pump loop **1152** and thus circulate for many passes through main nonlinear optical waveguide **1110**.

(236) In this illustrative example, the phase shift of the pump light **1112** due to propagation in the pump loop extension **1102** can be described by the expression:

$$\phi_{\text{sub.1p}} = 2\pi n_{\text{sub.1p}} L_{\text{sub.1}} / \lambda_{\text{sub.p}}$$
where $n_{\text{sub.1p}}$ is a net or equivalent effective refractive index of the wave-guided pump light in the pump loop extension **1102**; $L_{\text{sub.1}}$ is the length of pump loop extension **1102**; and $\lambda_{\text{sub.p}}$ is the wavelength of pump light **1112**.

(237) The phase shift of pump light **1112** from propagation through the main nonlinear optical waveguide **1110** can be described as follows:

$$\phi_{\text{sub.Mup}} = 2\pi n_{\text{sub.Mp}} L_{\text{sub.M}} / \lambda_{\text{sub.p}}$$
where $n_{\text{sub.Mp}}$ is the effective refractive index of the wave-guided pump mode in main nonlinear optical waveguide **1110**, $L_{\text{sub.M}}$ is the length of main nonlinear optical waveguide **1110**, which is located between first wavelength-selective coupler **1120** and second wavelength-selective coupler **1122**; and $\lambda_{\text{sub.p}}$ is the wavelength of pump light **1112**.

(238) Next, the resonance requirement for pump light **1112** can be given by the expression:

$$\phi_{\text{sub.RTp}} = \phi_{\text{sub.1p}} + \phi_{\text{sub.1Mp}} + \phi_{\text{sub.Mup}} + \phi_{\text{sub.M1p}} + \Delta\phi_{\text{sub.MEp}} = 2\pi P$$
where P is an integer. In an illustrative example, P can have values that also result in phase matching to maintain constructive generation of signal and idler from pump light.

(239) This resonance requirement can be met by designing optical waveguide structure **1100** to have suitable values for the length $L_{\text{sub.1}}$ and the phase shift $\phi_{\text{sub.1p}}$. The phase shift $\phi_{\text{sub.M1p}}$ is due to the first wavelength-selective coupler and the phase shift $\phi_{\text{sub.1Mp}}$ is due to the second wavelength-selective coupler.

(240) In this illustrative example, signal loop **1154** extends through main nonlinear optical waveguide **1110**. More specifically signal loop **1154** extends through main nonlinear optical waveguide **1110**, first wavelength-selective coupler **1120** (in its thru state) and second wavelength-selective coupler **1122** (in its thru state); segment **1107** and segment **1109** of secondary optical waveguide **1108**, in which both signal light **1114** and idler light **1116** propagate; third wavelength-selective coupler **1124** (in its thru state) and fourth wavelength-selective coupler **1126** (in its thru state); and signal loop extension **1104**. As depicted, only signal light **1114** propagates through signal loop extension **1104**.

(241) In this example, main nonlinear optical waveguide **1110** can have length $L_{\text{sub.Mu}}$. Pump light **1112**, signal light **1114**, and idler light **1116** propagate through main nonlinear optical waveguide **1110**. Segment **1107** and segment **1109** have a total length of $L_{\text{sub.Mc}}$. In this illustrative example, signal loop extension **1104** has a total length of $L_{\text{sub.2}}$.

(242) Signal loop **1154** is a resonator loop in which the signal light **1114** travels. The round-trip phase shift $\phi_{\text{sub.RTs}}$ of signal light **1114** traveling in signal loop **1154** can be given by:

$$\phi_{\text{sub.RTs}} = 2\phi_{\text{sub.1s}} + \phi_{\text{sub.Mus}} + \Delta\phi_{\text{sub.MEs}} + 2\phi_{\text{sub.2s}} + \phi_{\text{sub.Mcs}} + \phi_{\text{sub.Ss}} + \Delta\phi_{\text{sub.SEs}} = 2\pi S$$
where $n_{\text{sub.Ms}}$ is the effective refractive index of signal light **1114** in the main nonlinear optical waveguide **1110**; $L_{\text{sub.Mu}}$ is the length of main nonlinear optical waveguide **1110**; and $\lambda_{\text{sub.s}}$ is the wavelength of signal light **1114**.

(243) For signal light **1114** to remain circulating for many round-trips in signal loop **1154** and thus circulate for many passes through main nonlinear optical waveguide **1110**, the round-trip phase shift should be as close as possible to a multiple of 2π , that is, with S being an integer.

(244) The phase shift of signal light **1114** propagating in main nonlinear optical waveguide **1110** can be described by the expression:

$$\phi_{\text{sub.Mus}} = 2\pi n_{\text{sub.Ms}} L_{\text{sub.Mu}} / \lambda_{\text{sub.s}}$$
where $n_{\text{sub.Ms}}$ is the effective refractive index of signal light **1114** in the main nonlinear optical waveguide **1110**; $L_{\text{sub.Mu}}$ is the length of main nonlinear optical waveguide **1110**; and $\lambda_{\text{sub.s}}$ is the wavelength of signal light **1114**.

(245) Each of the two wavelength-selective couplers coupled to main nonlinear optical waveguide **1110**, first wavelength-selective coupler **1120** and second wavelength-selective coupler **1122**, in signal loop **1154** operate in its “cross” state for the signal wavelength and produces a phase shift of $\phi_{\text{sub.1s}}$ for the signal wavelength. In a similar fashion, each of the two wavelength-selective couplers coupled to signal

loop extension **1104**, third wavelength-selective coupler **1126**, in signal loop **1154** operate in its “thru” state for the signal wavelength and produces a phase shift of $\phi_{\text{sub.2s}}$ for the signal light **1114**. The net phase shift from the two corner portions, segment **1107** and segment **1109** of the secondary optical waveguide **1108** in signal loop **1154**, in which both signal light **1114** and idler light **1116** propagate can be given by $\phi_{\text{sub.Mcs}}$. The phase shift from signal loop extension **1104** in signal loop **1154**, in which only the signal light propagates, can be given by $\phi_{\text{sub.Ss}}$.

(246) In an illustrative example, tuning electrode **1160** used to adjust the phase shift for pump light **1112** in its resonator loop also produces a phase shift for signal light **1114** of $\Delta\phi_{\text{sub.MEs}}$. However, tuning electrode **1164** in signal loop extension **1104** affects only signal light **1114**. Tuning electrode **1166** produces a phase shift of $\Delta\phi_{\text{sub.SEs}}$.

(247) Idler loop **1156** in which idler light **1116** extends through main nonlinear optical waveguide **1110** and idler loop extension **1106**. In this depicted example, idler loop **1156** comprises segment **1107** in secondary optical waveguide **1108**, first wavelength-selective coupler **1120** and third wavelength-selective coupler **1124**; segment **1109** in secondary optical waveguide **1108**, fourth wavelength-selective coupler **1126** and second wavelength-selective coupler **1122**; and idler loop extension **1106**.

(248) Each of the two wavelength-selective couplers, first wavelength-selective coupler **1120** and second wavelength-selective coupler **1122**, in idler loop **1156** have a “cross” state for the pump wavelength and a “thru” state for the idler wavelength and produces a phase shift of $\phi_{\text{sub.1i}}$ at the “thru” state output of the wavelength-selective coupler. Likewise, each of the two wavelength-selective couplers, third wavelength-selective coupler **1124** and fourth wavelength-selective coupler **1126**, have a “cross” state for the idler wavelength and produces a phase shift of $\phi_{\text{sub.2i}}$ at its “cross” state output for idler light **1116**.

(249) The total phase shift of idler light **1116** from the two corner portions, segment **1107** and segment **1109**, in which both signal light **1114** and idler light **1116** propagate, can be given by $\phi_{\text{sub.Mci}}$. The phase shift from idler loop extension **1106**, in which only idler light **1116** propagates, can be given by $\phi_{\text{sub.Iei}}$.

(250) In this illustrative example, tuning electrode **1160** for main nonlinear optical waveguide **1110** used to adjust the phase shift for pump light **1112** will also produce a phase shift for idler light **1116** of $\Delta\phi_{\text{sub.MEi}}$. Tuning electrode **1166** for idler loop extension **1106** affects only idler light **1116**. Tuning electrode **1166** can produce a phase shift of $\Delta\phi_{\text{sub.IEi}}$.

(251) Thus, the round-trip phase shift $\phi_{\text{sub.RTi}}$ of idler light **1116** can be given by:

$$\phi_{\text{sub.RTi}} = 2\phi_{\text{sub.1i}} + \phi_{\text{sub.Mui}} + \Delta\phi_{\text{sub.MEi}} + 2\phi_{\text{sub.2i}} + \phi_{\text{sub.Mci}} + \phi_{\text{sub.Ii}} + \Delta\phi_{\text{sub.IEi}} = 2\pi I$$

(252) For idler light **1116** to remain circulating for many roundtrips in idler loop **1156** extending through main nonlinear optical waveguide **1110** and thus making many passes through main nonlinear optical waveguide **1110**, the round-trip phase shift should be as close as possible to a multiple of 2π , that is, with I being an integer. The length and waveguide cross-sectional structure in main nonlinear optical waveguide **1110** can be designed to achieve phase matching for the nonlinear optical interaction.

(253) Thus, the value for $\phi_{\text{sub.Mui}}$ can be determined by the design of the waveguide cross-sectional structure in main nonlinear optical waveguide **1110**. However, the length $L_{\text{sub.3}}$ of idler loop extension **1106** can be selected to achieve the desired resonance condition for the idler wavelength in its resonator loop, idler loop **1156**. Also, the additional phase shift $\Delta\phi_{\text{sub.IEi}}$ produced by the tuning electrode **1166** in the idler loop extension **1106** can be used to further adjust that round-trip phase shift for idler light **1116**.

(254) In the illustrative example, main nonlinear optical waveguide **1110** is the location where the desired nonlinear optical photon generation occurs in optical waveguide structure **1100**. Main nonlinear optical waveguide **1110** can be designed to achieve a phase matched condition for the nonlinear optical process. This phase matched condition can be achieved through the selection of dimensions of the cross-sectional waveguide structure.

(255) The cross-sectional structure of main nonlinear optical waveguide **1110** as well as the propagation direction of the light determines the effective refractive index of the pump light **1112**, signal light **1114** and idler light **1116** in a given portion of main nonlinear optical waveguide **1110**. The propagation direction for light guided in main nonlinear optical waveguide **1110**, in which the desired nonlinear optical interaction occurs, can be chosen to increase the nonlinear optical generation. For example, a waveguide comprising x-cut lithium niobate could be aligned parallel to the material Y-axis. Thus, the propagation direction would be in the +y direction or the -y direction of the lithium niobate crystalline material.

(256) For the nonlinear optical process to occur constructively over a long interaction distance so that the generation rate or generation efficiency of the signal photons and idler photons from the pump photons continues to increase as the physical interaction distance is increased, the phase matching condition of the nonlinear optical process also should be maintained. This condition includes the round-trip phase shift of pump light **1112** traveling in the main nonlinear optical waveguide **1110** as well as in pump loop extension **1102**, the round-trip phase shift of signal light **1114** traveling in main nonlinear optical waveguide **1110**, in segment **1107** and segment **1109** of secondary optical waveguide **1108**, as well as in signal loop extension **1104**, and the round-trip phase shift of idler light **1116** traveling in main nonlinear optical waveguide **1110**, in segment **1107** and segment **1109** of secondary optical waveguide **1108**, as well as in idler loop extension **1106**.

(257) Thus:

$$\phi_{\text{sub.RTp}} - \phi_{\text{sub.RTs}} - \phi_{\text{sub.RTi}} = 2\pi A$$

where A is an integer, and can be zero.

(258) Furthermore, to increase the nonlinear optical generation of signal and idler light that occurs in a given round-trip, meeting another phase matching condition is desirable for propagation of the three wavelengths of light through main nonlinear optical waveguide **1110**, which is the portion where the nonlinear optical generation occurs. This phase matching condition can be described as follows:

$$(259) \quad 0 \leq \phi_{\text{sub.Mup}} - \phi_{\text{sub.Mus}} - \phi_{\text{sub.Mui}} \leq \pi, \text{ or } -\pi \leq \phi_{\text{sub.Mup}} - \phi_{\text{sub.Mus}} - \phi_{\text{sub.Mui}} \leq 0, \text{ and is close to zero.}$$

(260) The additional phase shifts that can be achieved by applying bias voltages to the tuning electrodes for optical waveguide structure **1100** can be used to adjust the round-trip phase shifts for pump light **1112** (by adjusting $\Delta\phi_{\text{sub.MEp}}$), for the signal light **1114** (by adjusting $\Delta\phi_{\text{sub.SEs}}$) and for idler light **1116** (by adjusting $\Delta\phi_{\text{sub.IEi}}$). These adjustments can be used to correct or to compensate for departures of the other parameters from their as-designed values in actually fabricated and operating devices.

(261) The phase shift that can be obtained for a given electric field in the electro-optic material (due to a voltage applied to a set of tuning electrodes) can be described by the relation:

$$\Delta\phi_{\text{sub.KEj}} = 2\pi r_{\text{sub.jn}} \cdot j_{\text{sub.j}} \cdot \sup{3E\Gamma_{\text{sub.j}}L_{\text{sub.E}}/\lambda_{\text{sub.j}}}$$

where $j = p, s, i$, and where p indicates pump light **1112**, s indicates signal light **1114**, and i indicates idler light **1116**. Also, $K = M, S$ or P and indicates the optical waveguide with the tuning electrode, such as $K = M$ for main nonlinear optical waveguide **1110**, $K = S$ for signal loop extension **1104** and $K = I$ for idler loop extension **1106**. Other parameters in this expression are: the electric field E , the electro-optic coefficient $r_{\text{sub.j}}$, the refractive index $n_{\text{sub.j}}$, the overlap of the optical field of pump light **1112**, signal light **1114**, or idler light **1116** with the electro-optic material $\Gamma_{\text{sub.j}}$, the electrode length (or electro-optic interaction distance) $L_{\text{sub.E}}$, and the wavelength $\lambda_{\text{sub.j}}$ of the pump

light **1112**, signal light **1114**, or idler light **1116**. As an example, for an electro-optic material such as lithium niobate and for an electric field applied across the waveguide of 10×10^6 V/m, the electrode length needed to achieve a phase shift of 2π is about 3-10 mm.

(262) Turning to FIG. **14**, an illustration of an optical waveguide structure with five optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **1000** is an example of an implementation for optical waveguide structure **100** as shown in FIGS. **1-6**.

(263) In this illustrative example, optical waveguide structure **1000** comprises optical waveguides. These waveguides include main nonlinear optical waveguide **1010**, secondary optical waveguide **1008** having segment **1007** and segment **1009**, pump loop extension **1002**, signal loop extension **1004**, and idler loop extension **1006**.

(264) Main nonlinear optical waveguide **1010** is an example of main nonlinear optical waveguide **106** in FIG. **3** and main nonlinear optical waveguide **106** in FIG. **2**. Secondary optical waveguide **1008** is an example of an implementation for secondary optical waveguide **113** in FIG. **3** and secondary optical waveguide **113** in FIG. **2**.

(265) Pump loop extension **1002** is an example of an implementation for first extension optical waveguide **108** in FIG. **3** and first extension optical waveguide **108** in FIG. **2**. Signal loop extension **1004** and idler loop extension **1006** are examples of second extension optical waveguide **110** and third extension optical waveguide **119**, respectively, in FIGS. **3-6**.

(266) In this illustrative example, main nonlinear optical waveguide **1010** is comprised of a nonlinear optical material, such as nonlinear optical material **104**. Secondary optical waveguide **1008** can be comprised of a nonlinear optical material, such as nonlinear optical material **104** or a non-nonlinear optical material, such as non-nonlinear optical material **105**.

(267) As depicted, pump loop extension **1002** is comprised of a non-nonlinear optical material. Signal loop extension **1004** is comprised of both a nonlinear optical material and an electro-optic material, such as electro-optic material **103**, in this illustrative example. Idler loop extension **1006** has portions comprised of a nonlinear optical material **104** and other portions comprised of a non-nonlinear optical material. In this example, a taper **1049** can join an optical waveguide portion comprising nonlinear optical material and an optical waveguide portion comprising a non-nonlinear optical material. In this illustrative example, section **1043** and section **1045** of idler loop extension **1006** are comprised of a non-nonlinear optical material. Section **1046** of idler loop extension **1006** is comprised of a nonlinear optical material that also is an electro-optic material. Examples of material that have a large second-order nonlinear optical coefficient as well as a large electro-optic coefficient include lithium niobate and gallium arsenide.

(268) In this illustrative example, segment **1007** and segment **1009** of secondary optical waveguide **1008** is comprised of a nonlinear optical material that also is an electro-optic material. In this example, signal loop extension **1004** likewise is comprised of a nonlinear optical material that also is an electro-optic material.

(269) As depicted, optical waveguide structure **1000** also includes pump input optical waveguide **1032** that inputs pump light **1012**. Optical waveguide structure **1000** also includes signal output optical waveguide **1034** and idler output optical waveguide **1036**. Signal output optical waveguide **1034** can output signal light **1014**. Idler output optical waveguide **1036** can output idler light **1016**.

(270) As shown in this figure, first wavelength-selective coupler **1020** and second wavelength-selective coupler **1022** connect pump loop extension **1002** to main nonlinear optical waveguide **1010**. In this illustrative example, third wavelength-selective coupler **1024** and fourth wavelength-selective coupler **1026** connect idler loop extension **1006** to segment **1007** and segment **1009** of secondary optical waveguide **1008**. Third wavelength-selective coupler **1024** and fourth wavelength-selective coupler **1026** also connect signal loop extension **1004** to segment **1007** and segment **1009** of secondary optical waveguide **1008**.

(271) In this illustrative example, pump optical coupler **1031** couples pump input optical waveguide **1032** to pump loop extension **1002**. Signal optical coupler **1035** couples signal output optical waveguide **1034** to signal loop extension **1004**. Idler optical coupler **1037** couples idler output optical waveguide **1036** to idler loop extension **1006**.

(272) In this illustrative example, pump light **1012** travels in pump loop **1052** which extends through main nonlinear optical waveguide **1010** and pump loop extension **1002**. Signal light **1014** travels in signal loop **1054** which extends through main nonlinear optical waveguide **1010**, segment **1007** and segment **1009** in secondary optical waveguide **1008**, and signal loop extension **1004**. Idler light **1016** travels in idler loop **1056**, which extends through main nonlinear optical waveguide **1010**, segment **1007** and segment **1009** of secondary optical waveguide **1008**, and idler loop extension **1006**.

(273) As depicted, optical waveguide structure **1000** also includes phase shifters in the form of tuning electrodes. In this illustrative example, tuning electrode **1060** is located adjacent to a portion of main nonlinear optical waveguide **1010**. In this example, the portion of main nonlinear optical waveguide **1010** is segment **1040**. Tuning electrode **1064** is located adjacent to a portion of signal loop extension **1004**. As depicted, the portion of signal loop extension **1004** is segment **1044**. Tuning electrode **1066** is located adjacent to section **1046** of idler loop extension **1006**. These tuning electrodes can apply voltages to obtain a desired level of resonance to achieve a resonant condition for the three wavelengths of light traveling within optical waveguide structure **1000**. For example, tuning electrode **1060** can adjust the phase for pump light **1012**. Tuning electrode **1064** can adjust the phase of signal light **1014**. Tuning electrode **1066** can adjust the phase of idler light **1016**.

(274) A nonlinear optical process for the generation of photons for signal light **1014** and idler light **1016** from photons of pump light **1012** occurs in main nonlinear optical waveguide **1010** in optical waveguide structure **1000**. In this example, the nonlinear optical process does not occur, or negligibly occurs, in other parts of optical waveguide structure **1000**. In this depicted example, pump light **1012** supplied through pump input optical waveguide **1032** travels only through main nonlinear optical waveguide **1010**, first wavelength-selective coupler **1020**, second wavelength-selective coupler **1022** and pump loop extension **1002**. Nonlinear optical generation of signal photons and idler photons from pump photons occurs only where pump light travels and interacts with nonlinear optical material in a waveguide. Thus, both pump light and nonlinear optical material must be present for nonlinear optical generation of signal photons and idler photons from pump photons to occur.

(275) In this illustrative example, pump loop extension **1002** is comprised of a material having a negligible second order nonlinear optical coefficient such as Si₃N₄ and SiO₂. The other portions of optical waveguide structure **1000** through which pump light **1012** does not propagate can contain a material such as lithium niobate, which has a large electro-optic coefficient and also has a large second-order nonlinear optical coefficient. This material is useful for electro-optic tuning.

(276) Additionally, signal light **1014** travels in signal loop **1054** that traverses through main nonlinear optical waveguide **1010**, segment **1007** and segment **1009** of secondary optical waveguide **1008** and signal loop extension **1004**, as well as through first wavelength-selective coupler **1020** and second wavelength-selective coupler **1022** and third wavelength-selective coupler **1024** and fourth wavelength-selective coupler **1026**. In this example, this combination of optical waveguides can also serve as a resonator for signal light **1014**. Tuning electrode **1064** for signal loop extension **1004** is located along signal loop **1054** and can operate to achieve electrically controlled optical phase shifting for signal light **1014**.

(277) In this depicted example, idler light **1016** travels in idler loop **1056** extends through idler loop extension **1006**, and tuning electrode **1066** for idler loop extension **1006** can operate to achieve an electrically controlled optical phase shifting for idler light **1016**. Lithium niobate is an electro-optic material for which the material refractive index can be changed by applying an electrical field. A material such as lithium niobate can be used in the segment **1044** of signal loop extension **1004** adjacent to tuning electrode **1064** and in the section **1046** of idler loop extension **1006** adjacent to tuning electrode **1066**.

(278) In this illustrative example, pump loop extension **1002** does not have a tuning electrode. Tuning electrode **1060** can be used adjacent to main nonlinear optical waveguide **1010** and can operate to achieve some electrical control of the optical phase shift for pump light **1012**. However, the use of tuning electrode **1060** can affect the round-trip phase shift of pump light **1012**, as well as the round-trip phase shifts of signal light **1014** and idler light **1016**.

(279) These tuning electrodes in optical waveguide structure **1000** can apply voltages to obtain desired levels of phase shifts for the pump light **1012**, signal light **1014** and idler light **1016** to achieve resonance matching **300** in FIG. **5** for those three wavelengths of light traveling within optical waveguide structure **1000**. These tuning electrodes in optical waveguide structure **1000** also can apply voltages to obtain desired levels of phase shifts for the pump light **1012**, signal light **1014** and idler light **1016** to achieve roundtrip phase matching **302** for the combination of those three wavelengths of light traveling within optical waveguide structure **1000**.

(280) With reference now to FIG. **15**, an illustration of an optical waveguide structure with five optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **1400** is an example of an implementation for optical waveguide structure **100** as shown in FIGS. **1-6**. As depicted, optical waveguide structure **1400** comprises optical waveguides in the form of main nonlinear optical waveguide **1410**, segments of secondary optical waveguide **1408**, pump loop extension **1402**, signal loop extension **1404**, and idler loop extension **1406**.

(281) In this illustrative example, main nonlinear optical waveguide **1410** is comprised of a nonlinear optical material **104**.

(282) As depicted, pump loop extension **1402** is comprised of a non-nonlinear optical material, such as non-nonlinear optical material **105**. In this example, idler loop extension **1406** is comprised of an electro-optic material **103** that also can have a large second-order nonlinear optical coefficient. In this example, signal loop extension **1404** is comprised of a nonlinear optical material, such as nonlinear optical material **104**, that also has a large electro-optic coefficient.

(283) As depicted, optical waveguide structure **1400** includes pump input optical waveguide **1432** that inputs pump light **1412**. Optical waveguide structure **1400** also includes signal output optical waveguide **1434** and idler output optical waveguide **1436**. Signal output optical waveguide **1434** can output signal light **1414**. Idler output optical waveguide **1436** can output idler light **1416**.

(284) As depicted, first wavelength-selective coupler **1420** and second wavelength-selective coupler **1422** connect pump loop extension **1402** to main nonlinear optical waveguide **1410**. In this illustrative example, third wavelength-selective coupler **1424** and fourth wavelength-selective coupler **1426** connect idler loop extension **1406** to segment **1407** and segment **1409** of secondary optical waveguide **1408**. In this illustrative example, third wavelength-selective coupler **1424** and fourth wavelength-selective coupler **1426** also connect signal loop extension **1404** to segment **1407** and segment **1409** of secondary optical waveguide **1408**.

(285) In this illustrative example, pump input coupler **1431** couples pump input optical waveguide **1432** to pump loop extension **1402**. Signal output coupler **1435** couples signal output optical waveguide **1434** to signal loop extension **1404**. Idler output coupler **1437** couples idler output optical waveguide **1436** to idler loop extension **1406**.

(286) In this depicted example, pump loop **1452** is present for pump light **1412**. This pump loop **1452** is a resonator loop in which pump light **1412** travels in pump loop extension **1402** and in main nonlinear optical waveguide **1410**.

(287) In this example, signal light **1414** travels in signal loop **1454**. As depicted, signal loop **1454** extends through main nonlinear optical waveguide **1410**, through segments **1407** and **1409** of secondary optical waveguide **1408**, and through signal loop extension **1404**. As shown in the figure, idler light **1416** travels in idler loop **1456**. Further, in this example, idler loop **1456** extends through main nonlinear optical waveguide **1410**, through segments **1407** and segment **1409** of secondary optical waveguide **1408**, and through idler loop extension **1406**.

(288) As depicted, optical waveguide structure **1400** also includes phase shifters in the form of tuning electrodes. In this illustrative example, tuning electrode **1460** is located adjacent to main nonlinear optical waveguide **1410**. Tuning electrode **1464** is located adjacent to signal loop extension **1404**. Tuning electrode **1466** is located adjacent to idler loop extension **1406**.

(289) In this illustrative example, a nonlinear optical process occurs in main nonlinear optical waveguide **1410** in optical waveguide structure **1400**. Main nonlinear optical waveguide **1410** is constructed using a material such as x-cut lithium niobate, which can have both a large second order nonlinear optical coefficient and a large electro-optic coefficient.

(290) As depicted, main nonlinear optical waveguide **1410** has a straight segment **1470** and two corner segments, corner segment **1471** and corner segment **1473**. In this illustrative example, straight segment **1470** is aligned parallel to the y-axis of the x-cut lithium niobate crystal. Segment **1407** and segment **1409** are part of secondary optical waveguide **1408**. In this illustrative example, segment **1407** and segment **1409** are aligned parallel to the z-axis of the x-cut lithium niobate crystal.

(291) In this depicted example, transverse-electric (TE) polarized light propagating in main nonlinear optical waveguide **1410** can encounter the largest electro-optic coefficient **733** when the light travels in straight segment **1470** in main nonlinear optical waveguide **1410**. TE polarized light also encounters the largest electro-optic coefficient **733** of x-cut lithium niobate when the light travels in segment **1474** of signal loop extension **1404** adjacent to tuning electrode **1464** and when the light travels in the portion of idler loop extension **1406** adjacent to tuning electrode **1466**.

(292) As depicted, light travels in a clockwise direction through main nonlinear optical waveguide **1410**, pump loop extension **1402**, signal loop extension **1404**, and idler loop extension **1406**. This direction is selected by the configuration of the input and output couplers, such as pump input coupler **1431**, signal output coupler **1435**, and idler output coupler **1437**. However, these three input and output couplers could be configured to have the light travel in a counter-clockwise direction through main nonlinear optical waveguide **1410**, pump loop extension **1402**, signal loop extension **1404**, and idler loop extension **1406**, and by where pump light **1412** is supplied to pump input optical waveguide **1432**. Counter-clockwise travel is established by supplying pump light into the opposite end of pump input coupler **1431**, extracting signal light out from the opposite end of signal output coupler **1435**, and extracting idler light out from the opposite end of idler output coupler **1437**.

(293) As depicted, first wavelength-selective coupler **1420** connects corner segment **1471** of main nonlinear optical waveguide **1410** to segment **1407** of secondary optical waveguide **1408**. Second wavelength-selective coupler **1422** connects segment **1409** of secondary optical waveguide **1408** to corner segment **1473** of main nonlinear optical waveguide **1410**.

(294) As depicted, third wavelength-selective coupler **1424** and fourth wavelength-selective coupler **1426** operate to establish a resonator loop, idler loop **1456**, for idler light **1416** and also to establish a resonator loop, signal loop **1454**, for signal light **1414**. In this illustrative

example, third wavelength-selective coupler **1424** extracts idler light **1416** away from segment **1407** of secondary optical waveguide **1408** and into the idler loop extension **1406**. Fourth wavelength-selective coupler **1426** returns idler light **1416** back into segment **1409** of secondary optical waveguide **1408** after idler light **1416** has propagated through idler loop extension **1406** while traveling in idler loop **1456**.

(295) In this illustrative example, third wavelength-selective coupler **1424** also extracts signal light **1414** away from segment **1407** of secondary optical waveguide **1408** and into the signal loop extension **1404**. Fourth wavelength-selective coupler **1426** also returns signal light **1414** back into segment **1409** of secondary optical waveguide **1408** after signal light **1414** has propagated through signal loop extension **1404** while traveling in idler loop **1456**. Signal light **1414** travels to a thru-state output of third wavelength-selective coupler **1424** and travels to a thru-state output of fourth wavelength-selective coupler **1426**. Idler light **1416** travels to a cross-state output of third wavelength-selective coupler **1424** and travels to a cross-state output of fourth wavelength-selective coupler **1426**, as discussed before with reference to FIG. **10**.

(296) In this illustrative example, first wavelength-selective coupler **1420** and second wavelength-selective coupler **1422** operate to establish a resonator loop, pump loop **1452** for pump light **1412**. As depicted, first wavelength-selective coupler **1420** extracts pump light **1412** away from main nonlinear optical waveguide **1410** and into pump loop extension **1402** to travel in pump loop **1452**. Second wavelength-selective coupler **1422** returns pump light **1412** to main nonlinear optical waveguide **1410** after pump light **1412** has propagated through pump loop extension **1402** while traveling in pump loop **1452**.

(297) In this illustrative example, the material for idler loop extension **1406** and the material for signal loop extension **1404** can be a material such as lithium niobate for which the electro-optic coefficient is large. The large electro-optic coefficient allows the phase shifters in the signal loop extension and the idler loop extension to be more efficient, producing a larger phase shift for a given applied voltage. But for lithium niobate, the second order nonlinear optical coefficient also is large. However, pump light **1412** is not supplied to these portions of optical waveguide structure **1400**, resulting in an absence of undesired nonlinear optical generation of additional signal or idler photons in these portions. In this illustrative example, pump loop extension **1402** is comprised of a non-nonlinear optical material.

(298) As depicted, pump light **1412** propagates primarily only in main nonlinear optical waveguide **1410** and pump loop extension **1402**. The second order nonlinear optical coefficient is largest d.sub.33 for light propagating in straight segment **1470** of main nonlinear optical waveguide **1410** and is smaller for light propagating in corner segment **1471** and corner segment **1473**. Also, the sign of a component d.sub.22 of the second order nonlinear optical coefficient in corner segment **1471** is opposite from the sign of that component of the second order nonlinear optical coefficient in corner segment **1473**. As a result, the generation of signal light **1414** and idler light **1416** occurs mainly in straight segment **1470** and occurs much less in other portions of optical waveguide structure **1400** because of the manner in which pump light **1412** is introduced and removed from main nonlinear optical waveguide **1410**.

(299) In this illustrative example, pump light **1412** can be extracted from main nonlinear optical waveguide **1410** before idler light **1416** is extracted from main nonlinear optical waveguide **1410** through secondary optical waveguide **1408** into idler loop extension **1406**. Also in this example, pump light **1412** is re-supplied to main nonlinear optical waveguide **1410** from pump loop extension **1402** after idler light **1416** is re-supplied to main nonlinear optical waveguide **1410** from idler loop extension **1406** through secondary optical waveguide **1408**. A similar arrangement applies for the pump light **1412** in relation to the signal light **1414**.

(300) As a result, although the nonlinear optical material is present along the entire length of the signal loop **1054** for signal light **1414** and idler loop **1456** for idler light **1416**, the nonlinear optical generation of photons for signal light **1414** and idler light **1416** from photons for pump light **1412** occurs only in main nonlinear optical waveguide **1410**. Nonlinear optical generation of signal light **1414** and idler light **1416** is absent in secondary optical waveguide **1408**, idler loop extension **1406** and signal loop extension **1404**. The absence of nonlinear optical generation is because pump light **1412** is supplied only to main nonlinear optical waveguide **1410**.

(301) A nonlinear optical generation process can result in generation of lower intensity light from higher intensity light. A nonlinear optical generation process also can operate in reverse and result in the generation of a higher intensity light from a lower intensity light. The efficiency of the nonlinear optical generation process depends on the intensity of the source light involved in that generation process, or the intensities of the source light of several different wavelengths if source light of multiple wavelengths is involved in that process. For spontaneous parametric down conversion as an illustrative example of a nonlinear optical generation process, the pump light, which is the input or source light, has an intensity that is at least twice the intensity of the generated signal light and at least twice the intensity of the generated idler light.

(302) In many examples of spontaneous parametric down conversion, the intensity of the pump light is at least ten times greater than the intensity of the signal light or of the idler light. Thus, even when a phase-matched condition is present, if the pump light is absent from an optical waveguide comprising nonlinear optical material and only signal and idler light are present, the reverse process in which pump light, or light at the pump wavelength, is generated from the weaker source light at the signal and idler wavelengths is much less efficient and may produce very little or possibly even negligible light at the pump wavelength.

(303) With reference next to FIG. **16**, an illustration of an optical waveguide structure with five optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **1200** is an example of an implementation for optical waveguide structure **100** as shown in FIGS. **1-6**.

(304) As depicted, optical waveguide structure **1200** comprises optical waveguides in the form of main nonlinear optical waveguide **1210**, pump loop extension **1202**, secondary optical waveguide **1208**, signal loop extension **1204**, and idler loop extension **1206**. Main nonlinear optical waveguide **1210** is an example of main nonlinear optical waveguide **106** in FIG. **3** and main nonlinear optical waveguide **106** in FIG. **2**. Pump loop extension **1202** is an example of an implementation for first extension optical waveguide **108** in FIG. **3** and first extension optical waveguide **108** in FIG. **2**. Idler loop extension **1206** and signal loop extension **1204** are optical waveguides that can be coupled to secondary optical waveguide **113** in FIG. **3** or coupled to secondary optical waveguide **113** in FIG. **2**.

(305) In this illustrative example, first loop **1252** through main nonlinear optical waveguide **1210** and pump loop extension **1202** has a rectangular shape with curved corners and may also be referred to as a racetrack shape. First loop **1252** for the pump light is a closed path route.

(306) As depicted in this example, first loop **1252** for pump light **1212** through main nonlinear optical waveguide **1210** and through pump loop extension **1202** traverses segments of waveguide comprised of nonlinear optical material **104** and segments of waveguide comprised of non-nonlinear optical material **105**. The nonlinear optical material is present in main nonlinear optical waveguide **1210**, which includes straight segment **1270** corner segment **1271**, and corner segments **1273**. The nonlinear optical material also is present in portions of corner segment **1275** and corner segment **1277** of pump loop extension **1202**. A non-nonlinear optical material **105** is present in segment **1272** of pump loop extension **1202**.

(307) A non-nonlinear optical material also can be present in corner segment **1275** and corner segment **1277** of pump loop extension **1202**

instead of the nonlinear optical material. As depicted in this figure, a tapered transition **1247** can be present between the portion of corner segment **1275** and corner segment **1277** that contains a nonlinear optical material and the portion of corner segment **1275** and corner segment **1277** that does not contain a nonlinear optical material but rather comprises only non-nonlinear optical material.

(308) In this illustrative example, both signal loop extension **1204** and idler loop extension **1206** have portions that comprise an electro-optic material **103** that also is a nonlinear optical material **104** and other portions that comprise a non-nonlinear optical material **105**. The electro-optic material is located in section **1244** of signal loop extension **1204** and in section **1246** of idler loop extension **1206**. To reduce optical losses and reflections, there can be a tapered transition **1249** between a waveguide portion comprising an electro-optic material and a waveguide portion comprising a non-nonlinear optical material.

(309) As depicted, optical waveguide structure **1200** also includes pump input optical waveguide **1232** that inputs pump light **1212**. Optical waveguide structure **1200** also includes signal output optical waveguide **1234** and idler output optical waveguide **1236**. Signal output optical waveguide **1234** can output signal light **1214**. Idler output optical waveguide **1236** can output idler light **1216**.

(310) In this illustrative example, pump optical coupler **1231** couples pump input optical waveguide **1232** to pump loop extension **1202**. Signal optical coupler **1235** couples signal output optical waveguide **1234** to signal loop extension **1204**. Idler optical coupler **1237** couples idler output optical waveguide **1236** to idler loop extension **1206**.

(311) In this illustrative example, first wavelength-selective coupler **1220** and second wavelength-selective coupler **1222** connect pump loop extension **1202** to main nonlinear optical waveguide **1210**. Pump light **1212** is coupled via the thru-state outputs of first wavelength-selective coupler **1220** and second wavelength-selective coupler **1222**. As depicted, first wavelength-selective coupler **1220** and second wavelength-selective coupler **1222** also connect signal light **1214** and idler light **1216** between main nonlinear optical waveguide **1210** and segments of secondary optical waveguide **1208**. Signal light **1214** and idler light **1216** are coupled via the cross-state outputs of first wavelength-selective coupler **1220** and second wavelength-selective coupler **1222**. In this illustrative example, third wavelength-selective coupler **1224** and fourth wavelength-selective coupler **1226** connect idler loop extension **1206** to segment **1207** and segment **1209** of secondary optical waveguide **1208**. Third wavelength-selective coupler **1224** and fourth wavelength-selective coupler **1226** also connect signal loop extension **1204** to segment **1207** and segment **1209** of secondary optical waveguide **1208**. Signal light **1214** is coupled via the thru-state outputs of third wavelength-selective coupler **1224** and fourth wavelength-selective coupler **1226**. Idler light **1216** is coupled via the cross-state outputs of third wavelength-selective coupler **1224** and fourth wavelength-selective coupler **1226**.

(312) In this illustrative example, first loop **1252** is present for pump light **1212**. This first loop is a resonator loop in which pump light **1212** travels in main nonlinear optical waveguide **1210** and in pump loop extension **1202**. Signal light **1214** travels in second loop **1254**. As depicted, second loop **1254** extends through main nonlinear optical waveguide **1210**, through segment **1207** and segment **1209** of secondary optical waveguide **1208** and through signal loop extension **1204**. In this illustrative example, idler light **1216** travels in third loop **1256**. As depicted, third loop **1256** extends through main nonlinear optical waveguide **1210**, through segment **1207** and segment **1209** of secondary optical waveguide **1208**, and through idler loop extension **1206**.

(313) As depicted, optical waveguide structure **1200** also includes phase shifters in the form of tuning electrodes. In this illustrative example, tuning electrode **1260** is located adjacent to section **1240** in main nonlinear optical waveguide **1210**. Tuning electrode **1264** is located adjacent to section **1244** in signal loop extension **1204** and tuning electrode **1266** is located adjacent to section **1246** in idler loop extension **1203**. These tuning electrodes can apply voltages to obtain a desired level of resonance to achieve a resonant condition for light traveling within optical waveguide structure **1200**.

(314) In this illustrative example of optical waveguide structure **1200** having triple partially overlapping loop resonators for entanglement with direction dependent material, pump light **1212**, signal light **1214**, and idler light **1216** travel in different resonator loops, first loop **1252**, second loop **1254**, and third loop **1256**, respectively. First loop **1252** is a resonator loop for the pump light **1212** and extends through main nonlinear optical waveguide **1210**, first wavelength-selective coupler **1220** (via its thru-state output), second wavelength-selective coupler **1222** (via its thru-state output), pump loop extension **1202**, and pump optical coupler **1231** (via its thru-state output).

(315) In this illustrative example, second loop **1254** is a resonator loop for signal light **1214**. Second loop **1254** extends through main nonlinear optical waveguide **1210**, first wavelength-selective coupler **1220** (via its cross-state output), segment **1207** of secondary optical waveguide **1208**, third wavelength-selective coupler **1224** (via its thru-state output), signal loop extension **1204**, and signal optical coupler **1235** (via its thru-state output) fourth wavelength-selective coupler **1226** (via its thru-state output), segment **1209** of secondary optical waveguide **1208**, and second wavelength-selective coupler **1222** (via its cross-state output), looping again to main nonlinear optical waveguide **1210**.

(316) As depicted, third loop **1256** is a resonator loop for idler light **1216**. This third loop **1256** extends through main nonlinear optical waveguide **1210**, first wavelength-selective coupler **1220** (via its cross-state output); segment **1207** of secondary optical waveguide **1208** located between first wavelength-selective coupler **1220** and third wavelength-selective coupler **1224**; third wavelength-selective coupler **1224** (via its cross-state output); idler loop extension **1206**; idler optical coupler **1237** (via its thru-state output); fourth wavelength-selective coupler **1226** (via its cross-state output); segment **1209** of secondary optical waveguide **1208** located between fourth wavelength-selective coupler **1226** and second wavelength-selective coupler **1222**; and second wavelength-selective coupler **1222** (via its cross-state output); looping back to main nonlinear optical waveguide **1210**.

(317) In this illustrative example of optical waveguide structure **1200** having triple partially overlapping loop resonators for entanglement constructed from a direction dependent material, main nonlinear optical waveguide **1210** is common to and overlaps all three loop resonators. Also, first wavelength-selective coupler **1220** and second wavelength-selective coupler **1222** are encountered by the light in all three loops. However, first loop **1252** for pump light **1212** encounters the thru-state of these couplers. In this example, second loop **1254** and third loop **1256** for signal light **1214** and idler light **1216**, respectively, encounter the cross-state of these couplers.

(318) In this illustrative example, a second-order nonlinear optical process such as spontaneous parametric down conversion occurs in optical waveguide structure **1200**. Nonlinear optical generation of signal photons and idler photons from pump photons, which is a result of spontaneous parametric down conversion, occurs when pump light propagates in an optical waveguide comprising nonlinear optical material such as lithium niobate which has a large second-order nonlinear optical coefficient. Optical waveguide structure **1200** includes main nonlinear optical waveguide **1210**. Main nonlinear optical waveguide **1210** is the primary part of optical waveguide structure **1200** for which pump light **1212** is present and propagates in a waveguide comprising nonlinear optical material. As result, most of the generation of signal photons and idler photons from pump photons occurs in main nonlinear optical waveguide **1210**. Essentially, negligible generation of signal photons and idler photons occurs in other portions of optical waveguide structure **1200**. As depicted, main nonlinear optical waveguide **1210** comprises a nonlinear optical material. Most of the pump loop extension **1202**, such as portion or segment **1272** of pump loop extension **1202** does not comprise a nonlinear optical material.

(319) The various optical waveguides in optical waveguide structure **1200** can be fabricated using x-cut lithium niobate and in particular,

from x-cut thin-film lithium niobate 1200 in straight segment 1270 in main nonlinear optical waveguide 1210 and segment 1272 in pump loop extension 1202 can be considered long legs of a rectangular-shaped path with curved corners or of a racetrack shaped path. These two segments are oriented to be aligned parallel to the y-axis of the x-cut lithium niobate crystal. As depicted, corner segments 1271 and 1273 of main nonlinear optical waveguide 1210 together with corner segment 1275 and corner segment 1277 of pump loop extension 1202 are the short legs of this rectangular-shaped or racetrack shaped path. The straight portions of corner segment 1271 and corner segment 1275 closest to first wavelength-selective coupler 1220 and the straight portions of corner segment 1273 and corner segment 1277 closest to second wavelength-selective coupler 1222 are aligned parallel to the z-axis of the x-cut lithium niobate crystal. In this example, transverse-electric (TE) polarized light propagating in main nonlinear optical waveguide 1210 encounters the largest second order nonlinear optical coefficient $d_{\text{sub.33}}$ when the light travels in straight segment 1270 in main nonlinear optical waveguide 1210.

(320) In this example, when phase matching is achieved, most of the nonlinear optical generation of signal light 1214 and idler light 1216 occurs in straight segment 1270 of main nonlinear optical waveguide 1210. Some nonlinear optical generation of signal and idler photons also occurs in corner segments 1271 and 1273 of main nonlinear optical waveguide 1210. Some generation of signal light 1214 and idler light 1216 also can occur in portions of corner segment 1275 and corner segment 1277 of pump loop extension 1202 because these portions comprise nonlinear optical material, as depicted in FIG. 16. However, the second order nonlinear optical coefficient $d_{\text{sub.22}}$ for transverse-electric (TE) polarized light in these portions is more than one order of magnitude smaller than the second order nonlinear optical coefficient $d_{\text{sub.33}}$ for transverse-electric (TE) polarized light in straight segment 1270 in this illustrative example. Moreover, the nonlinear optical generation of signal and idler photons that occurs in corner segment 1275 is partially counter-acted by the nonlinear optical generation of signal and idler photons that occurs in corner segment 1277. This is because the second order nonlinear optical coefficient $d_{\text{sub.22}}$ in these two segments have opposite sign. Segment 1272 in pump loop extension 1202 comprises a non-nonlinear optical material. Thus, no generation of signal and idler photons occurs in that segment.

(321) Turning next to FIG. 17, an illustration of an optical waveguide structure with five optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure 1300 is an example of an implementation for optical waveguide structure 100 as shown in FIGS. 1-6. As depicted, optical waveguide structure 1300 comprises optical waveguides in the form of main nonlinear optical waveguide 1310, secondary optical waveguide 1308 having segment 1307 and segment 1309, pump loop extension 1302, signal loop extension 1304, and idler loop extension 1306.

(322) In this illustrative example, main nonlinear optical waveguide 1310 and pump loop extension 1302 form a path that has a rectangular shape with curved corners and may also be referred to as a racetrack shape. Main nonlinear optical waveguide 1310 is comprised of a nonlinear optical material 104. Pump loop extension 1302 also is comprised of a nonlinear optical material 104. Optical waveguide structure 1300 is similar to optical waveguide structure 1200 in FIG. 16 but with the entire length of pump loop extension 1302 being comprised of nonlinear optical material 104 rather than having a portion of its length being comprised of a non-nonlinear optical material 105. In this example, main nonlinear optical waveguide 1310 has at least portion 1340 that comprises an electro-optic material 103. Also, pump loop extension 1302 has at least portion 1342 that comprises an electro-optic material.

(323) As depicted in this illustrative example, both signal loop extension 1304 and idler loop extension 1306 have a portion of their length comprising a nonlinear optical material 104 and another portion of their length comprising a non-nonlinear optical material 105. Nonlinear optical material 104 is included in these waveguides because nonlinear optical material 104 is electro-optic material 103 that is efficient with a large electro-optic coefficient. The portion of waveguide with the electro-optic (and nonlinear optical) material is located in section 1344 of signal loop extension 1304 and in section 1346 of idler loop extension 1306.

(324) As depicted, optical waveguide structure 1300 includes pump input optical waveguide 1332 that inputs pump light 1312. Optical waveguide structure 1300 also includes signal output optical waveguide 1334 and idler output optical waveguide 1336. Signal output optical waveguide 1334 can output signal light 1314. Idler output optical waveguide 1336 can output idler light 1316.

(325) In this illustrative example, pump optical coupler 1331 couples pump input optical waveguide 1332 to pump loop extension 1302. Signal optical coupler 1335 couples signal output optical waveguide 1334 to signal loop extension 1304. Idler optical coupler 1337 couples idler output optical waveguide 1336 to idler loop extension 1306.

(326) As depicted, first wavelength-selective coupler 1320 and second wavelength-selective coupler 1322, operated in their thru-state, connect pump loop extension 1302 to main nonlinear optical waveguide 1310. In this illustrative example, first wavelength-selective coupler 1320 and second wavelength-selective coupler 1322 operated in their cross-state connect the segments of secondary optical waveguide 1308 to main nonlinear optical waveguide 1310. In this illustrative example, third wavelength-selective coupler 1324 and fourth wavelength-selective coupler 1326 operated in their thru-state connect signal loop extension 1304 to segments of secondary optical waveguide 1308. In this illustrative example, third wavelength-selective coupler 1324 and fourth wavelength-selective coupler 1326 operated in their cross-state connect idler loop extension 1306 to segments of secondary optical waveguide 1308.

(327) In this illustrative example, first loop 1352 is present for pump light 1312. This first loop is a resonator loop in which pump light 1312 travels in a route that extends through main nonlinear optical waveguide 1310 and pump loop extension 1302. Signal light 1314 travels in second loop 1354. As depicted, second loop 1354 extends through main nonlinear optical waveguide 1310, through segments of secondary optical waveguide 1308 and through signal loop extension 1304. In this illustrative example, idler light 1316 travels in third loop 1356. As depicted, third loop 1356 extends through main nonlinear optical waveguide 1310, through segments of secondary optical waveguide 1308, and through idler loop extension 1306.

(328) In this illustrative example, first wavelength-selective coupler 1320 operating in its thru-state connects segment 1371 of main nonlinear optical waveguide 1310 and segment 1375 of pump loop extension 1302, and second wavelength-selective coupler 1322 connects segment 1377 of pump loop extension 1302 and segment 1373 of main nonlinear optical waveguide 1310. As depicted, first wavelength-selective coupler 1320, operating in its thru-state, couples pump light 1312 away from main nonlinear optical waveguide 1310 and into pump loop extension 1302 and second wavelength-selective coupler 1322, operating in its thru-state, couples pump light 1312 away from pump loop extension 1302 and into main nonlinear optical waveguide 1310 such that pump light 1312 travels in first loop 1352.

(329) In this illustrative example, first wavelength-selective coupler 1320, operating in its cross-state, extracts signal light 1314 and idler light 1318 away from main nonlinear optical waveguide 1310 and into segment 1307 of secondary optical waveguide 1308 such that signal light 1314 generated in main nonlinear optical waveguide 1310 does not travel in first loop 1352 but instead travels in second loop 1354 and idler light 1316 generated in main nonlinear optical waveguide 1310 does not travel in first loop 1352 but instead travels in third loop 1356. In this illustrative example, second wavelength-selective coupler 1322, operating in its cross-state, returns signal light 1314 traveling in second loop 1354 and idler light 1316 traveling in third loop 1356 back through main nonlinear optical waveguide 1310.

(330) In this illustrative example, signal light 1314 reaches signal loop extension 1304 by passing through a segment 1307 of secondary optical waveguide 1308 before being coupled by third wavelength-selective coupler 1324, operating in its thru-state, into signal loop

extension **1304**, signal light **1314** is returned from signal loop extension **1304** into a segment **1309** of secondary optical waveguide **1308** by fourth wavelength-selective coupler **1326**, operating in its thru-state. In this example, signal light **1314** passes through another portion, segment **1309**, of secondary optical waveguide **1308** before being coupled back into main nonlinear optical waveguide **1310** by second wavelength-selective coupler **1322**, operating in its cross-state.

(331) In this illustrative example, idler light **1316** reaches idler loop extension **1306** by passing through a segment **1307** of secondary optical waveguide **1308** before being coupled by third wavelength-selective coupler **1324**, operating in its cross-state, into the idler loop extension **1306**. Additionally, idler light **1316** is returned from idler loop extension **1306** into another segment **1309** of secondary optical waveguide **1308** by fourth wavelength-selective coupler **1326**, operating in its cross-state. In this example, idler light **1316** passes through another portion of secondary optical waveguide **1308** before being coupled back into main nonlinear optical waveguide **1310** by second wavelength-selective coupler **1322**, operating in its cross-state.

(332) In this illustrative example, pump light **1312**, signal light **1314**, and idler light **3016** travel in different resonator loops. In this illustrative example, first loop **1352** is a resonator loop for pump light **1312**. First loop **1352** extends through main nonlinear optical waveguide **1310**, pump loop extension **1302**, first wavelength-selective coupler **1320**, and second wavelength-selective coupler **1322**.

(333) Second loop **1354** is resonator loop for signal light **1314**. This second loop extends through main nonlinear optical waveguide **1310**, first wavelength-selective coupler **1320** and second wavelength-selective coupler **1322**; segments **1307**, **1309** of secondary optical waveguide **1308**, third wavelength-selective coupler **1324**; fourth wavelength-selective coupler **1326**; and signal loop extension **1304**.

(334) Third loop **1356** is a resonator loop for idler light **1316**. Third loop **1356** comprises main nonlinear optical waveguide **1310**; first wavelength-selective coupler **1320**; a segment **1307** of secondary optical waveguide **1308** between first wavelength-selective coupler **1320** and third wavelength-selective coupler **1324**; third wavelength-selective coupler **1324**; idler loop extension **1306**; fourth wavelength-selective coupler **1326**; a segment **1309** of secondary optical waveguide **1308** between fourth wavelength-selective coupler **1326** and second wavelength-selective coupler **1322**.

(335) As depicted, optical waveguide structure **1300** also includes phase shifters in the form of tuning electrodes. In this illustrative example, tuning electrode **1360** is located adjacent to a portion **1340** of main nonlinear optical waveguide **1310**. Tuning electrode **1362** is located adjacent to a portion **1342** of pump loop extension **1302**. Tuning electrode **1364** is located adjacent to section **1344** of signal loop extension **1304** and tuning electrode **1366** is located adjacent to section **1346** of idler loop extension **1306**. These tuning electrodes can apply voltages to obtain desired level of resonance to achieve a resonant condition for light traveling within optical waveguide structure **1300**. These tuning electrodes also can apply voltages to obtain a desired round-trip phase matching condition for the nonlinear optical generation process that occurs in optical waveguide structure **1300**.

(336) Compared to optical waveguide structure **1200** of FIG. **16**, optical waveguide structure **1300** has four tuning electrodes rather than three tuning electrodes. The additional tuning electrode (or set of tuning electrodes) provides greater flexibility for simultaneously achieving resonance conditions for all three wavelengths of light—pump light **1312**, signal light **1314**, and idler light **1316** in their respective resonator loops, first loop **1352**, second loop **1354** and third loop **1356** as well as to achieve round-trip phase matching. For example, tuning electrode **1360** can be used to adjust the round-trip phase $\phi_{\text{sub.RTp}}$ of pump light **1312** in first loop **1352**. Tuning electrode **1364** can be used to adjust the round-trip phase $\phi_{\text{sub.RTs}}$ of signal light **1314** in second loop **1354**, which is a signal loop. Tuning electrode **1366** can be used to adjust the round-trip phase $\phi_{\text{sub.RTi}}$ of idler light **1316** in third loop **1356**, which is an idler loop. Tuning electrode **1362** can be used to further adjust the round-trip phase $\phi_{\text{sub.RTp}}$ of pump light **1312** in order to achieve round-trip phase matching for the nonlinear optical process that occurs in main nonlinear optical waveguide **1310**. Using the terminology defined with reference to optical waveguide structure **1100** shown in FIG. **13**, the round-trip phase matching condition is achieved when:

$$\phi_{\text{sub.RTp}} - \phi_{\text{sub.RTs}} - \phi_{\text{sub.RTi}} = 2\pi A$$

(337) where A is an integer, and can be zero. This means: $P - S - I = A$ with the integers P , S and I defined earlier with reference to optical waveguide structure **1100** shown in FIG. **13**. Thus, for the example of optical waveguide structure **1300**, the four conditions for achieving integer values for the parameters P , S , I and A can be satisfied by adjusting the four tuning electrodes, tuning electrode **1360**, tuning electrode **1364**, tuning electrode **1366** and tuning electrode **1362**.

(338) In this illustrative example, electrically controlled phase shifts are provided in optical waveguide structure **1300**. In this illustrative example, portions of optical waveguide structure **1300** can be fabricated in x-cut lithium niobate. As depicted, the main nonlinear optical waveguide **1310** and pump loop extension **1302** through which the first loop **1352** extends form a rectangular shape with rounded corners. The orientation of optical waveguide structure **1300** can be such that segment **1370** in main nonlinear optical waveguide **1310** and segment **1372** in pump loop extension **1302** are aligned parallel to the y-axis of the lithium niobate crystal in the x-cut lithium niobate. These two segments—segment **1370** of main nonlinear optical waveguide **1310** and segment **1372** of pump loop extension **1302**—can be referred to as the long legs of the rectangular shape.

(339) The other portions of optical waveguides in the rectangular shaped waveguide structure defined by first loop **1352** include segment **1371** and segment **1373** of main nonlinear optical waveguide **1310** as well as segment **1375** and segment **1377** of pump loop extension **1302**. These segments are part of what can be referred to as the corners and short legs of the rectangular shaped or race-track shaped path traversed by first loop **1352**. In this illustrative example, segment **1371**, segment **1373**, segment **1375** and segment **1377** together with first wavelength-selective coupler **1320** and second wavelength-selective coupler **1322** are aligned mainly parallel with the z-axis of the x-cut lithium niobate crystal.

(340) In this illustrative example, orientation for optical waveguide structure **1300**, transverse-electric (TE) polarized light propagating in the optical waveguides traversed by first loop **1352** encounters the largest electro-optic coefficient of x-cut lithium niobate when the light travels in portion **1340** and portion **1342** of main nonlinear optical waveguide **1310** and pump loop extension **1302**, respectively. Portion **1340** and portion **1342** portions in which tunable phase shifts can occur. As depicted, the light travels in a clockwise direction around first loop **1352**. Furthermore, TE polarized signal light traversing portion in section **1344** of signal loop extension **1304** and TE polarized idler light traversing portion in section **1346** of idler loop extension **1306** also encounter the largest electro-optic coefficient of x-cut lithium niobate. Thus, the orientation depicted in FIG. **17** for optical waveguide structure **1300** can achieve efficient voltage-controlled electro-optic phase shifting.

(341) In this illustrative example, a nonlinear optical light generation process occurs in main nonlinear optical waveguide **1310**. Furthermore, to increase the nonlinear optical generation of signal and idler light that occurs in a given round-trip, it is desirable to meet another phase matching condition for propagation of the three wavelengths of light through segment **1370** of main nonlinear optical waveguide **1310**, which is the portion where most of the desired nonlinear optical generation occurs. This phase matching can be as follows:

(342) $0 \leq \phi_{\text{sub.Mup}} - \phi_{\text{sub.Mus}} - \phi_{\text{sub.Mui}} \leq \pi$, or $-\pi \leq \phi_{\text{sub.Mup}} - \phi_{\text{sub.Mus}} - \phi_{\text{sub.Mui}} \leq 0$, and is close to zero.

(343) Many materials such as lithium niobate that have a large electro-optic coefficient for a certain orientation also have a large second-

order nonlinear optical coefficient. In this illustrative example, a transverse-electric (TE) polarized light propagating in the optical waveguides traversed by first loop **1352**, which is a pump loop, encounters the largest second order nonlinear optical coefficient when the light travels in segment **1370** of main nonlinear optical waveguide **1310** and in segment **1372** of pump loop extension **1302**. In this illustrative example, the entire length of the optical waveguides traversed by the light in first loop **1352**, which includes main nonlinear optical waveguide **1310** and pump loop extension **1302**, comprises a nonlinear optical material. As a result, photons for signal light **1314** and idler light **1316** can be generated both in segment **1370** of main nonlinear optical waveguide **1310** and in segment **1372** of pump loop extension **1302**. Some, albeit typically less, generation of signal and idler light also occurs in the corner segments, segment **1371**, segment **1373**, segment **1375** and segment **1377**.

(344) In this illustrative example, the optical fields of signal light **1314** and idler light **1316** generated in an optical waveguide segment that comprises nonlinear optical material can be described by expressions such as:

$$(345) A_i(L) = \frac{2}{k_i c^2} \int_A^B \frac{2id_{eff} A_p A_s}{1} e^{i k z} dz \sim \frac{2id_{eff}}{k_i c^2} \frac{2 A_p A_s L}{i \left(\frac{M_{up}}{M_{up}} - \frac{M_{us}}{M_{us}} - \frac{M_{ui}}{M_{ui}} \right) - 1} \text{ and}$$

$$A_s(L) = \frac{2}{k_s c^2} \int_A^B \frac{2id_{eff} A_p A_i}{1} e^{i k z} dz \sim \frac{2id_{eff}}{k_s c^2} \frac{2 A_p A_i L}{i \left(\frac{M_{up}}{M_{up}} - \frac{M_{us}}{M_{us}} - \frac{M_{ui}}{M_{ui}} \right) - 1}.$$

(346) In these expression, A and B are the starting and ending points of a segment, such as segment **1370** of main nonlinear optical waveguide **1310** or segment **1372** of pump loop extension **1302**, with L being the length of that segment. The subscripts i, s, and p indicate pump, signal, and idler, respectively. The second order nonlinear optical coefficient d.sub.eff in segment **1370** has the opposite sign from the second order nonlinear optical coefficient d.sub.eff in segment **1372**. As a result, the contributions to the signal and idler optical fields from segment **1370** of main nonlinear optical waveguide **1310** and segment **1372** of pump loop extension **1302** can counteract each other, or the optical fields can interfere in a destructive manner, if the optical fields from these two segments are combined together, assuming the phase matching is perfect.

(347) Optical waveguide structure **1300** avoids the interaction of signal and idler light generated in segment **1370** with signal and idler light generated in segment **1372**. First wavelength-selective coupler **1320** functions to couple signal light **1314** and idler light **1316** generated in segment **1370** away from pump loop extension **1302** and thus away from segment **1372** by diverting that light into segment **1307** of secondary optical waveguide **1308**. Similarly, second wavelength-selective coupler **1322** functions to couple signal light **1314** and idler light **1316** generated in segment **1372** away from main nonlinear optical waveguide **1310** and thus away from segment **1370**, as shown by arrow **1380** into output optical waveguide **1305**. This coupling function done by second wavelength-selective coupler **1322** is performed in addition to coupling signal light **1314** in second loop **1354** and idler light **1316** in third loop **1356** from segment **1309** of secondary optical waveguide **1308** into main nonlinear optical waveguide **1310**. Thus, the signal light **1314** and idler light **1316** coupled back into main nonlinear optical waveguide **1310** through second wavelength-selective coupler **1322** is generated in a prior pass through main nonlinear optical waveguide **1310** and is not generated in the pump loop extension **1302**.

(348) As a result, any destructive interaction between signal light **1314** and idler light **1316** generated in segment **1370** and generated in segment **1372** is absent. Thus, signal light **1314** and idler light **1316** that result from circulation through many round-trips in the optical waveguide structure **1300** are those photons for signal light **1314** and idler light **1316** generated primarily in segment **1370** in main nonlinear optical waveguide **1310**.

(349) Next, FIG. **18** is an illustration of an optical waveguide structure with ten optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **1500** is an example of an implementation for optical waveguide structure **100** as shown in FIGS. **1-6**. More specifically, FIG. **18** is an implementation for optical waveguides **102** as depicted in FIG. **2**.

(350) As depicted, optical waveguide structure **1500** comprises optical waveguides in the form of first main nonlinear optical waveguide segment **1510A**, second main nonlinear optical waveguide segment **1510B**, first pump bypass optical waveguide **1502A**, second pump bypass optical waveguide **1502B**, first secondary optical waveguide portion **1508A**, second secondary optical waveguide portion **1508B**, first signal loop extension **1504A**, second signal loop extension **1504B**, first idler loop extension **1506A**, and second idler loop extension **1506B**. First main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510** are examples of main nonlinear optical waveguide **106** in FIG. **2**. First pump bypass optical waveguide **1502A** and second pump bypass optical waveguide **1502B** are examples of an implementation for first extension optical waveguide **108** in FIG. **2**. First secondary optical waveguide portion **1508A** and second secondary optical waveguide portion **1508B** are examples of an implementation of secondary optical waveguide **113** in FIG. **2**.

(351) As depicted in the detailed illustrative example of FIG. **18**, main nonlinear optical waveguide **1510** comprises two separate segments, first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B**. Secondary optical waveguide **1508** comprises multiple segments that are part of first secondary optical waveguide portion **1508A** and second secondary optical waveguide portion **1508B**. In this example, extension optical waveguide **1502** has two distinct portions, called pump bypass waveguides. In this illustrative example, first pump bypass optical waveguide **1502A** and second pump bypass optical waveguide **1502B** are connected to optical couplers at each of the two ends of each of those optical waveguides. These optical waveguides are comprised of a non-nonlinear optical material **105** in this example.

(352) First secondary optical waveguide portion **1508A** is connected to first signal loop extension **1504A** and first idler loop extension **1506A**. Second secondary optical waveguide portion **1508B** is connected to second signal loop extension **1504B** and second idler loop extension **1506B**. These connections from the secondary optical waveguide portions to the various signal loop extensions and idler loop extensions are made through wavelength-selective couplers such as first signal loop coupler **1594A**, first idler loop coupler **1596A**, second signal loop coupler **1594B**, and second idler loop coupler **1596B**. Connections between first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B** of the main nonlinear optical waveguide and first secondary optical waveguide portion **1508A** and second secondary optical waveguide portion **1508B** of the secondary optical waveguide are made through wavelength selective optical couplers, such as first wavelength-selective coupler **1580**, second wavelength-selective coupler **1586**, third wavelength-selective coupler **1584**, and fourth wavelength-selective coupler **1582**.

(353) In this illustrative example, optical waveguide structure **1500** also includes pump input optical waveguide **1532**, signal output optical waveguide **1534**, and idler output optical waveguide **1536**. Pump input optical waveguide **1532** can input pump light **1512** into second pump bypass optical waveguide **1502B**. Signal output optical waveguide **1534** can output signal light **1514** from second signal loop extension **1504B**. Idler output optical waveguide **1536** can output idler light **1516** from second idler loop extension **1506B**.

(354) In this illustrative example, pump optical coupler **1531** couples pump input optical waveguide **1532** to second pump bypass optical waveguide **1502B**. Signal optical coupler **1535** couples second signal loop extension **1504B** to signal output optical waveguide **1534**. Idler optical coupler **1537** couples second idler loop extension **1506B** to idler output optical waveguide **1536**.

(355) As depicted, first wavelength-selective coupler **1580** and second wavelength-selective coupler **1586** connect pump bypass optical waveguide **1502A** to two different segments, first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B**, of main nonlinear optical waveguide **1510**. In this illustrative example, third wavelength-selective coupler **1584** and fourth wavelength-selective coupler **1582** connect second pump bypass optical waveguide **1502B** to the opposite ends of those two segments, first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B**, of main nonlinear optical waveguide **1510**.

(356) In this illustrative example, pump light **1512** travels in pump loop **1552**. Pump loop **1552** is a resonator loop that extends through first main nonlinear optical waveguide segment **1510A** of main nonlinear optical waveguide; first wavelength-selective coupler **1580**; first pump bypass optical waveguide **1502A**; second wavelength-selective coupler **1586**; second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide; third wavelength-selective coupler **1584**; second pump bypass optical waveguide **1502B**; and fourth wavelength-selective coupler **1582**; and continues again through first main nonlinear optical waveguide segment **1510A** of main nonlinear optical waveguide **1510**. The lengths of the various waveguides through which pump light **1512** of pump wavelength travels in pump loop **1552** can be selected so that pump wavelength matches a resonance condition for pump loop **1552**.

(357) Pump light **1512**, signal light **1514** and idler light **1516** all travel through first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510**. As depicted, first wavelength-selective coupler **1580** couples pump light **1512** into first pump bypass optical waveguide **1502A**. First wavelength-selective coupler **1580** also couples signal light **1514** and idler light **1516** into first secondary optical waveguide portion **1508A**. Thus, only signal light **1514** and idler light **1516** travel through second secondary optical waveguide portion **1508B**. First signal loop coupler **1594A** couples signal light from segment **1571** of first secondary optical waveguide portion **1508A** into first signal loop extension **1504A**. First signal loop coupler **1594A** also couples signal light that has propagated through first signal loop extension **1504A** into segment **1573** of first secondary optical waveguide portion **1508A**. Signal light **1514** then continues to propagate through first secondary optical waveguide portion **1508A**, being coupled by first idler loop coupler **1596A** from segment **1573** to segment **1575** of first secondary optical waveguide portion **1508A**. Second wavelength-selective coupler **1586** couples signal light **1514** from first secondary optical waveguide portion **1508A** into second main nonlinear optical waveguide segment **1510B**. Second wavelength-selective coupler **1586** also couples pump light from first pump bypass optical waveguide **1502A** into second main nonlinear optical waveguide segment **1510B**.

(358) As with first main nonlinear optical waveguide segment **1510A** of main nonlinear optical waveguide **1510**, pump light **1512**, signal light **1514** and idler light **1516** all travel through second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510**. As depicted, third wavelength-selective coupler **1584** couples pump light **1512** into second pump bypass optical waveguide **1502B**. Third wavelength-selective coupler **1584** also couples signal light **1514** and idler light **1516** into second secondary optical waveguide portion **1508B**. Thus, only signal light **1514** and idler light **1516** travel through second secondary optical waveguide portion **1508B**.

(359) Second signal loop coupler **1594B** couples signal light from segment **1576** of second secondary optical waveguide portion **1508B** into second signal loop extension **1504B**. Second signal loop coupler **1594B** also couples signal light that has propagated through second signal loop extension **1504B** into segment **1574** of second secondary optical waveguide portion **1508B**. Signal light **1514** then continues to propagate through second secondary optical waveguide portion **1508B**, being coupled by second idler loop coupler **1596B** from segment **1574** to segment **1572** of second secondary optical waveguide portion **1508B**. Fourth wavelength-selective coupler **1582** couples signal light **1514** from second secondary optical waveguide portion **1508B** again into first main nonlinear optical waveguide segment **1510A**. Fourth wavelength-selective coupler **1582** also couples pump light **1512** from first pump bypass optical waveguide **1502A** into first main nonlinear optical waveguide segment **1510A**.

(360) In this illustrative example, signal light **1514** travels in signal loop **1554**. Signal loop **1554** is a resonator loop that can be thought of as comprising two halves. One half of signal loop **1554** extends through first main nonlinear optical waveguide segment **1510A** of main nonlinear optical waveguide **106**; first wavelength-selective coupler **1580** (in its thru state); segment **1571** of first secondary optical waveguide portion **1508A**; first signal loop coupler **1594A** (in its cross state); first signal loop extension **1504A**; a second pass through first signal loop coupler **1594A** (again in its cross state); segment **1573** of first secondary optical waveguide portion **1508A**; first idler loop coupler **1596A** (in its thru state); segment **1575** of first secondary optical waveguide portion **1508A**; and second wavelength-selective coupler **1586** (in its thru state). A second half of signal loop **1554** extends through second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510**; third wavelength-selective coupler **1584** (in its thru state); segment **1576** of second secondary optical waveguide portion **1508B**; second signal loop coupler **1594B** (in its cross state); second signal loop extension **1504B**; a second pass through second signal loop coupler **1594B** (again in its cross state); segment **1574** of second secondary optical waveguide portion **1508B**; second idler loop coupler **1596B** (in its thru state); segment **1572** of second secondary optical waveguide portion **1508B**; and fourth wavelength-selective coupler **1582** (in its thru state). The lengths of the various waveguides through which signal light **1514** of a signal wavelength travels in signal loop **1554** can be selected so that signal wavelength matches a resonance condition for signal loop **1554**.

(361) Additionally, besides coupling signal light **1514**, first wavelength-selective coupler **1580** also couples idler light **1516** into first secondary optical waveguide portion **1508A**. Thus, only signal light **1514** and idler light **1516** travel through first secondary optical waveguide portion **1508A**. Idler light **1516** then continues to propagate through first secondary optical waveguide portion **1508A**, being coupled by first signal loop coupler **1594A** from segment **1571** to segment **1573** of first secondary optical waveguide portion **1508A**.

(362) In this illustrative example, first idler loop coupler **1596A** couples idler light **1516** from segment **1573** of first secondary optical waveguide portion **1508A** into first idler loop extension **1506A**. First idler loop coupler **1596A** also couples idler light that has propagated through first idler loop extension **1506A** into segment **1575** of first secondary optical waveguide portion **1508A**. Second wavelength-selective coupler **1586** couples idler light **1516** from first secondary optical waveguide portion **1508A** into second main nonlinear optical waveguide segment **1510B**. Pump light **1512**, signal light **1514** and idler light **1516** all travel through second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510**. Besides coupling pump light **1512** into second pump bypass optical waveguide **1502B**, third wavelength-selective coupler **1584** also couples idler light **1516** and signal light **1514** into second secondary optical waveguide portion **1508B**. Thus, only idler light **1516** and signal light **1514** travel through second secondary optical waveguide portion **1508B**. Idler light **1516** then continues to propagate through second secondary optical waveguide portion **1508B**, being coupled by second signal loop coupler **1594B** from segment **1576** to segment **1574** of second secondary optical waveguide portion **1508B**.

(363) As depicted, second idler loop coupler **1596B** couples idler light **1516** from segment **1574** of second secondary optical waveguide portion **1508B** into second idler loop extension **1506B**. Second idler loop coupler **1596B** also couples idler light that has propagated through second idler loop extension **1506B** into segment **1572** of second secondary optical waveguide portion **1508B**. Fourth wavelength-selective coupler **1582** couples idler light **1516** from second secondary optical waveguide portion **1508B** into first main nonlinear optical waveguide

segment **1510A**.

(364) In this illustrative example, idler light **1516** travels in idler loop **1556**. Idler loop **1556** is a resonator loop that can be thought of as comprising two halves. One half of idler loop **1556** extends through first main nonlinear optical waveguide segment **1510A** of main nonlinear optical waveguide **1510**; first wavelength-selective coupler **1580** (in its thru state); segment **1571** of first secondary optical waveguide portion **1508A**; first signal loop coupler **1594A** (in its thru state); segment **1573** of first secondary optical waveguide portion **1508A**; first idler loop coupler **1596A** (in its cross state); first idler loop extension **1506A**; a second pass through first idler loop coupler **1596A** (again in its cross state); segment **1575** of first secondary optical waveguide portion **1508A**; and second wavelength-selective coupler **1586** (in its thru state). A second half of idler loop **1556** extends through second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510**; third wavelength-selective coupler **1584** (in its thru state); segment **1576** of second secondary optical waveguide portion **1508B**; second signal loop coupler **1594B** (in its thru state); segment **1574** of second secondary optical waveguide portion **1508B**; second idler loop coupler **1596B** (in its cross state); second idler loop extension **1506B**; a second pass through second idler loop coupler **1596B** (again in its cross state); segment **1572** of second secondary optical waveguide portion **1508B**; and fourth wavelength-selective coupler **1582** (in its thru state). The lengths of the various waveguides through which idler light **1516** of idler wavelength travels in idler loop **1556** can be selected so that the idler wavelength matches a resonance condition for idler loop **1556**.

(365) As depicted, the resonator loops, pump loop **1552**, signal loop **1554**, and idler loop **1556**, have portions that overlap each other and portions that do not overlap each other. All three loops include nonlinear optical waveguide segments, such as first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B**. Signal loop **1554** and idler loop **1556** further overlap each other through portions, such as first secondary optical waveguide portion **1508A** and second secondary optical waveguide portion **1508B**. Phase shifters can be placed at the non-overlapping portions of pump loop **1552**, signal loop **1554**, and idler loop **1556** to produce phase shifts for pump light **1512**, idler light **1516**, and signal light **1514** that can be adjusted separately from each other.

(366) The signal loop **1554**, idler loop **1556**, and pump loop **1552** can each be considered as having two halves. These halves can be distinguished in the illustration of FIG. **18** by their location relative to the reference line **1595**. A first half includes the components to the right of reference line **1595**. A second half includes the components to the left of reference line **1595**.

(367) As depicted, optical waveguide structure **1500** includes phase shifters in the form of tuning electrodes. In this example, tuning electrode **1564A** and tuning electrode **1565A** are located adjacent to first signal loop extension **1504A**. These tuning electrodes enable adjustment of the phase of signal light **1514** in the first half of signal loop **1554**. Tuning electrode **1566A** and tuning electrode **1567A** are located adjacent to first idler loop extension **1506A**. These tuning electrodes enable adjustment of the phase of idler light **1516** in the first half of idler loop **1556**. Tuning electrode **1564B** and tuning electrode **1565B** are located adjacent to second signal loop extension **1504B**. These tuning electrodes enable adjustment of the phase of signal light **1514** in the second half of signal loop **1554**. Tuning electrode **1566B** and tuning electrode **1567B** are located adjacent to second idler loop extension **1506B**. These tuning electrodes enable adjustment of the phase of idler light **1516** in the second half of idler loop **1556**.

(368) Tuning electrode **1560A** is located adjacent to first main nonlinear optical waveguide segment **1510A**. Tuning electrode **1560A** can be used to adjust the phase of pump light **1512** in the first half of pump loop **1552**. Tuning electrode **1560B** is located adjacent to second main nonlinear optical waveguide segment **1510B**. Tuning electrode **1560B** can be used to adjust the phase of pump light **1512** in the second half of pump loop **1552**. Since signal light **1514** and idler light **1516** also propagate through first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B** for main nonlinear optical waveguide **1510**, tuning electrode **1560A** and tuning electrode **1560B** also affect the phase of signal light **1514** and idler light **1516**. The use of tuning electrodes to accomplish resonance matching and round-trip phase matching was described with reference to FIG. **13**, as an example.

(369) In optical waveguide structure **1500**, nonlinear optical generation of signal light **1514** and idler light **1516** from pump light **1512** occurs only in first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510**. First main nonlinear optical waveguide segment **1510A** can be considered as part of the first half of optical waveguide structure **1500**. Second main nonlinear optical waveguide segment **1510B** can be considered as part of the second half of optical waveguide structure **1500**.

(370) In this example, signal light **1514** and idler light **1516** propagate in first secondary optical waveguide portion **1508A** and second secondary optical waveguide portion **1508B** of secondary optical wavelength and in first signal loop extension **1504A** and second signal loop extension **1504B** as well as in first idler loop extension **1506A** and second idler loop extension **1506B** without further nonlinear optical generation of signal photons or idler photons. Pump light **1512**, from which the signal light **1514** and idler light **1516** are generated, is absent from those waveguides.

(371) In this illustrative example, the phases of the pump light **1512**, signal light **1514** and idler light **1516** in the two halves of optical waveguide structure **1500** can be adjusted to achieve a constructive interaction between the signal light and idler light generated in the first half of optical waveguide structure **1500** and the signal light and idler light generated in the second half of optical waveguide structure **1500**. This constructive interaction can be achieved even though the nonlinear optical coefficient can have a first sign in first main nonlinear optical waveguide segment **1510A** of the first half and a second sign, opposite to the first sign, in second main nonlinear optical waveguide segment **1510B** of the second half.

(372) In this illustrative example, the nonlinear optical coefficient for light propagating in the first main nonlinear optical waveguide segment **1510A** of the upper-right half-structure **1591** of optical waveguide structure **1500** has one sign for the nonlinear optical coefficient **112**. The light propagating in second main nonlinear optical waveguide segment **1510B** in lower-left half-structure **1592** of optical waveguide structure **1500** has an opposite sign for the nonlinear optical coefficient.

(373) In other words, the two segments, first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B**, of main nonlinear optical waveguide **1510** can be considered as part of two half-structures, upper-right half-structure **1591** and lower-left half-structure **1592**. As depicted, these two half-structures are separated by reference line **1595** extending from the upper left corner of optical waveguide structure **1500** to the lower right corner of optical waveguide structure **1500**. As shown, reference line **1595** intersects optical waveguide structure **1500** at a location A between second wavelength-selective coupler **1586** for reinserting pump light **1512** in second main nonlinear optical waveguide segment **1510B** of main nonlinear optical waveguide **1510** and the tuning electrode **1560B** in second main nonlinear optical waveguide segment **1510B** and at another location B between fourth wavelength-selective coupler **1582** for reinserting pump light **1512** into first main nonlinear optical waveguide segment **1510A** of main nonlinear optical waveguide **1510** and tuning electrode **1560A** in first main nonlinear optical waveguide segment **1510A**.

(374) For the upper-right half-structure **1591**, the relative phase walk-off for travel from upper left to lower right (i.e., from location A to location B) of upper-right half-structure **1591** should preferably be an odd multiple of π radians. Also, the relative phase walk-off from travel through first main nonlinear optical waveguide segment **1510A** of main nonlinear optical waveguide **1510**, where the nonlinear

optical generation occurs, is preferably less than π radians and as close to zero as possible. The cross-sectional structure of first main nonlinear optical waveguide segment **1510A** can be designed to achieve the desired phase match (and minimal relative phase walk-off) for travel through first main nonlinear optical waveguide segment **1510A**. Similarly, for lower-left half-structure **1592**, the relative phase walk-off for travel from lower right to upper left (i.e., from location B to location A) of lower-left half-structure **1592** should be an odd multiple of π radians. Also, the relative phase walk-off from travel through the second main nonlinear optical waveguide segment **1510B**, where the nonlinear optical generation occurs, is less than π radians and as close to zero as possible. The cross-sectional structure of second main nonlinear optical waveguide segment **1510B** can be designed to achieve the desired phase match (and minimal relative phase walk-off) for travel through second main nonlinear optical waveguide segment **1510B**.

(375) Thus, the lengths of the pump loop **1552**, the signal loop **1554**, and idler loop **1556** in each of the upper-right half-structure **1591** and the lower-left half-structure **1592**, as well as the cross-sectional structures of the waveguides in each of those two half-structures can be designed to achieve the desired relative phase walk-off that is preferably an odd multiple of π radians. Also, the relative phase walk-off from travel through first main nonlinear optical waveguide segment **1510A** for main nonlinear optical waveguide **1510**, where additional nonlinear optical generation occurs, is preferably less than π radians and ideally is zero. Similarly, the relative phase walk-off from travel through second main nonlinear optical waveguide segment **1510B**, where additional nonlinear optical generation again occurs, is preferably less than π radians and ideally is zero. Furthermore, the lengths and the cross-sectional structure of the waveguides traversed in both upper-right half-structure **1591** and the lower-left half-structure **1592** can be selected to also achieve round-trip phase matching for the nonlinear optical generation that occurs in the combination of two halves of optical waveguide structure **1500**. Thus, the round-trip phase for the nonlinear optical interaction of the pump, signal and idler light is preferably a multiple of 360 degrees or 2π radians.

(376) Making the phase walk-off for each half-structure, such as upper-right half-structure **1591** and lower-left half-structure **1592**, have a value that is an odd multiple of 180 degrees or π radians compensates for the reversal in sign of the nonlinear optical coefficient of the nonlinear optical material in the nonlinear optical waveguide segments, such as first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B**, of those two half-structures. First main nonlinear optical waveguide segment **1510A** is in upper-right half-structure **1591** and second main nonlinear optical waveguide segment **1510B** is in lower-left half-structure **1592**. The nonlinear optical coefficient **112** in first main nonlinear optical waveguide segment **1510A** has a first sign **107** and the nonlinear optical coefficient **112** in second main nonlinear optical waveguide segment **1510B** has a second sign **109** that is opposite from the first sign. The configuration of two half-structures is especially useful for optical waveguide structures **1500** that comprise second-order nonlinear optical material. An example of such material is x-cut lithium niobate. This configuration of two half-structures is especially useful when the nonlinear optical waveguide segments, such as first main nonlinear optical waveguide segment **1510A** and second main nonlinear optical waveguide segment **1510B** containing x-cut lithium niobate are oriented parallel to the material Y-axis, with the propagating optical fields of the pump light **1512**, signal light **1514** and idler light **1516** having transverse electric (TE) components that are aligned parallel to the material X-axis.

(377) Besides meeting the phase matching conditions for the two half-structures, upper-right half-structure **1591** and lower-left half-structure **1592** that form optical waveguide structure **1500**, the other optical waveguides in optical waveguide structure **1500** can be designed to enable the pump light **1512**, signal light **1514**, and idler light **1516** to match resonances of their respective resonator loops, pump loop **1552**, signal loop **1554**, and idler loop **1556**.

(378) Turning next to FIG. **19**, an illustration of a flowchart of a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart can be implemented in optical waveguide structure **100** in FIG. **1** as well the other optical waveguide structures depicted in other figures.

(379) The process begins by routing a first-wavelength light in a main nonlinear optical waveguide having a first length (operation **1900**). The process optically couples, by a first wavelength-selective coupler, a first main location in the main nonlinear optical waveguide and a first extension location in the extension optical waveguide to each other such that the first-wavelength light is coupled from the main nonlinear optical waveguide at the first main location to the extension optical waveguide at the first extension location (operation **1902**). The process does not optically couple a second-wavelength light from the main nonlinear optical waveguide at the first main location to the extension optical waveguide at the first extension location (operation **1903**). The process does not optically couple a third-wavelength light from the main nonlinear optical waveguide at the first main location to the extension optical waveguide at the first extension location (operation **1904**).

(380) The process optically couples, by a second wavelength-selective coupler, a second main location in the main nonlinear optical waveguide and a second extension location in the extension optical waveguide to each other such that the first wavelength-light is coupled from the extension optical waveguide at the second extension location to the main nonlinear optical waveguide at the second main location (operation **1905**). The process does not optically couple the second-wavelength light from the extension optical waveguide at the second extension location to the main nonlinear optical waveguide at the second main location (operation **1906**). The process does not optically couple the third-wavelength light from the extension optical waveguide at the second extension location to the main nonlinear optical waveguide at the second main location (operation **1908**). The process terminates thereafter.

(381) With reference to FIG. **20**, an illustration of a flowchart of additional operations for a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart depicts additional operations that can be performed in addition to the operations in FIG. **19**.

(382) The process optically couples, by the first wavelength-selective coupler the first main location in the main nonlinear optical waveguide and a first secondary location in the secondary optical waveguide to each other such that the second-wavelength light is coupled from the main nonlinear optical waveguide at the first main location to the secondary optical waveguide at the first secondary location (operation **2000**). The process optically couples, by a third wavelength-selective coupler, a third secondary location in the secondary optical waveguide and a primary second extension location in the second extension optical waveguide to each other such that the second-wavelength light is coupled from the secondary optical waveguide at the third secondary location to the second extension optical waveguide at the primary second extension location (operation **2001**). The process does not optically couple a third-wavelength light from the secondary optical waveguide at the third secondary location to the second extension optical waveguide at the primary second extension location (operation **2003**). The process optically couples, by a fourth wavelength-selective coupler, a fourth secondary location in the secondary optical waveguide and a secondary second extension location in the second extension optical waveguide to each other such that the second-wavelength light is coupled from the second extension optical waveguide at the secondary second extension location to the secondary optical waveguide at the fourth secondary location (operation **2004**). The process does not optically couple the third-wavelength light from the second extension optical waveguide at the secondary second extension location to the secondary optical waveguide at the fourth secondary location (operation **2006**). The process optically couples, by the second wavelength-selective coupler, a second secondary

location in the second optical waveguide and the second main location in the main nonlinear optical waveguide to each other such that the second-wavelength light is coupled from the secondary optical waveguide at the second secondary location to the main nonlinear optical waveguide at the second main location (operation **2008**). The process terminates thereafter.

(383) Turning to FIG. **21**, an illustration of a flowchart of additional operations for a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart depicts additional operations that can be performed in addition to the operations in FIG. **19**.

(384) The process optically couples, by the first wavelength-selective coupler, the first main location in the main nonlinear optical waveguide and a first secondary location in the secondary optical waveguide to each other such that the third-wavelength light is coupled from the main nonlinear optical waveguide at the first main location to the secondary optical waveguide at the first secondary location (operation **2100**). Also, the process optically couples, by the third wavelength-selective coupler, a third secondary location in the secondary optical waveguide and a primary third extension location in the third extension optical waveguide to each other such that the third-wavelength light is coupled from the secondary optical waveguide at the third secondary location to the third extension optical waveguide at the primary third extension location (operation **2101**). The process does not optically couple the second-wavelength light from the secondary optical waveguide at the third secondary location to the third extension optical waveguide at the primary third extension location (operation **2102**). The process optically couples, by the fourth wavelength-selective coupler, a fourth secondary location in the secondary optical waveguide and a secondary third extension location in the third extension optical waveguide to each other such that the third-wavelength light is coupled from the third extension optical waveguide at the secondary third extension location to the secondary optical waveguide at the fourth secondary location (operation **2103**). The process does not optically couple the second-wavelength light from the third extension optical waveguide at the secondary third extension location to the secondary optical waveguide at the fourth secondary location (operation **2104**). The process optically couples, by the second wavelength-selective coupler, the second secondary location in the secondary optical waveguide and the second main location in the main nonlinear optical waveguide to each other such that the third-wavelength light is coupled from the secondary optical waveguide at the second secondary location to the main nonlinear optical waveguide at the second main location (operation **2106**). The process terminates thereafter.

(385) Turning to FIG. **22**, an illustration of a flowchart of an additional operation for a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart depicts additional operations that can be performed in addition to the operations in FIGS. **19-21**.

(386) The process applies an activation to a portion of the main nonlinear optical waveguide such that a phase shifts in the first-wavelength light to achieve a resonance condition for the first-wavelength light (operation **2200**). The process terminates thereafter.

(387) With reference next to FIG. **23**, an illustration of a flowchart of an additional operation for a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart depicts additional operations that can be performed in addition to the operations in FIG. **21** and FIG. **22**.

(388) The process applies an activation to a portion of the second extension waveguide such that a phase shifts in the second-wavelength light to achieve a resonance condition for the second-wavelength light (operation **2300**). The process terminates thereafter.

(389) Turning now to FIG. **24**, an illustration of a flowchart of an additional operation for a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart depicts additional operations that can be performed in addition to the operations in FIG. **21** and FIG. **23**.

(390) The process applies an activation to a portion of the third extension optical waveguide such that a phase shifts in the third-wavelength light to achieve a resonance condition for the third-wavelength light (operation **2400**). The process terminates thereafter.

(391) Turning now to FIG. **25**, an illustration of a flowchart of an additional operation for a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart depicts additional operations that can be performed in addition to the operations in FIGS. **22-24**.

(392) The process applies at least one of an activation to a portion of the main nonlinear optical waveguide such that a phase shifts in the first-wavelength light, an activation to a portion of the second extension waveguide such that a phase shifts in the second-wavelength light, and an activation to a portion of the third extension optical waveguide such that a phase shifts in the third-wavelength light to achieve a round-trip phase matching condition for the nonlinear optical process involving the first-wavelength light, the second-wavelength light, and the third-wavelength light (operation **2500**). The process terminates thereafter.

(393) To achieve phase matching, the activation does not necessarily need to be applied to all three of the main nonlinear optical waveguide, the second extension waveguide, and the third extension waveguide. The activation can be applied to one of some combination of the three waveguides or waveguide portions.

(394) Turning now to FIG. **26**, an illustration of a flowchart of an additional operation for a process for a non-linear optical process is depicted in accordance with an illustrative embodiment. The process in this flowchart depicts additional operations that can be performed in addition to the operations in FIGS. **22-25**.

(395) The process applies an activation to a portion of the first extension optical waveguide such that a phase shifts in the first-wavelength light to achieve a resonance condition for the first-wavelength light and to achieve a round-trip phase matching condition for the nonlinear optical process involving the first-wavelength light, the second-wavelength light, and the third-wavelength light (operation **2600**). The process terminates thereafter. In operation **2600**, this activation can be accomplished by tuning electrode **1362** in FIG. **17**.

(396) The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams can represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks can be implemented as program code, hardware, or a combination of the program code and hardware. When implemented in hardware, the hardware can, for example, take the form of integrated circuits that are manufactured or configured to perform one or more operations in the flowcharts or block diagrams. When implemented as a combination of program code and hardware, the implementation may take the form of firmware. Each block in the flowcharts or the block diagrams can be implemented using special purpose hardware systems that perform the different operations or combinations of special purpose hardware and program code run by the special purpose hardware.

(397) In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram.

(398) Turning now to FIG. **27**, an illustration of a block diagram of a product management system is depicted in accordance with an

illustrative embodiment. Product management system **2700** is a physical hardware system. In this illustrative example, product management system **2700** includes at least one of manufacturing system **2702** or maintenance system **2704**.

(399) Manufacturing system **2702** is configured to manufacture products. As depicted, manufacturing system **2702** includes manufacturing equipment **2706**. Manufacturing equipment **2706** includes at least one of fabrication equipment **2708** or assembly equipment **2710**.

(400) Fabrication equipment **2708** is equipment that used to fabricate the nonlinear optical waveguide structure. Multiple copies or multiple versions of nonlinear optical waveguide structures can be fabricated on a substrate wafer.

(401) The substrate wafer can comprise a material such as silicon, lithium niobate, quartz, sapphire, silicon carbide, or some other suitable substrate. Fabrication equipment **2708** can be used to fabricate at least one of optical waveguide structures, nonlinear optical waveguides, optical couplers, optical waveguide segments, laser transmitters, ultraviolet transmission systems, point-to-point communication devices, laser infrared countermeasure sources, through water optical communication devices, or other suitable devices, antennas, or other suitable types of parts. For example, fabrication equipment **2708** can include machines and tools.

(402) With respect to fabricating semiconductor components and optical waveguide components, fabrication equipment **2708** can comprise at least one of an epitaxial reactor, an oxidation system, a diffusion system, an etching system, a cleaning system, a bonding machine, a dicing machine, a wafer saw, an ion implantation system, a physical vapor deposition system, a chemical vapor deposition system, a photolithography system, an electron-beam lithography system, a plasma etcher, a die attachment machine, a wire bonder, a die overcoat system, molding equipment, a hermetic sealer, an electrical tester, a burn-in oven, a retention bake oven, a UV erase system, or other suitable types of equipment that can be used to manufacture semiconductor structures.

(403) Assembly equipment **2710** is equipment used to assemble parts to form a product such as a chip, an integrated circuit, a multi-chip module, a computer, a signal processor, an aircraft, or some other product. Assembly equipment **2710** also can include machines and tools. These machines and tools may be at least one of a robotic arm, a spinner system, a sprayer system, and elevator system, a rail-based system, or a robot.

(404) In this illustrative example, maintenance system **2704** includes maintenance equipment **2712**. Maintenance equipment **2712** can include any equipment needed to perform maintenance on and evaluation of a product. Maintenance equipment **2712** may include tools for performing different operations on parts on a product. These operations can include at least one of disassembling parts, refurbishing parts, inspecting parts, reworking parts, manufacturing replacement parts, or other operations for performing maintenance on the product. These operations can be for routine maintenance, inspections, upgrades, refurbishment, or other types of maintenance operations.

(405) In the illustrative example, maintenance equipment **2712** may include optical inspection devices, electron-beam imaging systems, x-ray imaging systems, surface-profile measurement systems, drills, vacuum leak checkers, and other suitable devices. In some cases, maintenance equipment **2712** can include fabrication equipment **2708**, assembly equipment **2710**, or both to produce and assemble parts that needed for maintenance.

(406) Product management system **2700** also includes control system **2714**. Control system **2714** is a hardware system and may also include software or other types of components. Control system **2714** is configured to control the operation of at least one of manufacturing system **2702** or maintenance system **2704**. In particular, control system **2714** can control the operation of at least one of fabrication equipment **2708**, assembly equipment **2710**, or maintenance equipment **2712**.

(407) The hardware in control system **2714** can be implemented using hardware that may include computers, circuits, networks, and other types of equipment. The control may take the form of direct control of manufacturing equipment **2706**. For example, robots, computer-controlled machines, and other equipment can be controlled by control system **2714**. In other illustrative examples, control system **2714** can manage operations performed by human operators **2716** in manufacturing or performing maintenance on a product. For example, control system **2714** can assign tasks, provide instructions, display models, or perform other operations to manage operations performed by human operators **2716**. In these illustrative examples, the different processes for fabricating semiconductor structures, optical structures, nonlinear optical waveguides, laser transmitters, photon generators, photon transmitters, photon detectors, ultraviolet transmission systems, point-to-point communication devices, laser infrared countermeasure sources, through water optical communication devices, or other suitable devices can be manufactured using processes implemented in control system **2714**.

(408) In the different illustrative examples, human operators **2716** can operate or interact with at least one of manufacturing equipment **2706**, maintenance equipment **2712**, or control system **2714**.

(409) This interaction can occur to manufacture semiconductor structures and other components for products such as semiconductor devices or components for use in products such as aircraft, spacecraft, communications systems, computation systems, and sensor systems.

(410) Further, control system **2714** can be used to adjust manufacturing of nonlinear optical waveguides, optical waveguides, optical couplers, phase shifters, and other components dynamically in or by the waveguides during the manufacturing process. For example, many points in the process of fabricating the optical waveguide structure including the nonlinear optical waveguide as well as other components are present at which adjustments can be made to control characteristics of components in an optical waveguide structure.

(411) Some features of the illustrative examples are described in the following clauses. These clauses are examples of features not intended to limit other illustrative examples.

(412) Clause 1

(413) An optical waveguide structure comprising: a main nonlinear optical waveguide, wherein a first-wavelength light and a second-wavelength light travel in the main nonlinear optical waveguide; a first extension optical waveguide; a secondary optical waveguide; a first wavelength-selective coupler that optically couples the main nonlinear optical waveguide and the first extension optical waveguide to each other such that the first-wavelength light is coupled from the main nonlinear optical waveguide to the first extension optical waveguide, and that optically couples the main nonlinear optical waveguide and the secondary optical waveguide to each other such that the second-wavelength light is coupled from the main nonlinear optical waveguide to the secondary optical waveguide; and a second wavelength-selective coupler that optically couples the main nonlinear optical waveguide and the first extension optical waveguide to each other such that the first-wavelength light is coupled from the first extension optical waveguide to the main nonlinear optical waveguide, and that optically couples the main nonlinear optical waveguide and the secondary optical waveguide to each other such that the second-wavelength light is coupled from the secondary optical waveguide to the main nonlinear optical waveguide.

Clause 2

(414) The optical waveguide structure according to clause 1 further comprising: a second extension optical waveguide; a third wavelength-selective coupler that optically couples the secondary optical waveguide and the second extension optical waveguide to each other such that the second-wavelength light is coupled from the secondary optical waveguide to the second extension optical waveguide; and a fourth wavelength-selective coupler that optically couples the secondary optical waveguide and the second extension optical waveguide to each other such that the second-wavelength light is coupled from the second extension optical waveguide to the secondary optical waveguide.

Clause 3

(415) The optical waveguide structure according to clause 2 further comprising: a third extension optical waveguide; wherein the third wavelength-selective coupler optically couples the secondary optical waveguide and the third extension optical waveguide to each other such that a third-wavelength light is coupled from the secondary optical waveguide to the third extension optical waveguide and the second-wavelength light is not coupled into the third extension optical waveguide; and wherein the fourth wavelength-selective coupler optically couples the secondary optical waveguide and the third extension optical waveguide to each other such that the third-wavelength light is coupled from the third extension optical waveguide to the secondary optical waveguide.

Clause 4

(416) The optical waveguide structure according to any of clauses 2-4, wherein the first-wavelength light travels in a first loop through a main segment between a first main location and a second main location within the main nonlinear optical waveguide, through the first extension optical waveguide, and through the first wavelength-selective coupler and the second wavelength-selective coupler, in which the first loop has a first length, and wherein the second-wavelength light travels in a second loop through the main segment between the first main location and the second main location within the main nonlinear optical waveguide, through a secondary segment in the secondary optical waveguide, through the second extension optical waveguide, and through the first wavelength-selective coupler and the second wavelength-selective coupler, in which the second loop has a second length for the second-wavelength light.

Clause 5

(417) The optical waveguide structure according to clause 3, wherein the first-wavelength light travels in a first loop through a main segment within the main nonlinear optical waveguide, through and a first extension segment, through the first wavelength-selective coupler and the second wavelength-selective coupler, in which the first loop has a first length; wherein the second-wavelength light travels in a second loop through a secondary segment in the secondary optical waveguide, through the second extension optical waveguide, through the first wavelength-selective coupler and the second wavelength-selective coupler, through the third wavelength-selective coupler and the fourth wavelength-selective coupler, and through the main segment in the nonlinear optical waveguide, in which the second loop has a second length for the second-wavelength light; and wherein the third-wavelength light travels in a third loop through the secondary segment in the secondary optical waveguide, through the third extension optical waveguide, through the first wavelength-selective coupler and the second wavelength-selective coupler, through the third wavelength-selective coupler and the fourth wavelength-selective coupler, and through the main segment in the nonlinear optical waveguide, in which the third loop has a third length for the third-wavelength light.

Clause 6

(418) The optical waveguide structure according to any of clauses 2-6, wherein the first-wavelength light is a pump light and the second-wavelength light is one of a signal light and an idler light, and wherein an intensity of the first-wavelength light is greater than an intensity of the second-wavelength light.

(419) Clause 7

(420) The optical waveguide structure according to any of clauses 3 or 5 wherein the first-wavelength light is a pump light, the second-wavelength light is a signal light, and the third-wavelength light is an idler light; and wherein an intensity of the first-wavelength light is greater than an intensity of the second-wavelength light and is greater than an intensity of the third-wavelength light.

(421) Clause 8

(422) The optical waveguide structure according to any of clauses 1-7, wherein the main nonlinear optical waveguide is comprised of an electro-optic material.

(423) Clause 9

(424) The optical waveguide structure according to any of clauses 2-8, wherein the second extension optical waveguide is comprised of at least one of an electro-optic material, a nonlinear optical material or a non-nonlinear optical material

(425) Clause 10

(426) The optical waveguide structure according to any of clauses 3, 5, or 7, wherein the third extension optical waveguide is comprised of at least one of an electro-optic material, a nonlinear optical material or a non-nonlinear optical material

(427) Clause 11

(428) The optical waveguide structure according to any of clauses 1-10, wherein the main nonlinear optical waveguide is comprised of a nonlinear optical material.

(429) Clause 12

(430) The optical waveguide structure according to any of clauses 2-11, wherein the second extension optical waveguide is comprised of an electro-optic material.

(431) Clause 13

(432) The optical waveguide structure according to any of clauses 3, 5, 7, or 10, wherein the third extension optical waveguide is comprised of an electro-optic material.

(433) Clause 14

(434) The optical waveguide structure according to any of clauses 2-13, wherein the first wavelength-selective coupler, the second wavelength-selective coupler, the third wavelength-selective coupler, and the fourth wavelength-selective coupler are selected from at least one of a two-waveguide coupler, a multi-mode interference coupler, a pulley coupler, a Mach-Zehnder interferometer, or a 4-port micro-optical waveguide resonator coupler.

(435) Clause 15

(436) The optical waveguide structure according to any of clauses 3, 5, 7, 10, or 13 further comprising: a set of output optical waveguides that outputs output light out of at least one of the first extension optical waveguide, the second extension optical waveguide, or the third extension optical waveguide.

Clause 16

(437) The optical waveguide structure according to any of clauses 3, 5, 7, 10, 13, or 15 further comprising: a set of input optical waveguides that inputs input light into at least one of the first extension optical waveguide, the second extension optical waveguide, or the third extension optical waveguide.

Clause 17

(438) The optical waveguide structure according to clause 4 further comprising: a phase shifter located adjacent to a portion of the main nonlinear optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the first-wavelength light to achieve a resonance condition for the first-wavelength light.

Clause 18

(439) The optical waveguide structure according to any of clauses 4 or 17 further comprising: a phase shifter located adjacent to a portion of the second extension optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the second-wavelength light to achieve a resonance condition for the second-wavelength light.

Clause 19

(440) The optical waveguide structure according to clause 5 further comprising: a phase shifter located adjacent to a portion of the third extension optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the third-wavelength light to achieve the resonance condition for the third-wavelength light.

Clause 20

(441) The optical waveguide structure according to any of clauses 1-19 further comprising: a phase shifter located adjacent to a portion of the main nonlinear optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the first-wavelength light to achieve a round-trip phase matching condition for a nonlinear optical process involving the first-wavelength light.

Clause 21

(442) The optical waveguide structure according to any of clauses 2-20 further comprising: a phase shifter located adjacent to a portion of the second extension optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the second-wavelength light to achieve a round-trip phase matching condition for a nonlinear optical process involving the second-wavelength light.

Clause 22

(443) The optical waveguide structure according to any of clauses 3, 5, 7, 10, 13, 15, or 16 further comprising: a phase shifter located adjacent to a portion of the third extension optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the third-wavelength light to achieve a round-trip phase matching condition for a nonlinear optical process involving the third-wavelength light.

Clause 23

(444) The optical waveguide structure according to any of clauses 1-24 further comprising: a phase shifter located adjacent to a portion of the main nonlinear optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the first-wavelength light to achieve a phase walk-off that is an odd multiple of 180 degrees.

Clause 24

(445) The optical waveguide structure according to any of clauses 2-23 further comprising: a phase shifter located adjacent to a portion of the second extension optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the second-wavelength light to achieve a phase walk-off that is an odd multiple of 180 degrees.

Clause 25

(446) The optical waveguide structure according to any of clauses 35, 7, 10, 13, 15, 16, or 22 further comprising: a phase shifter located adjacent to a portion of the third extension optical waveguide, wherein the phase shifter applies an activation such that a phase shifts in the third-wavelength light to achieve a phase walk-off that is an odd multiple of 180 degrees.

Clause 26

(447) A method for facilitating a non-linear optical process comprising: routing a first-wavelength light and a second-wavelength light in a main nonlinear optical waveguide; optically coupling, by a first wavelength-selective coupler, the main nonlinear optical waveguide and an extension optical waveguide to each other such that the first-wavelength light is coupled from the main nonlinear optical waveguide to the extension optical waveguide and the second-wavelength light is not coupled from the main nonlinear optical waveguide to the extension optical waveguide but rather is coupled to a secondary optical waveguide; and optically coupling, by a second wavelength-selective coupler, the main nonlinear optical waveguide and the extension optical waveguide to each other such that the first-wavelength light is coupled from the extension optical waveguide to the main nonlinear optical waveguide.

Clause 27

(448) The method of according to clause 26 further comprising: optically coupling, by a third wavelength-selective coupler, the secondary optical waveguide and a second extension optical waveguide to each other such that the second-wavelength light is coupled from the secondary optical waveguide to the second extension optical waveguide, and such that a third-wavelength light is not coupled from the secondary optical waveguide to the second extension optical waveguide; and optically coupling, by a fourth wavelength-selective coupler, the secondary optical waveguide and the second extension optical waveguide to each other such that the second-wavelength light is coupled from the second extension optical waveguide to the secondary optical waveguide.

Clause 28

(449) The method of according to clause 27 comprising: routing the third-wavelength light in the main nonlinear optical waveguide; optically coupling, by the third wavelength-selective coupler, the secondary optical waveguide and a third extension optical waveguide to each other such that the third-wavelength light is coupled from the secondary optical waveguide to the third extension optical waveguide and the second-wavelength light is not coupled from the secondary optical waveguide to the third extension optical waveguide; and optically coupling, by the fourth wavelength-selective coupler, the secondary optical waveguide and the third extension optical waveguide to each other such that the third-wavelength light is coupled from the third extension optical waveguide to the secondary optical waveguide.

Clause 29

(450) The method according to clause 27, wherein the first-wavelength light travels in a first loop through a main segment between a first main location and a second main location within the main nonlinear optical waveguide and through a first extension segment, in which the first loop has a first length selected to achieve a resonance condition for the first-wavelength light; and wherein the second-wavelength light travels in a second loop through a secondary segment in the secondary optical waveguide, through the second extension optical waveguide, and through the main segment in the main nonlinear optical waveguide, in which the second loop has a second length for the second-wavelength light selected to achieve the resonance condition for the second-wavelength light.

Clause 30

(451) The method according to clause 27, wherein the first-wavelength light travels in a first loop through a main segment between a first main location and a second main location within the main nonlinear optical waveguide and through a first extension optical waveguide, in which the first loop has a first length selected to achieve a resonance condition for the first-wavelength light; wherein the second-wavelength light travels in a second loop through a segment in the secondary optical waveguide, through the second extension optical waveguide, and through the main segment in the nonlinear optical waveguide, in which the second loop has a second length for the second-wavelength light selected to achieve the resonance condition for the second-wavelength light; and wherein the third-wavelength light travels in a third loop through the segment in the secondary optical waveguide, through the third extension optical waveguide, and through the main segment in the nonlinear optical waveguide, in which the third loop has a third length selected to achieve a resonance condition for the third-wavelength light.

Clause 31

(452) The method according to any of clauses 27-30, wherein the first-wavelength light is a pump light, the second-wavelength light is one of a signal light and an idler light.

(453) Clause 32

(454) The method according to any of clauses 27-30, wherein the first-wavelength light is one of a signal light and an idler light and the second-wavelength light is a pump light.

(455) Clause 33

(456) The method according to any of clauses 27-30, wherein the first-wavelength light is a pump light, the second-wavelength light is a signal light, and the third-wavelength light is an idler light.

(457) Clause 34

(458) The method according to any of clauses 26-33 further comprising: applying an activation to a portion of the main nonlinear optical waveguide such that a phase shifts in the first-wavelength light to achieve a resonance condition for the first-wavelength light.

Clause 35

(459) The method according to any of clauses 27-34 further comprising: applying an activation such to a portion of the second extension optical waveguide such that a phase shifts in the second-wavelength light to achieve a resonance condition for the second-wavelength light.

Clause 36

(460) The method according to clause 30 further comprising: applying an activation to a portion of the third extension optical waveguide such that a phase shifts in the third-wavelength light to achieve a resonance condition for the third-wavelength light.

Clause 37

(461) The method according to any of clauses 27-36 further comprising: applying an activation to a portion of the second extension optical waveguide such that a phase shifts in the second-wavelength light.

Clause 38

(462) The method according to any of clauses 27-37 further comprising: applying an activation to a portion of the main nonlinear optical waveguide such that a phase shifts in the first-wavelength light to achieve a round trip phase matching condition for a nonlinear optical process involving the first-wavelength light, the second-wavelength light, and the third-wavelength light.

(463) Thus, the illustrative examples include the wavelength-selective couplers that enable selective coupling of light in a manner that establishes loops in which light of different wavelengths can travel. Additionally, optical waveguides in the illustrative examples are designed to manage a reversal in the sign of the nonlinear optical coefficient that occurs for the two halves of an optical waveguide structure for which the light travels in opposite directions in portions of those two halves. The optical waveguide structures in this optical waveguide structure can avoid undesired effects of the sign reversal in the nonlinear optical coefficient by removing the pump light or by having an absence of a non-linear optical material in part of the structure.

(464) In another illustrative embodiment, loops formed from optical waveguides is unnecessary. For example, the nonlinear optical waveguide structure in the different illustrative examples described above in FIGS. 1-27 can comprise triple partially overlapping loops for entanglement (TriPOLE). In this illustrative example, these partially overlapping loops for entanglement are unnecessary to obtain a desired level of performance.

(465) In this illustrative example, phase adjustment is provided to control the phases of different wavelengths of light traveling through optical waveguide structures that do not include closed loops. The illustrative embodiments recognize and take into account a number of different considerations. Some of these considerations are recognized and taken into account as described below.

(466) For example, maintenance of phase matching is important for achieving nonlinear optical processes that are efficient in terms of usage of input source power and that can produce high power of the light generated by the nonlinear optical process. Higher efficiency and higher generated power can be achieved by using a longer nonlinear optical waveguide. However, imperfect phase matching can limit the length of nonlinear optical waveguide for which the nonlinear optical generation process continues to be effective in producing additional light.

(467) Further, in a nonlinear optical process, such as parametric down conversion (PDC) or difference frequency generation (DFG) and parametric up conversion (PUC) or sum frequency generation (SFG), the phase of the idler light generated at a given location in the nonlinear optical waveguide is determined by the phases of the source pump wave and signal wave at that location. With perfect phase matching, the wave-vector of the generated idler light continues to be matched with the wave-vectors of the source pump light and signal light so that the phase of the idler light generated at a first location and then propagated to a second location match the phase of the idler light generated at the second location.

(468) As a result, a constructive nonlinear optical interaction occurs at the second location between the previously generated idler light and the pump and signal waves at that second location to produce additional generation of idler light at the second location. Thus, the intensity of the idler light can increase for longer nonlinear optical waveguides.

(469) Alternatively, the efficiency of nonlinear optical generation, such as by spontaneous parametric down conversion (SPDC), can increase as the nonlinear optical waveguide becomes longer. However, when the phases for the light are not matched, the phase walk-off between the signal and idler light generated at different locations can result in destructive nonlinear optical interaction at the second location between those components of the signal light and the idler light and the pump light.

(470) As a result, the additional generation of signal and idler light at the second location can be reduced or, for some values of the phase walk-off, the signal and idler light at the second location can even be consumed to produce pump light. This situation reverses the nonlinear optical generation process. Thus, increasing the length of the nonlinear optical waveguide does not continue to increase the nonlinear optical generation efficiency or the intensity of the generated signal and idler light.

(471) The illustrative embodiments recognize and take into account that in a nonlinear optical (NLO) generation process, the phase(s) of the generated light is determined by the phase(s) of the source light(s). A nonlinear optical waveguide structure can be designed to achieve a condition of matching of the phases for the source light and the previously generated light. This type of matching enables the nonlinear optical generation process to become more efficient and to produce more power in the generated light for longer nonlinear optical waveguides. However, current nonlinear optical waveguide structures are not fabricated with perfect matching of the wave vectors needed to sustain efficient nonlinear optical generation over a long nonlinear optical waveguide. When a nonlinear optical waveguide does not have the wave-vector match to achieve perfect phase matching for sustaining a nonlinear optical process, the phase walk-off of the light generated can result in destructive interference. The generated light can be, for example, signal and idler light generated in spontaneous parametric down conversion (SPDC) from different locations in the nonlinear optical waveguide. Thus, the nonlinear optical interaction between source and generated light can be constructive for producing more generated light at some locations and destructive at other locations

(472) Some approaches to compensate for imperfect phase matching are based on quasi-phase-matching (QPM). Quasi-phase-matching involves a periodic change in the structure of a nonlinear optical waveguide. Some examples of quasi-phase-matching involve changing a transverse dimension of the nonlinear optical waveguide, such as the waveguide width. Other examples of quasi-phase-matching involve changing the polarization direction of the nonlinear optical material in the nonlinear optical waveguide. With quasi-phase-matching the longitudinal period of the structural or material change in the nonlinear optical waveguide must exactly the longitudinal distance over which the imperfect phase match results in a phase walk-off of 2π radians or 360 degrees.

(473) Yet other examples involve using a nonlinear optical waveguide structure that comprises two parallel and optically coupled nonlinear optical waveguides. For these two-coupled-waveguide structures, either the generated light or the source light, but not both, is gradually and continuously coupled from a first waveguide of the two parallel waveguides to the second waveguide and then back from the second waveguide to the first waveguide. The length over which the round-trip coupling from first waveguide to second waveguide and back to first waveguide occurs is matched to the length over which phase walk-off of the generated light reaches 2π radians or 360 degrees.

(474) For one example with a two-coupled-waveguide structure, the light generated in a first location in the first waveguide does not propagate to a second location in the first waveguide at which the light generated in the first location and the light generated in the second location would have a phase walk-off of π radians (180 degrees) and would otherwise result in a fully destructive interaction. Instead, the light generated in the first location becomes fully coupled into the second waveguide by that point and does not interfere with the light generated at that second location in the first waveguide. The light generated in the first location becomes fully coupled back from the second waveguide to the first waveguide at a third location in the first waveguide for which the light generated in the first location would have a phase walk-off of 2π radians (360 degrees) and would result in a fully constructive interaction. This gradual and continuous coupling of the generated light from the first waveguide into the second waveguide and then back again from the second waveguide into the first waveguide can be designed, by designing the coupling coefficient of the two-guide coupling structure, to occur in a periodic manner with the longitudinal period selected to avoid a periodically destructive interaction in the nonlinear optical generation process and to reinforce a periodically constructive interaction in the nonlinear optical generation process. In practice, it is difficult to achieve accurate control of the coupling coefficient for a two-guide coupling structure that exactly matches the phase walk-off, especially when the actual values obtained for the various wave vectors and for the two-guide coupling coefficient can change depending on fabrication-related tolerances and variations in the operational environment.

(475) In the following examples of the illustrative embodiments, a nonlinear optical waveguide structure includes tuning optical waveguides that can operate to adjust the phase of light traveling through these tuning optical waveguides. In these illustrative examples, phase shifters are associated with the tuning optical waveguides. These phase shifters can adjust the phases of the light and the resulting phase match to enable the full length of the fabricated nonlinear optical waveguide to be useful for increasing the nonlinear optical generation efficiency and increasing the power in the light generated by the nonlinear optical process.

(476) In the illustrative examples of these illustrative embodiments, a nonlinear optical waveguide structure provides one or more benefits that are not present with nonlinear optical waveguide structures that do not use phase tuning optical waveguides. For example, without phase tuning optical waveguides, an ability to have controlled adjustment of the light and thus of the phase matching to compensate for fabrication or operational tolerances is not present.

(477) Further, when phase tuning optical waveguides are present for adjusting the phases of the generated light, the use of closed-loop nonlinear optical waveguide structures results in the source light cycling repeatedly through the closed-loop nonlinear optical waveguide. Also, the generated light propagates in other closed loop waveguides. These types of nonlinear optical waveguides can have challenges in using the phase shifters to both adjust the source and generated light to achieve phase matching of the nonlinear optical process as well as adjust the source and generated light to achieve a match to a spectral resonance of a closed-loop path. The need to match two constraints limits the flexibility of these types of nonlinear optical waveguide structures to compensate for different tolerances.

(478) Thus, this example of these illustrative embodiments, provides phase adjustments to control the phases of different wavelengths of light traveling through optical waveguide structures without a need for closed loops.

(479) With reference now to the figures describing this illustrative example and in particular with reference to FIG. 28, an illustration of a block diagram of an optical waveguide structure is depicted in accordance with an illustrative embodiment. In this example, optical waveguide structure **2800** comprises a number of different components. As depicted, optical waveguide structure **2800** comprises nonlinear optical waveguide **2809**, a set of tuning optical waveguides **2840**, a set of wavelength selective couplers **2820**, and a set of phase shifters **2860**.

(480) Nonlinear optical waveguide **2809** and the set of tuning optical waveguides **2840** are optical waveguides in optical waveguide structure **2800**. Nonlinear optical waveguide **2809** is comprised of nonlinear optical material **2897** and can include other types of materials. In some illustrative examples, a tuning optical waveguide, a portion of the tuning optical waveguide, or multiple tuning optical waveguides can be comprised of nonlinear optical material **2897**. With nonlinear optical material **2897** present, light generation can occur while light **2810** travels through nonlinear optical waveguide **2809**.

(481) In this illustrative example, light **2810** has a set of wavelengths **2899** and comprises first wavelength light **2811**. Further, light **2810** can also include at least one of second wavelength light **2813** or third wavelength light **2815**. One or both of second wavelength light **2813** and third wavelength light **2815** can have a different wavelength in wavelengths **2899** from the wavelength of first wavelength light **2811**. Further, second wavelength light **2813** and third wavelength light **2815** can have values for their wavelengths in wavelengths **2899** that are different from each other or second wavelength light **2813** and third wavelength light **2815** can the same value for their wavelength in wavelengths **2899**. This condition of second wavelength light **2813** and third wavelength light **2815** having the same wavelength occurs when degenerate down conversion is the nonlinear optical process. Further, first wavelength light **2811** and second wavelength light **2813** can have the same wavelength, or first wavelength light **2811** and third wavelength light **2815** can have the same wavelength in another example. This condition occurs in up-conversion processes such as second harmonic generation.

(482) At least one of second wavelength light **2813** or third wavelength light **2815** can be produced from first wavelength light **2811** through nonlinear optical interaction **2898** within nonlinear optical waveguide **2809** caused by nonlinear optical material **2897** in nonlinear optical waveguide **2809**. In one example, first wavelength light **2811** is pump light **2812** and second wavelength light **2813** can be signal light **2814** or idler light **2816**. In another example, first wavelength light **2811** is pump light **2812** and second wavelength light **2813** is signal light **2814**, and third wavelength light **2815** is idler light **2816**.

(483) A nonlinear optical process involves a nonlinear optical interaction **2898** between source light **2817** and nonlinear optical material **2897** in a portion of nonlinear optical waveguide **2809** that can produce generated light **2819**.

(484) In these examples, source light **2817** includes first wavelength light **2811**. When source light comprises only first wavelength light **2811**, such as in a spontaneous parametric down conversion (SPDC) process, generated light **2819** could include second wavelength light

2813 as well as third wavelength light **2815**. In some nonlinear optical processes such as parametric up-conversion or parametric down-conversion, source light could further include second wavelength light **2813** or third wavelength light **2815**. When source light further comprises second wavelength light **2813**, such as in a parametric frequency-conversion (up-conversion or down-conversion) process, generated light **2819** would comprise third wavelength light **2815**. Similarly, when source light **2817** further comprises third wavelength light **2815**, generated light **2819** would comprise second wavelength light **2813**.

(485) The set of wavelength selective couplers **2820** couples light **2810** between nonlinear optical waveguide **2809** and the set of tuning optical waveguides **2840** based on wavelengths **2899** of light **2810**. The set of wavelength selective couplers **2820** route first wavelength light **2811**, second wavelength light **2813**, and third wavelength light **2815** in light **2810** between segments of nonlinear optical waveguide **2809** and the set of tuning optical waveguides **2840** based on wavelengths **2899** of first wavelength light **2811**, second wavelength light **2813**, and third wavelength light **2815**.

(486) As a result, some components of light **2810** can be routed to one or more of the set of tuning optical waveguides **2840** while routing other components of light **2810** to continue to traverse through nonlinear optical waveguide **2809**. In the illustrative example, a component of light **2810** having a particular wavelength. For example, first wavelength light **2811** and second wavelength light **2813** are two components in light **2810**.

(487) In this example, first wavelength light **2811** can continue to travel in nonlinear optical waveguide **2809** while at least one of second wavelength light **2813** or third wavelength light **2815** can be routed between nonlinear optical waveguide **2809** and the set of tuning optical waveguides **2840**.

(488) The set of phase shifters **2860** can be located along the set of tuning optical waveguides **2840**. In other words, a single phase shifter or multiple phase shifters in the set of phase shifters **2860** can be associated with a tuning optical waveguide in the set of tuning optical waveguides **2840** or multiple phase shifters can be associated with one or more of tuning optical waveguides **2840**.

(489) The set of phase shifters **2860** can be located along nonlinear optical waveguide **2809** or the set of tuning optical waveguides **2840** by being at least one of adjacent to part of nonlinear optical waveguide **2809** or the set of tuning optical waveguides **2840**, connected to part of the nonlinear optical waveguide **2809** or the set of tuning optical waveguides **2840**, or integrated as part of the nonlinear optical waveguide **2809** or the set of tuning optical waveguides **2840**.

(490) The set of phase shifters **2860** can apply a set of activations **2890** to the nonlinear optical waveguide **2809** or to the set of tuning optical waveguides **2840** to change phase shift **2895** for different wavelengths of light **2810** in the nonlinear optical waveguide **2809** or the set of tuning optical waveguides **2840**. In the illustrative examples, activations **2890** as applied to tuning optical waveguide **2840** can change the refractive index of tuning optical waveguides **2840**. This change in the refractive index can change the phase shift of light traveling in tuning optical waveguides **2840**. Activations **2890** that change the refractive index can be for example, heat or an electric field.

(491) The particular wavelength of light **2810** for which phase shift occurs can be selected based on the set of activations **2890** applied to light **2810** traveling through the set of tuning optical waveguides **2840**. The set of activations **2890** can be applied to cause phase shift **2895** to light **2810** propagating in the set of the set of tuning optical waveguides **2840** that results in phase walk-off **2896** for nonlinear optical interaction **2898** having a desired value. In these illustrative examples, a phase shift is a change in the phase of the light, such as the phase of one component of the light wave. This component, such as first wavelength light **2811** or second wavelength light **2813**, can have a single wavelength.

(492) For example, the set of activations **2890** can be applied such that particular wavelengths of light **2810** have phase shift **2895** that result in the nonlinear optical interaction **2898** having values for the phase walk-off **2896** of zero or an even multiple of π radians. In other examples, phase walk-off **2896** can be close to zero or close to an even multiple of π radians. For example, the phase walk-off can be within 0.25π radians or 0.5π radians. The desired value of phase walk-off **2896** achieved using the set of activations **2890** can depend on the amount of light generation desired.

(493) For example, one or more phase shifters can also be associated with nonlinear optical waveguide **2809** in addition to being associated with the set of tuning optical waveguides **2840** or in place of being associated with the set of tuning optical waveguides **2840**.

(494) With reference to FIG. **29**, an illustration of an optical waveguide structure is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **2900** comprises nonlinear optical waveguide **2909**, tuning optical waveguide **2948**, a set of wavelength selective couplers **2920**, and a set of phase shifters **2960**.

(495) In this illustrative example, light **2910** can comprise first wavelength light **2911** and second wavelength light **2913**. Second wavelength light **2913** is produced from first wavelength light **2911** through nonlinear optical interaction **2998** occurring within nonlinear optical waveguide **2909**. In other words, second wavelength light **2913** is produced within nonlinear optical waveguide **2909** through nonlinear optical interaction **2998** of first wavelength light **2911** and second wavelength light **2913** with nonlinear optical material **2997** in nonlinear optical waveguide **2909**. The light generated in a preceding segment of a nonlinear optical waveguide can participate in the nonlinear optical interaction that produces additional generated light in a subsequent segment of the nonlinear optical waveguide. Thus, the phase of the previously generated second wavelength light should be considered in order to have constructive light generation when second wavelength light generated in different segments are included.

(496) In this example, the set of wavelength selective couplers **2920** couples light **2910** between nonlinear optical waveguide **2909** and a set of tuning optical waveguides **2940** based on wavelengths **2999** of light **2910** and in particular for wavelengths **2999** of first wavelength light **2911**, second wavelength light **2913**, and third wavelength light **2915** in light **2910**. In this example, the set of tuning optical waveguides **2940** includes first set of tuning optical waveguides **2941** and second set of tuning optical waveguides **2942**. A set of phase shifters **2960** is located along the set of tuning optical waveguides **2940**. For example, one or more phase shifters **2960**, such a phase shifter **2968**, in the set of phase shifters **2960** is located along tuning optical waveguide **2948** in the set of tuning optical waveguides **2940**.

(497) In this example, wavelength selective coupler **2928** in the set of wavelength selective couplers **2920** couples second wavelength light **2913** from exit location **2971** in nonlinear optical waveguide **2909** to starting point **2978** in tuning optical waveguide **2948**. Wavelength selective coupler **2928** also couples second wavelength light **2913** from ending point **2979** in tuning optical waveguide **2948** back into nonlinear optical waveguide **2909** at entry location **2930**. Wavelength selective coupler **2928** does not couple first wavelength light **2911** from the nonlinear optical waveguide **2909** into tuning optical waveguide **2948**. Instead, first wavelength light **2911** is coupled from exit location **2971** in nonlinear optical waveguide **2909** to a different location in nonlinear optical waveguide **2909**.

(498) Phase shifter **2968** in the set of phase shifters **2960** is located between starting point **2978** and ending point **2979** in the tuning optical waveguide **2948**. In this example, phase shifter **2968** applies activation **2994** to tuning optical waveguide **2948** to change phase shift **2995** of second wavelength light **2913** in tuning optical waveguide **2948**.

(499) In another illustrative example, optical waveguide structure **2900** can have another configuration that includes first wavelength selective coupler **2921**, second wavelength selective coupler **2922** in the set of wavelength selective couplers **2920**, tuning optical

waveguide **2948** in the set of tuning optical waveguides **2940**, and phase shifter **2968** in the set of phase shifters **2960**.

(500) In yet another illustrative example, optical waveguide structure **2900** can have another configuration that includes a combination of wavelength selective coupler **2928** with a pair of first wavelength selective coupler **2921** and second wavelength selective coupler **2922**.

(501) In this example, first wavelength selective coupler **2921** in the set of wavelength selective couplers **2920** couples second wavelength light **2913** from exit location **2971** in the nonlinear optical waveguide to starting point **2978** in tuning optical waveguide **2948**. Second wavelength selective coupler **2922** in the set of wavelength selective couplers **2920** couples second wavelength light **2913** from ending point **2979** in tuning optical waveguide **2948** to entry location **2930** in nonlinear optical waveguide **2909**.

(502) Further, phase shifter **2968** in the set of phase shifters **2960** is located between starting point **2978** and ending point **2979** in tuning optical waveguide **2948**. In this example, phase shifter **2968** applies activation **2994** to tuning optical waveguide **2948** to change phase shift **2995** of second wavelength light **2913** in tuning optical waveguide **2948**. Thus, optical waveguide structure **2800** in FIG. **28** and optical waveguide structure **2900** illustrated in FIG. **29** enable adjusting light traveling through optical waveguides such as tuning optical waveguides to obtain a desired level of light generation within the optical waveguide structures. The set of phase shifters **2960** provides the ability to adjust parameters such as phase shift **2995** of one or more components of light **2910**. As a result, adjusting phase walk-off **2996** for nonlinear optical interaction **2998** can enable the manufacture of devices such as optical sources that are more compact and can be produced at a lower cost as compared to current devices because these devices comprise nonlinear optical waveguides. In other words, the use of optical waveguides comprising nonlinear optical materials instead of propagating “unguided” light through nonlinear optical material enables the devices in the illustrative examples to be more compact and also to be produced at lower cost as compared to currently available devices. These devices can be designed without partially overlapping loops for entanglement when those loops are unnecessary to obtain a desired level of performance and physical size.

(503) With reference to FIGS. **30A** and **30B**, illustrations of an optical waveguide structure are depicted in accordance with an illustrative embodiment. This figure illustrates cascading of many segments in nonlinear optical waveguide **3009** that increases the physical length of nonlinear optical waveguide **3009**.

(504) In this depicted example, optical waveguide structure **3000** is comprised as first segment **3001** in nonlinear optical waveguide **3009**, second segment **3002** in nonlinear optical waveguide **3009**, third segment **3003** in nonlinear optical waveguide **3009**, first tuning optical waveguide **3041** in the set of tuning optical waveguides **3040**, first wavelength selective coupler **3021** in the set of wavelength selective couplers **3020**, and first phase shifter **3061** in a set of phase shifters **3060**. First phase shifter **3061** is located along first tuning optical waveguide **3041**.

(505) In this example, first segment **3001** and second segment **3002** in nonlinear optical waveguide **3009** are physically separated from each other by first wavelength selective coupler **3021**. In this example, second segment **3002** and third segment **3003** in nonlinear optical waveguide **3009** are physically separated from each other by second wavelength selective coupler **3022** and by third wavelength selective coupler **3023**. In the illustrative example, first wavelength light **3011** can be supplied to first segment **3001** at first entry location **3030**.

(506) With this example, first wavelength selective coupler **3021** couples first wavelength light **3011** from first segment **3001** in nonlinear optical waveguide **3009** into second segment **3002** in nonlinear optical waveguide **3009** at entry location **3032**. First wavelength selective coupler **3021** couples second wavelength light **3017**, which is generated in first segment **3001**, from first segment **3001** into first tuning optical waveguide **3041**. Further, second wavelength light **3017** in first tuning optical waveguide **3041** is coupled from first tuning optical waveguide **3041** to third segment **3003** by second wavelength selective coupler **3022**. In this example, second wavelength light **3017** is coupled away from third segment **3003** in nonlinear optical waveguide **3009** by second wavelength selective coupler **3022**.

(507) In this example, second wavelength light **3017** generated in first segment **3001** goes through first tuning optical waveguide **3041** and then to third segment **3003** after having its phase shift adjusted to again re-establish a phase-matching condition. Additionally, second wavelength light **3018** generated in second segment **3002** is intentionally not coupled into third segment **3003** but rather is diverted into second tuning optical waveguide **3042**.

(508) In this example, second wavelength light **3017** generated in first segment **3001** is coupled by second wavelength selective coupler **3022** into entry location **3034** of third segment **3003**.

(509) In this depicted example, second wavelength selective coupler **3022** in the set of wavelength selective couplers **3020** couples first wavelength light **3011** from the second segment **3002** into the third segment **3003**. Second wavelength light **3017** in first tuning optical waveguide **3041** is coupled from first tuning optical waveguide **3041** to third segment **3003** by second wavelength selective coupler **3022**.

(510) In the illustrative example, optical waveguide structure **3000** further comprises second tuning optical waveguide **3042** in the set of tuning optical waveguides **3040**; second phase shifter **3062** in the set of phase shifters **3060**, and third wavelength selective coupler **3023** in the set of wavelength selective couplers **3020**. Second phase shifter **3062** is located along second tuning optical waveguide **3042**.

(511) Second wavelength light **3018** generated in second segment **3002** in nonlinear optical waveguide **3009** is not coupled into third segment **3003** in nonlinear optical waveguide **3009** by second wavelength selective coupler **3022**. Instead, second wavelength light **3018** generated in second segment **3002** is coupled by third wavelength selective coupler **3023** into second tuning optical waveguide **3042**. Second wavelength light **3018** generated in second segment **3002** is coupled by third wavelength selective coupler **3023** from second segment **3002** into second tuning optical waveguide **3042**, and thus does not even enter second wavelength selective coupler **3022**.

(512) Second wavelength light **3017** is generated in first segment **3001** and second wavelength light **3018** is generated in second segment **3002**. Second wavelength light **3017** and second wavelength light **3018** can have different phases relative to each other. Second wavelength light **3017** and second wavelength light **3018** will experience an interference if they are combined into the same optical waveguide. This interference can result in an increase or a decrease in the intensity or power of the combined second wavelength light, depending, respectively, on whether the interference is constructive or destructive. In this example, second wavelength light **3017** and second wavelength light **3018** are intentionally kept separate until they both reach combiner **3188** in FIG. **31B** as described below. In the examples, different tuning optical waveguides and different phase shifters can be used to adjust the phases of these two components of second wavelength light **3017** and **3018** to reduce their destructive interference and enhance their constructive interference when second wavelength light **3017** and second wavelength light **3018** are combined at combiner **3188** in FIG. **31B** as described below.

(513) In this example, third wavelength selective coupler **3023** couples second wavelength light **3018**, generated in second segment **3002**, from second segment **3002** into second tuning optical waveguide **3042**. Third wavelength selective coupler **3023** also couples first wavelength light **3011** from second segment **3002** into third segment **3003** through second wavelength selective coupler **3022**.

(514) In this illustrative example, optical waveguide structure **3000** can further comprise fourth segment **3004** in nonlinear optical waveguide **3009** and fourth wavelength selective coupler **3024** in the set of wavelength selective couplers **3020**. With this example, fourth wavelength selective coupler **3024** couples second wavelength light **3018**, generated in second segment **3002**, from second tuning optical waveguide **3042** into fourth segment **3004**. Additionally, fourth wavelength selective coupler **3024** couples the first wavelength light **3011**

from third segment **3003** into fourth segment **3004**.

(515) In this illustrative example, optical waveguide structure **3000** can further comprise third tuning optical waveguide **3043** in the set of tuning optical waveguides **3040** and fifth wavelength selective coupler **3025** in the set of wavelength selective couplers **3020**. In this example, fifth wavelength selective coupler **3025** in the set of wavelength selective couplers **3020** couples second wavelength light **3017** from third segment **3003** into third tuning optical waveguide **3043** and couples first wavelength light **3011** from third segment **3003** into fourth segment **3004** through fourth wavelength selective coupler **3024**. In this illustrative example, third phase shifter **3063** in the set of phase shifters **3060** can be located along third tuning optical waveguide **3043** and can apply activation **3093** to second wavelength light **3017** in third tuning optical waveguide **3043**.

(516) In this example, additional second wavelength light can be produced as a result of the nonlinear optical interaction that occurs in third segment **3003**. Thus, the second wavelength light **3017** that is coupled from third segment **3003** by fifth wavelength selective coupler **3025** could comprise a combination of second wavelength light generated in first segment **3001** and additional second wavelength light generated in third segment **3003**. First phase shifter **3061** in first tuning optical waveguide **3041** adjusts the phase of the second wavelength light **3017** coupled into entry location **3034** of third segment **3003** in order to achieve a phase walk-off **3096** with a value of zero or a multiple of 2π radians for the nonlinear optical interaction **3098** occurring up through entry location **3034** in the third segment **3003**. As a result, the second wavelength light **3017** exiting the third segment and coupled through fifth wavelength selective coupler **3025** can be greater than the second wavelength light **3017** exiting the first segment **3001** and coupled into third segment **3003**.

(517) Further, optical waveguide structure **3000** can include other optical components. For example, optical waveguide structure **3000** can include source input coupler **3031** that supplies first wavelength light **3011**, such as pump light **3012**, from a location external to optical waveguide structure **3000** into first segment **3001** of nonlinear optical waveguide **3009**. The first wavelength light **3011** can be supplied, via source input coupler **3031**, to first entry location **3030** in first segment **3001**.

(518) The phase walk-off of interest at entry location **3034** in third segment **3003** is the phase walk-off for the cumulative nonlinear optical interaction that occurs through nonlinear optical waveguide **3009** from first entry location **3030** in first segment **3001** to entry location **3034** in third segment **3003**. The additional generation of second wavelength light **3017** that occurs in third segment **3003** depends on this phase walk-off, which is affected by the first phase shifter **3061** that adjusts the phase of the second wavelength light **3017** generated in first segment **3001**, as discussed above. However, second wavelength light **3018** generated in second segment **3002** is diverted away from third segment **3003** by third wavelength selective coupler **3023**. Thus, the phase of second wavelength light **3018** is not a factor in determining the phase walk-off of interest at entry location **3034** in third segment **3003**.

(519) In this illustrative example, optical waveguide structure **3000** can further comprise fifth segment **3005** in nonlinear optical waveguide **3009**, fourth tuning optical waveguide **3044** in the set of tuning optical waveguides **3040** and seventh wavelength selective coupler **3027** as well as sixth wavelength selective coupler **3026** in the set of wavelength selective couplers **3020**. In this example, seventh wavelength selective coupler **3027** in the set of wavelength selective couplers **3020** couples second wavelength light **3018** from fourth segment **3004** into fourth tuning optical waveguide **3044** and also couples first wavelength light **3011** from fourth segment **3004** into fifth segment **3005** through seventh wavelength selective coupler **3027** and then through sixth wavelength selective coupler **3026**. In this illustrative example, fourth phase shifter **3064** in the set of phase shifters **3060** can be located along fourth tuning optical waveguide **3044** and can apply activation **3094** to second wavelength light **3018** in fourth tuning optical waveguide **3044**.

(520) In this example, additional second wavelength light can be produced as a result of the nonlinear optical interaction that occurs in fourth segment **3004**. Thus, the second wavelength light **3018** that is coupled by seventh wavelength selective coupler **3027** could comprise a combination of second wavelength light generated in second segment **3002** and additional second wavelength light generated in fourth segment **3004**. Second phase shifter **3062** in second tuning optical waveguide **3042** adjusts the phase of the second wavelength light **3018** coupled into entry location **3036** of fourth segment **3004** in order to achieve a phase walk-off **3096** with a value of zero or a multiple of 2π radians for the nonlinear optical interaction **3098** occurring up through entry location **3036** in the fourth segment **3004**. As a result, the second wavelength light **3018** exiting the fourth segment and coupled through seventh wavelength selective coupler **3027** can be greater than the second wavelength light **3018** exiting the second segment **3002** and coupled into fourth segment **3004**.

(521) In this illustrative example, different phase shifters can apply different activations depending on the phase adjustment desired. In this depicted example, first phase shifter **3061** in set of phase shifters **3060** applies activation **3091** to second wavelength light **3017** in first tuning optical waveguide **3041** to change the phase shift for second wavelength light **3017** in first tuning optical waveguide **3041** such that the phase walk-off **3096** for the nonlinear optical interaction **3098** in nonlinear optical waveguide **3009** from first entry location **3030** in first segment **3001** where first wavelength light **3011** is supplied to first segment **3001** to entry location **3034** in third segment **3003** has a value of zero or an even multiple of π radians. In other examples, the phase walk-off **3096** can have a value close to zero or close to an even multiple of π radians. For example, the value for the phase walk-off can be within 0.25π radians or 0.5π radians. Entry location **3034** is where second wavelength selective coupler **3022** and third segment **3003** connect to each other.

(522) Further in this example, second phase shifter **3062** can apply activation **3092** to second wavelength light **3018** in second tuning optical waveguide **3042**. The second wavelength light **3018** in second tuning optical waveguide **3042** was generated in second segment **3002**.

Activation **3092** is applied to this second wavelength light **3018** in second tuning optical waveguide **3042** such that the phase walk-off **3096** for nonlinear optical interaction **3098** in nonlinear optical waveguide **3009** from entry location **3032**, at which first wavelength selective coupler **3021** connects to second segment **3002**, to entry location **3036**, at which fourth wavelength selective coupler **3024** connects to fourth segment **3004** has a value that is zero or is an even multiple of π radians.

(523) In other examples, the phase walk-off **3096** can have a value close to zero or close to an even multiple of π radians. For example, the value for the phase walk-off can be within 0.25π radians or 0.5π radians. Entry location **3036** is where fourth wavelength selective coupler **3024** and fourth segment **3004** connect to each other.

(524) In this example, the order of components in optical waveguide structure **3000** can be first segment **3001**, first wavelength selective coupler **3021**, second segment **3002**, third wavelength selective coupler **3023**, second wavelength selective coupler **3022**, and third segment **3003**. To continue with this order of components, additional components can be fifth wavelength selective coupler **3025**, fourth wavelength selective coupler **3024**, fourth segment **3004**, seventh wavelength selective coupler **3027**, sixth wavelength selective coupler **3026** and fifth segment **3005**.

(525) In this illustrative example, additional phase shifters can apply additional activations depending on the phase adjustment desired to achieve effective nonlinear optical interactions in segments of the nonlinear optical waveguide **3009** beyond the first segment **3001** and the second segment **3002**. In this depicted example, third phase shifter **3063** in set of phase shifters **3060** applies activation **3093** to second wavelength light **3017** in third tuning optical waveguide **3043** to change the phase shift for second wavelength light **3017** in third tuning optical waveguide **3043** such that the phase walk-off **3096** for the nonlinear optical interaction **3098** in nonlinear optical waveguide **3009**

from first entry location **3030** in first segment **3001** where first wavelength light **3011** is supplied to first segment **3001** to entry location **3038** in fifth segment **3005** has a value of zero or an even multiple of π radians. The second wavelength light **3017** in the third tuning optical waveguide **3043** can be generated in the first segment **3001** and in the third segment **3003**. In other examples, the phase walk-off **3096** can have a value close to zero or close to an even multiple of π radians. For example, the value for the phase walk-off can be within 0.25π radians or 0.5π radians. Entry location **3038** is where sixth wavelength selective coupler **3026** and fifth segment **3005** connect to each other. (526) Likewise, as depicted in FIGS. **30A** and **30B**, fourth phase shifter **3064** can apply activation **3094** to second wavelength light **3018** in fourth tuning optical waveguide **3044**. The second wavelength light **3018** in fourth tuning optical waveguide **3044** was generated in second segment **3002** and in fourth segment **3004**. Activation **3094** is applied to this second wavelength light **3018** in fourth tuning optical waveguide **3044** such that the phase walk-off **3096** for nonlinear optical interaction **3098** in nonlinear optical waveguide **3009** from entry location **3032**, at which first wavelength selective coupler **3021** connects to second segment **3002**, to an entry location (not shown), at which fourth wavelength selective coupler **3024** connects to a sixth segment **3006** through eighth wavelength selective coupler **3028** of nonlinear optical waveguide **3009** has a value that is zero or is an even multiple of π radians.

(527) In this example, first segment **3001** and second segment **3002** in nonlinear optical waveguide **3009** are physically separated from each other by first wavelength selective coupler **3021**. In this example, second segment **3002** and third segment **3003** in nonlinear optical waveguide **3009** are physically separated from each other by second wavelength selective coupler **3022** and by third wavelength selective coupler **3023**. In this example, third segment **3003** and fourth segment **3004** are physically separated from each other by fourth wavelength selective coupler **3024** and by fifth wavelength selective coupler **3025**. In the illustrative example, first wavelength light **3011** can be supplied to first segment **3001** at first entry location **3030**. In this example, first wavelength light **3011** can be pump light **3012**.

(528) With reference to FIGS. **31A** and **31B**, illustrations of an optical waveguide structure is depicted in accordance with an illustrative embodiment. This figure illustrates how the two components for generated light, such as second wavelength light **3117** and second wavelength light **3118** can be combined such that constructive interference of second wavelength light **3117** and second wavelength light **3118** occurs.

(529) In this illustrative example, optical waveguide structure **3100** comprises nonlinear optical waveguide **3109**, a set of wavelength selective couplers **3120**, a set of phase shifters **3160**, and a set of tuning optical waveguides **3140**. Like the example in FIGS. **30A** and **30B**, the example illustrated in FIGS. **31A** and **31B** has a first segment **3101** of nonlinear optical waveguide **3109** into which first wavelength light **3111** is supplied and in which second wavelength light **3117** can be generated by a nonlinear optical interaction occurring in the first segment **3101**. The example illustrated in FIGS. **31A** and **31B** further has a second segment **3102** of nonlinear optical waveguide **3109** into which first wavelength light is supplied, through first wavelength selective coupler **3121**, and in which second wavelength light **3118** can be generated by a nonlinear optical interaction occurring in the second segment **3102**.

(530) In this example, the nonlinear optical interaction can be extended to more and more segments of nonlinear optical waveguide **3109**. Second wavelength light **3117** generated in first segment **3101** is coupled through first wavelength selective coupler **3121** to first tuning optical waveguide **3141** and then through second wavelength selective coupler **3122** to third segment **3103**. In third segment **3103**, additional second wavelength light **3117** can be generated by nonlinear optical interaction of the first wavelength light **3111** and the second wavelength light present in the third segment **3103**, which can include the second wavelength light generated in the first segment.

(531) In this example, second wavelength light **3118** generated in second segment **3102** is coupled through third wavelength selective coupler **3123** to second tuning optical waveguide **3142** and then through fourth wavelength selective coupler **3124** to fourth segment **3104**. In fourth segment **3104**, additional second wavelength light **3118** can be generated by nonlinear optical interaction of the first wavelength light **3111** and the second wavelength light present in the fourth segment **3104**, which can include the second wavelength light generated in the second segment **3102**.

(532) The example illustrated in FIGS. **31A** and **31B** shows that more and more segments of nonlinear optical waveguide can be cascaded in this manner. Second wavelength light **3117** can be coupled from Nth segment **3107** through $(2N-1)$ th wavelength selective coupler **3127** and Nth tuning optical waveguide **3147**. $(2N-1)$ th wavelength selective coupler **3127** couples second wavelength light **3117** from Nth segment **3107** to Nth tuning optical waveguide **3147** and couples first wavelength light **3111** from Nth segment **3107** to $(2N-2)$ th wavelength selective coupler **3126**, which then couples first wavelength light **3111** to $(N+1)$ th segment **3108**. Second wavelength light **3118** can be coupled from $(N-1)$ th tuning optical waveguide (not shown) through $(2N-2)$ th tuning optical waveguide **3126** to $(N+1)$ th segment **3108**. Second wavelength light **3118** can be coupled through $(N+1)$ th segment **3108** of nonlinear optical waveguide **3109** and $(N+1)$ th tuning optical waveguide **3148**. $(2N+1)$ th wavelength selective coupler **3129** couples second wavelength light **3118** from $(N+1)$ th segment **3108** into $(N+1)$ th tuning optical waveguide **3148**. $(2N+1)$ th wavelength selective coupler **3129** also couples first wavelength light **3111** from $(N+1)$ th segment **3108** through $2N$ th wavelength selective coupler **3128** to final segment **3189**.

(533) The example illustrated in FIGS. **31A** and **31B** also shows a way in which the two cascades can be terminated. $2N$ th wavelength selective coupler **3128** couples second wavelength light **3117** from Nth tuning optical waveguide **3147** into final segment **3189** of optical waveguide structure **3100**. Second wavelength light **3117** travels through final segment **3189** is connected to combiner junction **3187** of combiner **3188**. As depicted, $(2N+1)$ th wavelength selective coupler **3129** couples second wavelength light **3118** from $(N+1)$ th segment **3108** into $(N+1)$ th tuning optical waveguide **3148** in tuning optical waveguides **3140**. Second wavelength light **3118** travels through $(N+1)$ th tuning optical waveguide **3148** to combiner junction **3187** of combiner **3188**.

(534) In a simplified example of optical waveguide structure **3100**, Nth tuning optical waveguide **3147** can be a third tuning optical waveguide and $(N+1)$ th tuning optical waveguide **3148** can be a fourth tuning optical waveguide. For this example, Nth segment would be third segment **3103**. For this example, $(N-1)$ th segment would be second segment **3102** and $(N+1)$ th segment would be fourth segment **3104**. For this example, $(2N-1)$ th wavelength selective coupler **3127** would be fifth wavelength selective coupler **3125** and $(2N-2)$ th wavelength selective coupler **3126** would be fourth wavelength selective coupler **3124**.

(535) In one example of optical waveguide structure **3100**, as illustrated in FIGS. **31A** and **31B**, source output coupler **3133** is connected to final segment **3189**. For this example, final segment **3189** does not comprise a substantial amount of nonlinear optical material **3197**, so that no nonlinear optical interaction would occur. Source output coupler **3133** removes first wavelength light **3111**, which could be pump light **2812** to a source out port or waveguide of the structure. Source output coupler **3133** also couples second wavelength light **3117** to combiner junction **3187** of combiner **3188**.

(536) In other examples (not depicted), final segment **3189** could comprise nonlinear optical material **3197**. For these other examples, source output coupler **3133** is located between $2N$ th wavelength selective coupler **3128** and final segment **3189**. In these cases, source output coupler **3133** couples first wavelength light **3111** from Nth tuning optical waveguide **3147** away from final segment **3189**. Thus, first wavelength light **3111** is removed and does not propagate through final segment **3189** of optical waveguide structure **3100** when final segment **3189** comprises nonlinear optical material **3197**. As a result, no additional generation of second wavelength light occurs in final

segment **3189** even if that final segment comprises nonlinear optical material, since the first wavelength light **3111**, which serves as the pump light for the nonlinear optical interactions that occur in the nonlinear optical waveguide **3109**, is absent.

(537) In these examples, no first wavelength light **3111** is supplied to optical waveguide structure **3100** besides first wavelength light **3111** supplied to first segment **3101** through source input coupler **3131**. First wavelength light **3111** in second segment **3102** is supplied through first segment **3101**. First wavelength light **3111** in third segment **3103** is supplied through first segment **3101**, and travels through second segment **3102** to third segment **3103**.

(538) In the illustrated example, optical waveguide structure **3100** further comprises combiner **3188**. Second wavelength light **3117** is supplied through final segment **3189** to a combiner junction **3187** of combiner **3188**. In addition, second wavelength light **3118** is supplied through (N+1)th tuning optical waveguide **3148** to the combiner junction **3187** of combiner **3188**. The phase of second wavelength light **3117** at the combiner junction **3187** and the phase of second wavelength light **3118** at the combiner junction **3187** are adjusted to produce a constructive interference condition for these two components of second wavelength light **3117** and second wavelength light **3118**. The combining of second wavelength light **3117** from the route through first segment **3101** and other odd-numbered segments and second wavelength light **3118** from the route through second segment **3102** and other even-numbered segments form combined second wavelength light **3113**. In this example, combined second wavelength light **3113** is an idler light.

(539) In the example of FIGS. **31A** and **31B**, an optional final phase shifter **3169** can be located along final segment **3189**. This final phase shifter **3169** of the set of phase shifters **3160** applies activation **3190** to adjust the phase of second wavelength light **3117** traveling in final segment **3189**.

(540) In the illustrated example, the optical waveguide structure **3100** also can comprise an optional auxiliary input segment **3176** into which third wavelength light **3115** is supplied. This third wavelength light **3115** serves as an auxiliary source light for nonlinear optical processes such as parametric up-conversion and parametric down-conversion that occur in nonlinear optical waveguide **3109**. The third wavelength light **3115** can be signal light or can be idler light, depending on whether second wavelength light is signal light or idler light. If second wavelength light is idler light, then third wavelength light would be signal light. Conversely, if second wavelength light is signal light, then third wavelength light would be idler light. When supplied as a source light, third wavelength light **3115** is coupled via source input coupler **3131** into nonlinear optical waveguide **3109**. In this example, third wavelength light **3115** is directed through the segments of nonlinear optical waveguide **3109** in the same route followed by first wavelength light **3111**. Optical waveguide structure **3100** can further comprise optional auxiliary output coupler **3135**. In this example, source output coupler **3133** couples third wavelength light **3115** from final segment **3189** to auxiliary output coupler **3135**. Auxiliary output coupler **3135** then couples third wavelength light **3115** away to an output waveguide or port of the structure. In some variations of this example, auxiliary output coupler **3135** can be located between source output coupler **3133** and combiner **3188**. In those variations, auxiliary output coupler also couples second wavelength light **3117** from source output coupler **3133** to combiner **3188**. As a result, only second wavelength light **3117** and second wavelength light **3118**, which would be components of combined second wavelength light **3113**, are supplied to combiner **3188**.

(541) In this illustrative example, different phase shifters can apply different activations depending on the phase adjustment desired. In this depicted example, first phase shifter **3161** in set of phase shifters **3160** applies activation **3191** to first tuning optical waveguide **3141** to change the phase shift for second wavelength light **3117** in first tuning optical waveguide **3141** such that the phase walk-off **3196** for the nonlinear optical interaction **3198** in nonlinear optical waveguide **3109** from first entry location **3130** in first segment **3101** where first wavelength light **3111** is supplied to first segment **3101** to entry location **3134** in third segment **3103** has a value of zero or an even multiple of n radians. In other examples, phase walk-off **3196** can have a value close to zero or close to an even multiple of π radians. For example, the value for the phase walk-off can be within 0.25π radians or 0.5π radians. Entry location **3134** is where second wavelength selective coupler **3122** and third segment **3103** connect to each other.

(542) Further in this example, second phase shifter **3162** can apply activation **3192** to second tuning optical waveguide **3142** to change the phase shift for second wavelength light **3118** in second tuning optical waveguide **3142**. The second wavelength light **3118** in second tuning optical waveguide **3142** was generated in second segment **3102**. Activation **3192** is applied to second tuning optical waveguide **3142** such that the phase walk-off **3196** for nonlinear optical interaction **3198** in nonlinear optical waveguide **3109** from entry location **3132**, at which first wavelength selective coupler **3121** connects to second segment **3102**, to entry location **3136**, at which fourth wavelength selective coupler **3124** connects to fourth segment **3104** has a value that is zero or is an even multiple of π radians. In other examples, the phase walk-off **3196** can have a value close to zero or close to an even multiple of π radians. For example, the value for the phase walk-off can be within 0.25π radians or 0.5π radians.

(543) FIGS. **31A** and **31B** illustrates additional functions of the embodiment of optical waveguide structure **3100** depicted in FIG. **31A**. With reference to FIGS. **31A** and **31B**, optical waveguide structure **3100** further comprises combiner **3188**. First component of combined second wavelength light **3113** is second wavelength light **3117**. This component is supplied through Nth tuning optical waveguide **3147** and final segment **3189** to the combiner junction **3187** of combiner **3188**. In addition, second component of combined second wavelength light **3113** is second wavelength light **3118**, which is supplied through (N+1)th tuning optical waveguide **3148** to the combiner junction **3187** of combiner **3188**. The phase of second wavelength light **3117** of combined second wavelength light **3113** at the combiner junction **3187** and the phase of the second wavelength light **3118** of combined second wavelength light **3113** at the combiner junction **3187** are adjusted to produce a constructive interference condition for second wavelength light **3117** and second wavelength light **3118**.

(544) In this example, Nth phase shifter **3167** applies activation **3193** to second wavelength light **3117** in Nth tuning optical waveguide **3147** to change the phase shift for second wavelength light **3117** in Nth tuning optical waveguide **3147**. Second wavelength light **3117** further traverses through final segment **3189**. Optional final phase shifter **3169** in final segment **3189** can apply activation **3190** to second wavelength light **3117** to further change the phase of second wavelength light **3117** presented at combiner junction **3187** of combiner **3188**.

(545) Further in this example, (N+1)th phase shifter **3168** can apply activation **3194** to second wavelength light **3118** in (N+1)th tuning optical waveguide **3148**. Second wavelength light **3118** is generated in the even-numbered segments of nonlinear optical waveguide **3109**. Activation **3194** is applied to second wavelength light **3118** in (N+1)th tuning optical waveguide **3148** to change the phase of second wavelength light **3118** presented at combiner junction **3187** of combiner **3188**.

(546) In this depicted example, combiner junction **3187** of combiner **3188** functions as an optical interferometer. Combined second wavelength light **3113**, the combined output from combiner **3188**, has the greatest intensity when the relative phases of second wavelength light **3117** and second wavelength light **3118** are the same or are an even multiple of π radians. Thus, for this example, Nth phase shifter **3167** and optional final phase shifter **3169** apply activation **3193** and activation **3190** to second wavelength light **3117** and (N+1)th phase shifter **3168** applies activation **3194** to second wavelength light **3118** such that, ideally, the relative phases of second wavelength light **3117** and second wavelength light **3118** are the same or are different by an even multiple of π radians when second wavelength light **3117** and second wavelength light **3118** are presented at combiner junction **3187** of combiner **3188**. In other examples, the relative phases of second

wavelength light **3117** and second wavelength light **3118** presented at the combiner junction **3187** can be within 0.25π radians or 0.5π radians. In other words, the phase difference between second wavelength light **3117** and second wavelength light **3118** can be as large as 0.25π radians or even as large as 0.5π radians.

(547) With reference to FIGS. **30A**, **30B**, **31A**, and **31B**, optical waveguide structure **3000** and optical waveguide structure **3100** can have as few as only two segments of nonlinear optical waveguide **3009** and nonlinear optical waveguide **3109**. These two segments would be first segment **3001** and first segment **3101** and second segment **3002** and second segment **3102**. Second wavelength light **3017** and second wavelength light **3117** is generated by nonlinear optical interaction of first wavelength light **3011** and first wavelength light **3111** in first segment **3001** and first segment **3101** and second wavelength light **3018** and second wavelength light **3118** is generated by nonlinear optical interaction of first wavelength light **3011** and first wavelength light **3111** in second segment **3002** and second segment **3102**.

(548) In one example, $N=1$. Thus, N th tuning optical waveguide **3147** is first tuning optical waveguide **3041**, $(2N-1)$ th wavelength selective coupler **3127** is first wavelength selective coupler **3021**, and $2N$ th wavelength selective coupler **3128** is second wavelength selective coupler **3022** in FIG. **30A**. Also, $(N+1)$ th tuning optical waveguide **3148** is second tuning optical waveguide **3042** in FIG. **30A**, $(2N+1)$ th wavelength selective coupler **3129** is third wavelength selective coupler **3023** in FIG. **30A**. Second wavelength light **3017** and second wavelength light **3117** from first tuning optical waveguide **3041** are coupled through final segment **3189** to combiner **3188**; and second wavelength light **3018** and second wavelength light **3118** from second tuning optical waveguide **3042** in FIGS. **30A** and **30B** also would be coupled to combiner **3188**.

(549) In another illustrative example, again with reference to FIGS. **30A** and **30B** and FIGS. **31A** and **31B**, optical waveguide structure **3000** and optical waveguide structure **3100** can have just four segments of nonlinear optical waveguide **3009** and nonlinear optical waveguide **3109**. These four segments would be first segment **3001**, second segment **3002**, third segment **3003** and fourth segment **3004**. Second wavelength light **3017** and second wavelength light **3117** are generated by nonlinear optical interaction of first wavelength light **3011** and first wavelength light **3111** in first segment **3001** and first segment **3101** and in third segment **3003** and third segment **3103**. Second wavelength light **3018** and second wavelength light **3118** is generated by nonlinear optical interaction of first wavelength light **3011** and first wavelength light **3111** in second segment **3002** and second segment **3102** and in fourth segment **3004** and fourth segment **3104**.

(550) In this example in FIGS. **31A** and **31B**, N is equal to 3. Thus, N th tuning optical waveguide **3147** is third tuning optical waveguide **3043** in FIG. **30B**, $(2N-1)$ th wavelength selective coupler **3127** can be a fifth wavelength selective coupler **3025** in FIG. **30A**, and $2N$ th wavelength selective coupler **3128** is a sixth wavelength selective coupler **3026**. Also, $(N+1)$ th tuning optical waveguide **3148** is fourth tuning optical waveguide **3044** in FIG. **30B**, $(2N+1)$ th wavelength selective coupler **3129** is seventh wavelength selective coupler **3027** in FIG. **30B**. Second wavelength light **3017** from third tuning optical waveguide **3043** and second wavelength light **3018** from fourth tuning optical waveguide **3044** are coupled to combiner **3188**.

(551) With reference to FIGS. **32A** and **32B**, illustrations of a block diagram of routes for light traveling through an optical waveguide structure is depicted in accordance with an illustrative embodiment. In this example, routes **3250** are depicted for light traveling through optical waveguide structure **3000** in FIG. **30A** and for light traveling through optical waveguide structure **3100** in FIG. **31A**.

(552) First wavelength light **3011** travels in route **3251** that traverses, in sequence, source input coupler **3031**, first segment **3001**, first wavelength selective coupler **3021**, second segment **3002**, third wavelength selective coupler **3023**, second wavelength selective coupler **3022**, third segment **3003**, fifth wavelength selective coupler **3025**, fourth wavelength selective coupler **3024**, and fourth segment **3004**. Route **3251** for first wavelength light **3011** can be extended to pass through additional segments of nonlinear optical waveguide **3109** and additional wavelength selective couplers of the set of wavelength selective couplers **3120** as illustrated in FIG. **31A**. These additional segments and couplers include, in sequence, N th segment **3107**, $(2N-1)$ th wavelength selective coupler **3127**, $(2N-2)$ th wavelength selective coupler **3126**, $(N+1)$ th segment **3108**, $(2N+1)$ th wavelength selective coupler **3129**, $2N$ th wavelength selective coupler **3128**, final segment **3189**, and source output coupler **3133**.

(553) In this illustrative example, odd index route **3257** and even index route **3258** are present in optical waveguide structure **3100**. Odd index route **3257** comprises odd numbered segments and their associated components. Even index route **3258** comprises even numbered segments and their associated components.

(554) Two different components of second wavelength light traverse these two different routes through optical waveguide structure **3000**. For example, second wavelength light **3017** can be generated in first segment **3001** as well in third segment **3003**. Odd index route **3257** for second wavelength light **3017** traverses, in sequence, first segment **3001**, first wavelength selective coupler **3021**, first tuning optical waveguide **3041**, second wavelength selective coupler **3022**, third segment **3003**, and fifth wavelength selective coupler **3025**.

(555) Odd index route **3257** for second wavelength light **3017** can be further extended, as illustrated in FIGS. **31A** and **31B**, to traverse through additional segments of nonlinear optical waveguide **3109**, additional wavelength selective couplers of the set of wavelength selective couplers **3120**, and additional tuning optical waveguides of the set of tuning optical waveguides **3140** as illustrated in FIG. **31A**. These additional segments and couplers can include, in sequence, N th segment **3107**, $(2N-1)$ th wavelength selective coupler **3127**, N th tuning optical waveguide **3147**, $2N$ th wavelength selective coupler **3128**, final segment **3189**, source output coupler **3133**, auxiliary output coupler **3135**, and combiner **3188**. As depicted, second wavelength light **3117** is generated in the odd-indexed segments of nonlinear optical waveguide **3109**.

(556) As depicted, second wavelength light **3018** is generated in second segment **3002**. Second wavelength light **3018** travels in even index route **3258** that is different from odd index route **3257** traveled by second wavelength light **3017**. In even index route **3258**, second wavelength light **3018** traverses, in sequence, second segment **3002**, third wavelength selective coupler **3023**, second tuning optical waveguide **3042**, fourth wavelength selective coupler **3024**, and fourth segment **3004**.

(557) In the illustrative example, even index route **3258** for second wavelength light **3018** can be further extended, as illustrated in FIGS. **31A** and **31B**, to traverse through additional segments of nonlinear optical waveguide **3109**, additional wavelength selective couplers of the set of wavelength selective couplers **3120**, and additional tuning optical waveguides of the set of tuning optical waveguides **3140** as illustrated in FIGS. **31A** and **31B**. These additional segments and couplers can include, in sequence, $(N+1)$ th segment **3108**, $(2N+1)$ th wavelength selective coupler **3129**, $(N+1)$ th tuning optical waveguide **3148**, and combiner **3188**. The second wavelength light **3018** of second wavelength light is generated in the even-numbered segments of nonlinear optical waveguide **3109**.

(558) Turning next to FIG. **33**, an illustration of graphs of the effect of waveguide cross-sectional dimensions on the phase walk-off associated with imperfect wave vector matching is depicted in accordance with an illustrative embodiment. Graphs **3300** are for a nonlinear optical waveguide structure comprising a nonlinear optical waveguide using cross-sections **3700** shown in FIGS. **37A-37G** and described below. As depicted, graphs **3300** show the dependence of the π phase walk-off distance on the values for different features in a waveguide. Graphs **3300** illustrate tolerances for variations in different dimensions in nonlinear optical waveguides. These dimensions shown in graphs **3300** have ranges in which normal operating regions are present that have a desired level of performance. Although the designs can have

dimensions within these normal operating ranges, the environment during operation of the waveguides and deviations in fabrication can result in the values of these dimensions being out of tolerance for the desired level of performance within an operating region.

(559) The length selected for a nonlinear optical waveguide segment can depend on the desired fabrication and operational variations that can be tolerated by the nonlinear optical waveguide structure. Graphs **3300** give some examples of expected values for the nonlinear optical interaction distance at which a phase walk-off of π radians is obtained and the dependence of this π phase walk-off distance on variations in some examples of different types of waveguide dimensions.

(560) Line **3301** in graph **3302** and line **3305** in graph **3306** show the distance for a nonlinear optical interaction at which the phase walk-off reaches π radians. Line **3301** in graph **3302** shows this phase walk-off distance for different values of the strip width. Line **3305** in graph **3306** shows this phase walk-off distance for different values of the strip height.

(561) In this example, the portion of line **3301** within desired fabrication tolerance **3307** of strip widths in graph **3302** is desired operating region **3309**. Likewise, the portion of line **3305** within desired operating region **3312** of strip heights in graph **3306** is desired operating region **3313**. This desired operating region constrains the tolerable variation in strip width or strip height that provides the desired operation for a waveguide in which the cross section is implemented. For example, if a nonlinear optical waveguide is 1 mm long, this degree of variation can be tolerated in the strip width or strip height. But, if greater nonlinear optical interaction distance than 1 mm is desired, phase shifters can be used to provide desired levels of constructive nonlinear optical interaction the result in an increased amount of the generated light.

(562) In other words, the portion of line **3305** within desired operating region **3312** constrains a desired operating region **3312** for the strip height. Likewise, the portion of line **3301** within desired operating region **3309** constrains desired fabrication tolerance **3307** for the strip width. In desired operating region **3309** and **3312**, the distance for nonlinear optical interaction can be 1 mm or greater before the phase walk-off of that nonlinear optical interaction reaches or exceeds π radians. As discussed above, when the phase walk-off has a value smaller than π radians, the nonlinear optical generation process continues to be constructive and produces more and more generated light from the source light. However, when the phase walk-off has a value greater than π radians and up to a value of 2π radians, the nonlinear optical generation process becomes destructive and reduces the amount of generated light, converting some previously generated light back to source light. Thus, it is desirable to keep the phase walk-off between zero and π radians, and as close to zero as possible.

(563) In these illustrative examples, fabrication tolerances of approximately ± 1 nm (or ± 0.001 μm) in waveguide dimensions such as the strip width and the strip height reduce the distance at which the phase walk-off reaches π radians from being infinite (for perfect wave vector matching) to being on the order of 1 mm. If the departure of the fabricated waveguide dimensions such as the strip width and the strip height from the specified values is greater than ± 1 nm, the maximum distance at which the phase walk-off reaches π radians can be much smaller than 1 mm.

(564) To achieve longer nonlinear optical interaction distances, the examples in FIGS. **29-31** as well as the following examples couple the generated light into a phase tuning path that contains a phase shifter. An activation applied by the phase shifter can change the phase of the light propagating through the phase tuning path. This change in the phase of the generated light that is diverted to propagate through the phase tuning path and then is coupled back into the nonlinear optical waveguide can be used to adjust the phase walk-off observed at the point where the generated light is coupled back into the nonlinear optical waveguide so that the phase walk-off has a value that is zero or an even multiple of π radians.

(565) To produce the desired amount of adjustable phase change, the length of a phase tuning path formed by a tuning optical waveguide can be selected to have a length to accommodate the desired applicable length of the phase shifter associated with that phase tuning path. For example, with phase shifters that are based on electro-optic (EO) control of the wave vector, the change in the effective index achieved by an electro-optic phase shifter for light such as idler light can be described by an expression such as

$\Delta n_{\text{sub.I}} = n_{\text{sub.I}} \cdot \text{r}_{\text{sub.eff}} \Delta E_{\text{sub.applied}}$. The change in phase for the idler light can be described by an expression such as

$$(566) \Delta \vartheta_i = \left(\frac{2\pi}{\lambda_{\text{sub.I}}}\right) n_{\text{sub.I}} D_{\text{sub.i}}$$

In these expressions, $\Delta E_{\text{sub.applied}}$ is the applied electric field, $r_{\text{sub.eff}}$ is the value of the relevant electro-optic coefficient, $n_{\text{sub.I}}$ is the refractive of the electro-optic material for the idler light, $\lambda_{\text{sub.I}}$ is the wavelength of the idler light, and $D_{\text{sub.i}}$ is the applicable length of the phase shifter, with i being an index that indicates a specific phase shifter and tuning optical waveguide.

(567) In this illustrative example, the “activation” applied by the phase shifter upon the tuning optical waveguide is the electric field. This applied electric field results in a change of the effective index of that waveguide that changes the phase of the light traveling through that waveguide.

(568) For practical widths of the electro-optic material in the phase shifter, the maximum E-field that can be applied before risking breakdown can be on the order of 10 V/ μm . For an electro-optic material such as x-cut lithium niobate and TE polarized light in the phase shifter, the value for $n_{\text{sub.I}} \cdot \text{r}_{\text{sub.eff}}$ is on the order of 3×10^{-4} $\mu\text{m/V}$. To achieve a maximum electrically controlled phase shift of $\pm \pi$ radians, the applicable length of the phase shifter should be at least

$$(569) \left(\frac{D_{\text{sub.i}}}{\lambda_{\text{sub.I}}}\right) \sim 1.5 \times 10^3$$

As an example, if the idler wavelength is 1.5 μm , the applicable length of the phase shifter should be approximately 3 mm.

(570) A phase shifter can comprise multiple sections. For appropriate supply of the electrical control signals to these multiple phase shifter sections, the applicable length of the phase shift adjustment can be as long as the total length of the multiple phase shifter sections. If the phase shifter comprises a material that has a large thermo-optic coefficient, such a thermo-optic phase shifter can have a shorter length than an electro-optic phase shifter.

(571) The phase shifters in the final tuning optical waveguide and the phase shifters in the next-to-final tuning optical waveguide are used to control the phases of the two components of generated light that are combined together. These two phase shifters can be operated in a push-pull manner, with one tuning optical waveguide producing a positive phase shift and the other tuning optical waveguide producing a negative phase shift. Thus, the resulting net phase shift between the outputs from these two tuning optical waveguides can be two times as large as the phase shift applied in tuning optical waveguide. As a result, the phase shifters of the final and the next-to-final tuning optical waveguide can have a smaller applicable length than the phase shifters in the preceding tuning optical waveguide of the nonlinear optical waveguide structure and still achieve a maximum electrically adjustable phase shift of $\pm \pi$ radians.

(572) Turning now to FIG. **34**, an illustration of an optical waveguide structure is depicted in accordance with an illustrative example. In this example, optical waveguide structure **3400** is an open-ended nonlinear optical waveguide structure. In other words, loops for recirculating light are not used.

(573) As depicted, optical waveguide structure **3400** comprises nonlinear optical waveguide **3409**, first idler tuning optical waveguide **3441A**, first signal tuning optical waveguide **3441B**, second idler tuning optical waveguide **3442A**, second signal tuning optical waveguide **3442B**, first idler wavelength selective coupler **3421A**, first signal wavelength selective coupler **3421B**, second idler wavelength selective

coupler **3422A**, second signal wavelength selective coupler **3422B**, pump input waveguide **3472**, signal input waveguide **3474**, pump output waveguide **3473**, signal output waveguide **3475**, idler output waveguide **3479**, pump input coupler **3432**, signal input coupler **3434**, pump output coupler **3433**, signal output coupler **3435**, idler output coupler **3439**, first idler phase shifter **3461A**, first signal phase shifter **3461B**, second idler phase shifter **3462A**, second signal phase shifter **3462B**.

(574) In this illustrative example, nonlinear optical waveguide **3409**, has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment **3401**, second segment **3402**, and third segment **3403**.

(575) As depicted, pump input coupler **3432** couples pump light **3412** introduced through pump input waveguide **3472** to first segment **3401**. Optional signal input coupler **3434** couples signal light **3414** introduced in signal input waveguide **3474** to first segment **3401**. Pump output coupler **3433** couples pump light **3412** from third segment **3403** to pump output waveguide **3473**. Signal output coupler **3435** couples signal light **3414** from third segment **3403** to signal output waveguide **3475**. Idler output coupler **3439** couples idler light **3416** from third segment **3403** to idler output waveguide **3479**.

(576) As depicted, nonlinear optical waveguide **3409** and other components are formed on a yz plane defined by z-axis **3493** and y-axis **3492** in which an x-axis **3491** is perpendicular to the plane. In an illustrative example, nonlinear optical waveguides can be fabricated from a nonlinear optical material such as x-cut lithium niobate. In this example, the x-axis of the nonlinear optical material is perpendicular to the yz plane of the structure of nonlinear optical waveguide **3409** and the other components.

(577) In this illustrative example, first idler wavelength selective coupler **3421A** couples idler light **3416** by extracting idler light **3416** from first segment **3401** into first idler tuning optical waveguide **3441A** and reinserting idler light **3416** into second segment **3402** after activations have been applied using first idler phase shifter **3461A** associated with first idler tuning optical waveguide **3441A**. Each phase shifter is comprised of three pairs of electrodes in this example. First signal wavelength selective coupler **3421B** couples signal light **3414** by extracting signal light **3414** from first segment **3401** into first signal tuning optical waveguide **3441B** and reinserting signal light **3414** into second segment **3402** after activations have been applied using first signal phase shifter **3461B** associated with first signal tuning optical waveguide **3441B**.

(578) In this illustrative example, second idler wavelength selective coupler **3422A** couples idler light **3416** by extracting idler light **3416** from second segment **3402** into second idler tuning optical waveguide **3442A** and reinserting idler light **3416** into third segment **3403** after activations have been applied using second idler phase shifter **3462A** associated with second idler tuning optical waveguide **3442A**. Second signal wavelength selective coupler **3422B** couples signal light **3414** by extracting signal light **3414** from second segment **3402** into second signal tuning optical waveguide **3442B** and reinserting signal light **3414** into third segment **3403** after activations have been applied using second signal phase shifter **3462B** associated with second signal tuning optical waveguide **3442B**.

(579) In this example, the phase shifters for the different tuning optical waveguides can apply activations to the light traveling through the tuning optical waveguides. The application of the activations can adjust the phase of the light to obtain a desired phase walk-off when at least one of the environment or fabrication inconsistencies result in a dimension of the nonlinear optical waveguide being out of tolerance for design level performance. In other words, the phase shifters can be used when the length of the tuning optical waveguides do not provide a desired phase walk-off for a nonlinear optical interaction involving light that has traveled through the tuning optical waveguides. In this example, both the length of a tuning optical waveguide and a phase shifter associated with the tuning optical waveguide can be used to obtain a desired phase walk-off. The phase shifter essentially adjusts the effective length of the tuning waveguide by applying an activation to change the tuning optical waveguide in a manner that changes the phase shift of light traveling through the tuning optical waveguide.

(580) Consider an illustrative implementation of optical waveguide structure **3400**, nonlinear optical waveguide **3409** that has strip width and strip height designed to achieve perfect phase matching, so that the π phase walk-off distance for nonlinear optical waveguide **3409** is essentially infinite. Realistic fabrication tolerances may cause the achieved π phase walk-off distance to be at least 1 mm. Thus, this implementation has the lengths of first segment **3401**, second segment **3402** and third segment **3403** no larger than 1 mm. This implementation also sets the length of first idler tuning optical waveguide **3441A** and the length of first signal tuning optical waveguide **3441B** so that the phase walk-off for the nonlinear optical interaction occurring in nonlinear optical waveguide **3409** from a starting point of first segment **3401** to a starting point of second segment **3402** has a value of zero or an even multiple of π radians. The starting point of first segment **3401** is where pump input coupler **3432** connects with first segment **3401**. The starting point of second segment **3402** is where first signal wavelength selective coupler **3421B** connects with second segment **3402**. For this illustrative implementation, first segment **3401** connects with first idler wavelength selective coupler **3421A**, which connects with first signal wavelength selective coupler **3421B**, which then connects with second segment **3402**.

(581) This implementation also sets the length of second idler tuning optical waveguide **3442A** and the length of second signal tuning optical waveguide **3442B** so that the phase walk-off for the nonlinear optical interaction occurring in nonlinear optical waveguide **3409** from the starting point of second segment **3402** to a starting point of third segment **3403** has a value of zero or an even multiple of π radians. The starting point of third segment **3403** is where second signal wavelength selective coupler **3422B** connects with third segment **3403**. For this illustrative implementation, second segment **3402** connects with second idler wavelength selective coupler **3422A**, which connects with second signal wavelength selective coupler **3422B**, which then connects with third segment **3403**.

(582) This illustrative implementation also has first idler phase shifter **3461A** designed to apply activations to first idler tuning optical waveguide **3441A** and has first signal phase shifter **3461B** designed to apply activations to first signal tuning optical waveguide **3441B** so that the phase walk-off for the nonlinear optical interaction occurring in nonlinear optical waveguide **3409** from the starting point of first segment **3401** to the starting point of second segment **3402** has a value of zero or an even multiple of π radians even when the strip width and strip height of first idler tuning optical waveguide **3441A**, first signal tuning optical waveguide **3441B** and first segment **3401** depart from their as-designed values as a result of fabrication tolerances or of a variation in an operating condition such as temperature.

(583) This implementation also has second idler phase shifter **3462A** designed to apply activations to second idler tuning optical waveguide **3442A** and has second signal phase shifter **3462B** designed to apply activations to second signal tuning optical waveguide **3442B** so that the phase walk-off for the nonlinear optical interaction occurring in nonlinear optical waveguide **3409** from the starting point of second segment **3402** to the starting point of third segment **3403** has a value of zero or an even multiple of π radians even when the strip width and strip height of second idler tuning optical waveguide **3442A**, second signal tuning optical waveguide **3442B** and second segment **3402** depart from their as-designed values as a result of fabrication tolerances or of a variation in an operating condition.

(584) Each segment in nonlinear optical waveguide **3409** except the last segment is associated with a tuning optical waveguide for the generated idler light and a different tuning optical waveguide for the generated signal light. In some examples of optical structures, the two wavelength selective couplers, for the signal light and for the idler light, are located near the end of a given segment and just before the start of the next segment. As depicted, pump removing coupler is located at the end of the last segment. Pump input coupler **3432** and the pump

output coupler **3433** establish the overall length of the nonlinear optical interaction that produces the generated light.

(585) Optical waveguide structure **3400** can be used for nonlinear optical processes such as spontaneous parametric down conversion (SPDC) by omitting the optional signal input coupler **3434** and by using both the idler-phase tuning paths and the signal-phase tuning paths. The structure of FIG. **34** also can be used for dual-source nonlinear optical processes such as difference frequency generation (DFG) with signal light **3414** as an auxiliary source or input light that is supplied to optical waveguide structure **3400** in addition to the pump light. In this case, optical waveguide structure can include optional signal input coupler **3434** and signal light can be supplied to the structure through signal input waveguide **3474**. The overall length of a segment and including the lengths of the two wavelength-selective couplers (for the idler light and for the signal light) associated with that segment is selected to be sufficiently small that the magnitude of the phase walk-off resulting from anticipated fabrication and operational tolerances is no greater than π radians.

(586) The phase shifters in a phase tuning path are configured to have a length sufficiently large to achieve an electrically controlled phase shift as much as $\pm\pi$ radians or greater. The overall length of a phase tuning path is selected to achieve a relative phase shift that is 0 or a multiple of 2π radians between the previously generated light reinserted from the phase tuning path into the subsequent segment and the newly generated light in the subsequent segment.

(587) In this example, the generation of light by a nonlinear optical process in optical waveguide structure can include difference-frequency generation (DFG) and spontaneous parametric down conversion (SPDC). However, the elements of optical waveguide structure **3400** described in this example can apply to generation of light by other nonlinear optical processes such as sum-frequency generation and four-wave mixing. These nonlinear optical processes in optical waveguide structure **3400** can involve second-order nonlinearity, such as for the examples discussed, as well as third order nonlinearity.

(588) Further in FIG. **34**, in optical waveguide structure **3400**, first route **3452** includes first segment **3401**, second segment **3402** and third segment **3403**. First route **3452** is traversed by the pump light **3412**. Second route **3456** includes first segment **3401**, first idler tuning optical waveguide **3441A**, second segment **3402**, second idler tuning optical waveguide **3442A**, and third segment **3403**. Second route **3456** is traversed by the idler light **3416**. Third route **3454** includes first segment **3401**, first signal tuning optical waveguide **3441B**, second segment **3402**, second signal tuning optical waveguide **3442B**, and third segment **3403**. Third route **3454** is traversed by signal light **3414**. In this example, the generated light is idler light **3416**.

(589) For the example depicted in FIG. **34**, the same wavelength selective coupler functions as both an extracting out-coupler from a nonlinear optical waveguide segment and an inserting in-coupler to a different nonlinear optical waveguide segment. For some examples of optical waveguide structure **3400**, the idler coupler is located as close as feasible to the signal coupler for the same index value. This proximity of the two couplers reduces the length of nonlinear optical waveguide **3409** between them so that the additional phase shifts applied to the idler light and the signal light are done at essentially the same point on the nonlinear optical waveguide.

(590) As depicted, nonlinear optical waveguide **3409** in waveguide structure **3400** has a series of nonlinear optical waveguide segments coupled to a series of tuning optical waveguides. As depicted, nonlinear optical waveguide **3409** has only one group of nonlinear optical waveguide segments and tuning optical waveguides. Optical waveguide structure **3400** is suitable for nonlinear optical processes such as ones involving TE polarized light in x-cut lithium niobate. As depicted in FIG. **34**, optical waveguide structure **3400** has a single wavelength-selective coupler for each tuning optical waveguide. In this example, each of these wavelength selective couplers functions both as an out-coupler to couple a generated idler or signal light from nonlinear optical waveguide **3409** into a tuning optical waveguide and also as an in-coupler to couple phase-shifted idler or signal light from the tuning optical waveguide back into nonlinear optical waveguide **3409**. In other examples, discussed below, the optical waveguide structure has two wavelength-selective couplers for each phase tuning path. In this example, a first wavelength selective coupler functions as an out-coupler to couple generated idler light or signal light into a tuning optical waveguide and a second wavelength selective coupler functions as an in-coupler to couple phase-shifted idler or phase shifted signal light from the tuning optical waveguide back into nonlinear optical waveguide **3409**.

(591) Turning to FIG. **35**, an illustration of an optical waveguide structure with phase shifters for tuning light is depicted in accordance with an illustrative example. In this illustrative example, optical waveguide structure **3500** comprises a number of different components. As depicted, optical waveguide structure **3500** comprises nonlinear optical waveguide **3509** first tuning optical waveguide **3541**, second tuning optical waveguide **3542**, third tuning optical waveguide **3543**, fourth tuning optical waveguide **3544**, first idler out wavelength selective coupler **3521**, second idler in wavelength selective coupler **3522**, third idler out wavelength selective coupler **3523**, fourth idler in wavelength selective coupler **3524**, fifth idler out wavelength selective coupler **3525**, sixth idler in wavelength selective coupler **3526**, seventh idler out wavelength selective coupler **3527**, and idler combiner **3528**, pump input waveguide **3572**, signal input waveguide **3574**, pump output waveguide **3573**, signal output waveguide **3575**, idler output waveguide **3579**, pump input coupler **3532**, signal input coupler **3534**, pump output coupler **3533**, signal output coupler **3535**, idler output coupler **3539**, first phase shifter **3561**, second phase shifter **3562**, third phase shifter **3563**, fourth phase shifter **3564**, and fifth phase shifter **3565**.

(592) In this illustrative example, nonlinear optical waveguide **3509**, has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment **3501**, second segment **3502**, third segment **3503**, fourth segment **3504**, and fifth segment **3505**. In this example, fifth segment **3505** can be considered a “final segment” having a nonlinear optical material.

(593) In this example, pump input coupler **3532** couples pump light **3512** introduced through pump input waveguide **3572** to first segment **3501**. Signal input coupler **3534** couples signal light **3514** introduced in signal input waveguide **3574** to first segment **3501**. Pump output coupler **3533** couples pump light **3512** from fifth segment **3505** to pump output waveguide **3573**. As a result, pump light **3512** does not travel through fifth segment **3505**. Idler output coupler **3539** couples a combination of idler light **3517** from fifth segment **3505** and idler light **3518** from fourth tuning optical waveguide **3544** to form idler light **3516**. Idler output coupler **3539** couples idler light **3516** from fifth segment **3505** to idler output waveguide **3579**. Optional signal output coupler **3535** couples signal light **3514** to signal output waveguide **3575**.

(594) In this example, signal output coupler **3535** can be an optional component because idler output coupler **3539** typically can be designed to separate signal light **3514** from idler light **3516**. Thus, the two outputs for idler output coupler **3539** can be connected to idler output waveguide **3579** and signal output waveguide **3575**.

(595) As depicted, nonlinear optical waveguide **3509** and other components are formed on a yz plane defined by z-axis **3593** and y-axis **3592** in which x-axis **3591** for the nonlinear optical material is perpendicular to the yz plane. In an illustrative example, nonlinear optical waveguide **3509** can be fabricated from a nonlinear optical material such as x-cut lithium niobate. In this example, nonlinear optical waveguide **3509** is aligned parallel to the y-axis **3592** of the nonlinear optical material and is aligned perpendicular to the z-axis **3593** of the nonlinear optical material, with the light propagating in nonlinear optical waveguide **3509** being in a TE mode of nonlinear optical waveguide **3509**.

(596) For this example, both pump light **3512** and signal light **3514** can be supplied at the input end of the nonlinear optical waveguide

3509 in optical waveguide structure 3500. The portion of nonlinear optical waveguide 3509 between the pump input coupler 3532 and first idler out wavelength selective coupler 3521 is considered first segment 3501. Idler light 3517 generated in first segment 3501 is diverted into first tuning optical waveguide 3541 by the first idler out wavelength selective coupler 3521. This wavelength selective coupler diverts idler light 3517 into first tuning optical waveguide 3541 but couples pump light 3512 and signal light 3514 to travel through second segment 3502 in nonlinear optical waveguide 3509.

(597) As depicted in this example, optical waveguide structure 3500 can be used in a difference-frequency generation (DFG) process or a sum-frequency generation (SFG) process, such as second harmonic generation, in which source light in the form of both pump light 3512 and signal light 3514 are used to produce generated light in the form of idler light 3516. In this example, idler light 3516 results from combining idler light 3517 and idler light 3518, which travel through the two different routes for the generated light.

(598) Pump light 3512 and signal light 3514 continue to propagate in the nonlinear optical waveguide 3509 and are not diverted into the tuning optical waveguides. In this example, only the generated idler light is diverted into the tuning optical waveguides.

(599) In this example, idler light 3517 propagates through first tuning optical waveguide 3541 and is inserted back into the third segment 3503 in nonlinear optical waveguide 3509 by second idler in wavelength selective coupler 3522. This idler in wavelength selective coupler, like the other idler in wavelength selective couplers in the nonlinear optical waveguide 3509, is a wavelength-selective coupler, and is similar to the idler out wavelength selective couplers. In this example second idler in wavelength selective coupler 3522 is located at the beginning of third segment 3503.

(600) The portion of nonlinear optical waveguide 3509 between first idler out wavelength selective coupler 3521, second idler in wavelength selective coupler 3522 is second segment 3502. Since pump light 3512, as well as signal light 3514, still travels in the nonlinear optical waveguide 3509, additional idler light can be generated in second segment 3502.

(601) In this example, idler light 3518 generated in second segment 3502 is diverted into second tuning optical waveguide 3542 by third idler out wavelength selective coupler 3523. This wavelength selective coupler keeps the generated idler light 3518 from interacting with the nonlinear optical process that occurs in third segment 3503. The diverted idler light 3518 propagates through the second tuning optical waveguide 3542 and is inserted back into nonlinear optical waveguide 3509 by fourth idler in wavelength selective coupler 3524. This fourth idler in wavelength selective coupler is located at the beginning of fourth segment 3504. As depicted, fourth idler in wavelength selective coupler 3524 functions as an idler in coupler and is connected to the beginning of fourth segment 3504 and couples idler light 3518 from second tuning optical waveguide 3542 into fourth segment 3504 at the beginning of fourth segment 3504.

(602) In this illustrative example, third segment 3503 and fifth idler out wavelength selective coupler 3525 are similar to first segment 3501 and the first idler out wavelength selective coupler 3521. Likewise, fourth segment 3504 and seventh idler out wavelength selective coupler 3527 are similar to second segment 3502 and third idler out wavelength selective coupler 3522. In this illustrative example, first segment 3501 and third segment 3503 form part of a first, odd index, group of segments. Second segment 3502 and fourth segment 3504 form part of a second, even index, group of segments.

(603) The final portion of the nonlinear optical waveguide 3509 is used to combine constructively the generated light of the odd indexed segments with the generated light of the even indexed segments. In this illustrative example, this final portion includes fifth segment 3505, fourth tuning optical waveguide 3544, and third tuning optical waveguide 3543. In this example, sixth idler in wavelength selective coupler 3526 re-inserts idler light 3517 from third tuning optical waveguide 3543 into nonlinear optical waveguide 3509 at the starting point of fifth segment 3505. Just prior to sixth idler in wavelength selective coupler 3526, seventh idler out wavelength selective coupler 3527 extracts idler light 3518 from fourth segment 3504, coupling idler light 3518 into fourth tuning optical waveguide 3544.

(604) Shortly following sixth idler in wavelength selective coupler 3526, pump output coupler 3533 removes pump light 3512 into pump output waveguide 3573. Thus, since pump light 3512 is absent, nonlinear optical generation of additional idler light in fifth segment 3505 does not occur. Idler light 3517 inserted into and propagating through the fifth segment 3505 was generated in third segment 3503 and in first segment 3501. Both third tuning optical waveguide 3543 and fifth segment 3505 conduct idler light 3517 generated in third segment 3503 and first segment 3501 with no additional generation of idler light 3517. Since no additional idler light is generated, fifth segment 3505 functions like an extension of the third tuning optical waveguide 3543.

(605) Light from the final segment, which in this case is fifth segment 3505, and light from the final tuning optical waveguide, which in this case is fourth tuning optical waveguide 3544, are combined together at the final in-coupling point. Idler combiner 3528 at the final in-coupling point couples together light that has been generated from two different routes. In optical waveguide structure 3500, first route 3557 includes first segment 3501, first tuning optical waveguide 3541, third segment 3503, the third tuning optical waveguide 3543, and fifth segment 3505. Second route 3558 includes second segment 3502, second tuning optical waveguide 3542, fourth segment 3504, and fourth tuning optical waveguide 3544.

(606) In the illustrative example, two differences are present between the generated idler light 3517 from first route 3557 and the generated idler light 3518 from second route 3558. The first difference is that the two routes have different values for the phases of the source light contributing to the generation of idler light 3517 and idler light 3518. Thus, the phase of idler light 3517 supplied from fifth segment 3505 into idler combiner 3528 is different from the phase of idler light 3518 supplied from fourth tuning optical waveguide 3544 into idler combiner 3528. Idler light 3517 is supplied through fifth idler out wavelength selective coupler 3525 and through third tuning optical waveguide 3543. Idler light 3518 is supplied through seventh idler out wavelength selective coupler 3527 and through fourth tuning optical waveguide 3544. The difference between the phase of idler light 3517 inserted by the sixth idler in wavelength selective coupler 3526 to idler combiner 3528 and idler light 3518 inserted from fourth tuning optical waveguide 3544 to idler combiner 3528 is approximately the difference between the phase of the source light at the end of first segment 3501 and the source light at the end of second segment 3502. The magnitude of this difference, modulo 2π , can be as large as π radians.

(607) The second difference is that first route 3557 has one segment more than second route 3558. In optical waveguide structure 3500, first route 3557 has three segments, but second route 3558 has only two segments. Thus, first route 3557 is longer than second route 3558, assuming the corresponding segments and phase tuning paths of the two routes are matched in their lengths. The additional length of first route 3557 is designed to at least partially compensate for the difference between the phases of the source light for the two routes, and thus the difference between the phases of the generated light.

(608) In this example, the length of the final segment, fifth segment 3505, and the lengths of the final tuning optical waveguide, fourth tuning optical waveguide 3544, and the next-to-final tuning optical waveguide, third tuning optical waveguide 3543 can be selected such that constructive interference is present between the light from the final segment, fifth segment 3505, and the light from the final tuning optical waveguide, fourth tuning optical waveguide 3544 when they are combined together at the final in-coupling point at coupler, also called idler combiner 3528, which couples together the light that has been generate in these two different routes.

(609) In optical waveguide structure **3500**, The final in-coupling point in nonlinear optical waveguide **3509** is unlike any of the other coupling points in nonlinear optical waveguide **3509**. As depicted, the fourth tuning optical waveguide **3544** terminates at idler combiner **3528**. Fifth segment **3505** also terminates at idler combiner **3528**. A wavelength selective coupler, like the idler out wavelength selective couplers, such as first idler out wavelength selective coupler **3521**, third idler out wavelength selective coupler **3523**, fifth idler out wavelength selective coupler **3525** and seventh idler out wavelength selective coupler **3527** and the idler in wavelength selective couplers, such as second idler in wavelength selective coupler **3522**, fourth idler in wavelength selective coupler **3524**, and sixth idler in wavelength selective coupler **3526**, is designed to be in a “cross” state for the wavelength of idler light **3517** or idler light **3518** but to be in a “thru” state for the wavelength of pump light **3512** and for the wavelength of signal light **3514**. The idler combiner **3528** in the final portion, following fifth segment **3505** of nonlinear optical waveguide **3509**, is designed to combine equally the idler light supplied to that combiner from the fourth tuning optical waveguide **3544** and from fifth segment **3505**. This idler combiner **3528** functions more like a 50-50 coupler that couples “in phase” light into one output of that coupler and couples “out of phase” light into another output of that coupler. The “in phase” idler light is coupled to an output segment of nonlinear optical waveguide **3509**. Idler output coupler **3539** can extract the combined idler light to idler output waveguide **3579** of optical waveguide structure **3500**, leaving residual signal light in nonlinear optical waveguide **3509**. In an alternative implementation, optional signal output coupler **3535** can extract signal light **3514** from nonlinear optical waveguide **3509** before the signal light **3514** reaches idler combiner **3528**.

(610) In the illustrative example of optical waveguide structure **3400** in FIG. **34** and optical waveguide structure **3500** in FIG. **35**, optical waveguide structure **3400** has a single route, second route **3456** for idler light **3416** that is generated. As depicted second route **3456** extends through all of the successive segments and idler tuning waveguides. In contrast, optical waveguide structure **3500** in FIG. **35** has two routes for idler light **3517** and **3518** that are generated and then combined to comprise idler light **3516**. These routes are first route **3557** and second route **3558**. Each of these two routes extends through every other segment and tuning waveguide in optical waveguide structure **3500**. In this example, idler light **3516** is present in optical waveguide structure **3500** only after idler combiner **3528**. Before idler combiner **3528**, idler light **3517** and idler light **3518** are present.

(611) Turning now to FIG. **36**, an illustration of a graph of light generation is depicted in accordance with an illustrative embodiment. In graph **3600**, an illustration of normalized light generation rates based on distances of nonlinear optical interactions are shown. This light generation can be for a light such as idler light or signal light traveling through an optical waveguide structure having a nonlinear optical waveguide with segments and tuning optical waveguides coupled to segments of nonlinear optical waveguide. In this example, the light generation depicted occurs in an optical waveguide structure such as optical waveguide structure **3500** in FIG. **35**.

(612) As depicted in graph **3600**, x-axis **3602** illustrates the nonlinear optical interaction distance for light and nonlinear optical waveguide while y-axis **3604** illustrates the normalized generation rate of light as a result of that nonlinear optical interaction. As depicted, line **3607** and line **3608** illustrates light generation for light traveling through different routes within the optical waveguide structure. In this illustrative example, line **3607** illustrates the light generation rates for light traveling through first route **3557** through odd indexed segments of optical waveguide structure **3500**. Line **3608** illustrates light generation rates for light traveling through second route **3558** using even index segments in optical waveguide structure **3500** in FIG. **35**. As can be seen, line **3607** and line **3608** have a stair stepped shape in which light generation rates increase as the light travels through segments and tuning optical waveguides in optical waveguide structure **3500**.

(613) Line **3607** and line **3608** both have sections that corresponding to structures in optical waveguide structure **3500**. These sections include horizontal sections and angled sections relative to x-axis **3602**. As depicted, line **3607** has angled section **3611**, horizontal section **3621**, angled section **3613**, horizontal section **3623**, and horizontal section **3615**. For line **3607**, angled section **3611** corresponds to light generation that occurs in first segment **3501**; horizontal section **3621** corresponds to light generation that occurs in first tuning optical waveguide **3541**, angled section **3613** corresponds to light generation that occurs in third segment **3503**, horizontal section **3623** corresponds to light generation that occurs in third tuning optical waveguide **3543**, and horizontal section **3615** corresponds to light generation that occurs in fifth segment **3505**, fifth segment **3505** can be a final segment. For line **3608**, angled section **3632** corresponds to light generation that occurs in second segment **3502**, horizontal section **3642** corresponds to light generation that occurs in second tuning optical waveguide **3542**, angled section **3634** corresponds to light generation that occurs in fourth segment **3504**, and horizontal section **3644** corresponds to light generation that occurs in fourth tuning optical waveguide **3544**. As indicated by the sections of lines **3607** and **3608**, light generation occurs only in the nonlinear optical waveguide segments, such as first segment **3501**, second segment **3502**, third segment **3503**, and fourth segment **3504**. These are segments comprised of nonlinear optical material and through which pump light **3512** also travels. Light generation does not occur in tuning optical waveguides, such as first tuning optical waveguide **3541**, second tuning optical waveguide **3542**, third tuning optical waveguide **3543**, and fourth tuning optical waveguide **3544**. Pump light **3512** does not travel through these tuning optical waveguides. Light generation also does not occur in fifth segment **3505** because pump light is removed at the start of fifth segment **3505** and thus does not travel through fifth segment **3505**.

(614) As depicted a jump occurs from horizontal section **3615** and horizontal section **3644** to horizontal section **3619** because of the idler light **3517** from first, odd index, first route **3557** being combined with the idler light **3518** from second, even index, second route **3558** by idler combiner **3528**, which is located after fifth segment **3505** and prior to idler output waveguide **3579**.

(615) With reference to FIGS. **37A-37G**, illustrations of cross-sections for nonlinear optical waveguide structures are depicted in accordance with an illustrative embodiment. As depicted, cross-sections **3700** can be used to implement optical waveguide structures such as nonlinear optical waveguides, tuning optical waveguides, optical couplers such as wavelength selective couplers and input or output couplers as well as combiners, and phase shifters.

(616) As depicted, cross-section **3701** (FIG. **37A**) can be used to implement a nonlinear optical waveguide. Cross-section **3701** comprises core region **3702** within cladding **3703** formed from silicon oxide (SiO_2) and cladding **3703** is located on silicon substrate **3704**. Core region **3702** comprises lithium niobate center section **3705** and side sections **3706** formed from silicon nitride. As depicted, center width **3708** is the width of lithium niobate center section **3705**. Strip width **3712** is the width of core region **3702**. The height of core region **3702** is strip height **3714**.

(617) As depicted, cross-section **3715** (FIG. **37B**) has core region **3716** comprised of silicon nitride located within cladding **3717** formed from silicon oxide. Cladding **3717** is located on silicon substrate **3718**. In this illustrative example, core region **3716** has strip width **3719** and strip height **3720**. Cross-section **3721** (FIG. **37D**) has core region **3722** comprised of lithium niobate (LiNbO_3) located within cladding **3723** formed from silicon oxide. Cladding **3723** is located on silicon substrate **3724**. As depicted, core region **3721** has strip width **3725** and strip height **3726**.

(618) These two cross-sections can be used to implement tuning optical waveguides. For example, tuning optical waveguides using cross-section **3715** and cross-section **3721** can have tapers that transition gradually between waveguide portions with the silicon nitride core and with the lithium niobate core in these cross-sections.

(619) In this illustrative example, cross-section 3727 (FIG. 37C) is a cross-section for an optical coupler between a nonlinear optical waveguide and another waveguide such as a tuning optical waveguide or an input waveguide or an output waveguide. As depicted, core region 3728 can be for the nonlinear optical waveguide while core region 3729 is for another optical waveguide. Core region 3728 is comprised of lithium niobate center section 3730 with silicon nitride sides 3731. Core region 3729 is comprised of silicon nitride.

(620) As depicted, core region 3728 has strip width 3732 and lithium niobate center section 3730 has center width 3733. Core region 3729 has strip width 3734. In this illustrative example, both core regions have strip height 3735. Gap 3736 is present between core region 3728 and core region 3729. These components are within cladding 3737 formed using silicon oxide, which is located on silicon substrate 3738.

(621) The coupler using cross-section 3727 can be, for example, a wavelength selective out-coupler, a wavelength selective in-coupler, or a combiner. This coupler also can be a pump input coupler, signal input coupler or idler input coupler, a pump removal coupler, a signal output coupler, or an idler output coupler. Further, lithium niobate can be present in core region 3728 when core region 3728 is for a nonlinear optical waveguide.

(622) In this example, cross-section 3739 (FIG. 37E) is an example of the cross-section that can be used for a phase shifter. In this illustrative example, cross-section 3739 comprises electrodes 3740 for the phase shifter located on each side of core region 3741. Core region 3741 is comprised of lithium niobate, which is an electro-optic material for an electro-optically activated optical waveguide phase shifter. In this example, silicon nitride structure 3742 is located on core region 3741. These components are located within cladding 3744 which is comprised of silicon oxide. Cladding 3744 is located on silicon substrate 3745. The phase shifter using cross-section 3739 is an electro-optic phase shifter.

(623) Cross-section 3747 (FIG. 37F) is an example of a cross section that can be used in a phase shifter for phase tuning for a nonlinear optical waveguide. Core region 3748 is for a nonlinear optical waveguide.

(624) In this example, core region 3748 is comprised of lithium niobate center section 3749 and silicon nitride sides 3750. These components are located within cladding 3753. Cladding 3753 is comprised of silicon oxide and is located on silicon substrate 3754.

(625) Cross-section 3756 (FIG. 37G) is a cross-section for an optical coupler between a nonlinear optical waveguide and a tuning optical waveguide. As depicted, core region 3728 can be for the nonlinear optical waveguide while core region 3729 is for another optical waveguide.

(626) As depicted, cross section 3756 is a cross-section for an optical coupler between a nonlinear optical waveguide and another waveguide. As depicted, core region 3757 can be for the nonlinear optical waveguide while core region 3758 is for a tuning optical waveguide. Core region 3757 is comprised of lithium niobate center section 3761 with silicon nitride sides 3762. Core region 3758 is comprised of lithium niobate.

(627) As depicted, core region 3757 has strip width 3764 and lithium niobate center section 3761 has center width 3765. Core region 3758 has strip width 3766. In this illustrative example, both core regions have strip height 3767. Gap 3768 is present between core region 3757 and core region 3758. These components are within cladding 3759 formed using silicon oxide, which is located on silicon substrate 3760 in cross-section 3756. Whether a wavelength selective coupler is implemented with cross-section 3756 or cross-section 3727 is dependent on the specific values for the wavelengths of the pump light, signal light and idler light as well as on whether the wavelength selective coupler couples only idler light (which typically has the longest wavelength) between a nonlinear optical waveguide segment and a tuning optical waveguide or couples both idler light and signal light, of differing wavelengths, between the nonlinear optical waveguide and the tuning optical waveguide. The wavelength selective coupler of these exemplary cross-sections would couple pump light from one nonlinear optical waveguide segment to another nonlinear optical waveguide segment.

(628) Although cross-sections 3700 depicted in FIGS. 37A-37G are for waveguides with core regions comprising lithium niobate and silicon nitride and cladding regions comprising silicon dioxide, other materials and combinations of materials could be used for the core regions and the cladding regions. For example, other materials can include at least one of gallium arsenide, aluminum gallium arsenide, silicon carbide, titanium dioxide, aluminum nitride, or gallium nitride.

(629) In the illustrative examples, the different examples of optical waveguide structures can be scaled to a large number of segments and tuning optical waveguides. The particular configurations used can depend on the desired length of components such as phase shifters used with the tuning optical waveguides. Design specifications can limit the length of a single pair of electrodes of a phase shifter associated with a tuning optical waveguide. As a result, tuning optical waveguide can have folds with phase shifters having electrodes on the folds. In some examples, folds are used in the tuning optical waveguides to provide the amount of activation desired from phase shifters associated with tuning optical waveguides.

(630) With reference to FIG. 38, an illustration of an optical waveguide structure is depicted in accordance with an illustrative embodiment. As depicted, optical waveguide structure 3800 comprises nonlinear optical waveguide 3809, first tuning optical waveguide 3841, second tuning optical waveguide 3842, third tuning optical waveguide 3843, fourth tuning optical waveguide 3844, first idler out wavelength selective coupler 3821, second idler in wave selective coupler 3822, third idler out wavelength selective coupler 3823, fourth idler in wavelength selective coupler 3824, fifth idler out wavelength selective coupler 3825, sixth idler in wavelength selective coupler 3826, seventh idler out wavelength selective coupler 3827, combiner 3828, pump input waveguide 3872, signal input waveguide 3874, pump output waveguide 3873, signal output waveguide 3875, idler output waveguide 3879, pump input coupler 3832, signal input coupler 3834, pump output coupler 3833, signal output coupler 3835, idler output coupler 3839, first phase shifter 3861, second phase shifter 3862, third phase shifter 3863, and fourth phase shifter 3864.

(631) In this illustrative example, nonlinear optical waveguide 3809 has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment 3801, second segment 3802, third segment 3803, fourth segment 3804, and fifth segment 3805.

(632) In this example, pump input coupler 3832 couples pump light 3812 introduced through pump input waveguide 3872 to first segment 3801. Signal input coupler 3834 couples signal light 3814 introduced in signal input waveguide 3874 to first segment 3801. Pump output coupler 3833 couples pump light 3812 from fifth segment 3805 to pump output waveguide 3873. Signal output coupler 3835 couples signal light 3814 from fifth segment 3805 to signal output waveguide 3875. Combiner 3828 combines idler light 3817 from fifth segment 3805 and idler light 3818 from fourth tuning optical waveguide 3844 to form idler light 3816. Idler output coupler 3839 couples idler light 3816 from fifth segment 3805 to idler output waveguide 3879.

(633) As depicted, nonlinear optical waveguide 3809 and other components are formed on a yz plane defined by z-axis 3893 and y-axis 3892, with x-axis 3891 perpendicular to the yz plane. In an illustrative example, nonlinear optical waveguides can be fabricated from a nonlinear optical material such as x-cut lithium niobate. In this example, y-axis 3892 of the nonlinear optical material is parallel to the direction in which the nonlinear optical waveguide segments are oriented and z-axis 3893 of the nonlinear optical material is perpendicular to the propagation direction of the light in the nonlinear optical waveguide segments. For nonlinear optical materials such as x-cut lithium

niobate, this orientation of a nonlinear optical waveguide allows a spontaneous parametric down conversion, or parametric down conversion or up conversion, or second harmonic generation to make use of the largest second order nonlinear optical coefficient $d_{\text{sub}33}$ of the lithium niobate material when the light is in a TE mode of the waveguide. For an accompanying phase shifter formed in a tuning optical waveguide comprising an electro-optic material such as x-cut lithium niobate, orienting the phase shifter parallel to the material's y-axis and perpendicular to the material's z-axis also allows an electro-optic phase shifter to make use of the largest electro-optic coefficient $r_{\text{sub}33}$ of the lithium niobate material when the light is in a TE mode of the waveguide.

(634) In this illustrative example, the phase shifters have more electrodes than used in optical waveguide structure **3500** in FIG. **35**. The additional electrodes can be placed on the folds of the tuning optical waveguides.

(635) In this example, a phase shifter has a pair of electrodes on the portion of tuning optical waveguide following a fold of the tuning optical waveguide. These folds allow for a longer length on the combined electrodes in each phase shifter while reducing the length needed for nonlinear optical waveguide **3809** along y-axis **3892**.

(636) As depicted, first tuning optical waveguide **3841** has eight folds, second tuning optical waveguide **3842** has eight folds, third tuning optical waveguide **3843** has four folds, and fourth tuning optical waveguide **3844** has four folds. For example, the eight folds of first tuning optical waveguide **3841** are first fold **3850**, second fold **3851**, third fold **3852**, fourth fold **3853**, fifth fold **3854**, sixth fold **3855**, seventh fold **3856** and eighth fold **3848**. In this example, first phase shifter **3861** comprises electrode pair **3880** located between first fold **3850** and second fold **3851**; electrode pair **3881** located between second fold **3851** and third fold **3852**; electrode pair **3882** located between third fold **3852** and fourth fold **3853**; electrode pair **3883** located between fourth fold **3853** and fifth fold **3854**; electrode pair **3884** located between fifth fold **3854** and sixth fold **3855**; electrode pair **3885** located between sixth fold **3855** and seventh fold **3856**; and electrode pair **3886** located between seventh fold **3856** and eighth fold **3848** of the first tuning optical waveguide **3841**. In this example, the four folds of third tuning optical waveguide **3843** are first fold **3857**, second fold **3858**, third fold **3859** and fourth fold **3849**. Third phase shifter **3863** comprises electrode pair **3887** located between first fold **3857** and second fold **3858**; electrode pair **3888** located between second fold **3858** and third fold **3859**; and electrode pair **3889** located between third fold **3859** and fourth fold **3849** of the third tuning optical waveguide **3843**.

(637) Thus, the phase shifters in optical waveguide structure **3800** can have a larger applicable length for a given segment length in a segment in nonlinear optical waveguide **3809**. The number of folds can be increased from those shown in optical waveguide structure **3800** as needed to provide an over length of a phase shifter with multiple electrode pairs placed in the folds of the tuning optical waveguide.

(638) The number of segments in a nonlinear optical waveguide in an optical waveguide structure can affect the configuration of tuning optical waveguides used in the optical waveguide structure. For example, if the number of segments is relatively small number, such as 11 or fewer segments, optical waveguide structure can have a configuration that comprises an offset arrangement of tuning optical waveguides that each have 4 folds.

(639) With reference to FIG. **39**, an illustration of an optical waveguide structure with offset tuning optical waveguides is depicted in accordance with an illustrative embodiment. As depicted, optical waveguide structure **3900** comprises nonlinear optical waveguide **3909**, first tuning optical waveguide **3941**, second tuning optical waveguide **3942**, third tuning optical waveguide **3943**, fourth tuning optical waveguide **3944**, fifth tuning optical waveguide **3945**, sixth tuning optical waveguide **3946**, first idler out wavelength selective coupler **3921**, second idler in wave selective coupler **3922**, third idler out wavelength selective coupler **3923**, fourth idler in wavelength selective coupler **3924**, fifth idler out wavelength selective coupler **3925**, sixth idler in wavelength selective coupler **3926**, seventh idler out wavelength selective coupler **3927**, eighth idler in wavelength selective coupler **3928**, ninth idler out wavelength selective coupler **3987**, tenth idler in wavelength selective coupler **3988**, eleventh idler out wavelength selective coupler **3989**, combiner **3980**, pump input waveguide **3972**, signal input waveguide **3974**, pump output waveguide **3973**, signal output waveguide **3975**, idler output waveguide **3979**, pump input coupler **3932**, signal input coupler **3934**, pump output coupler **3933**, signal output coupler **3935**, idler output coupler **3939**, first phase shifter **3961**, second phase shifter **3962**, third phase shifter **3963**, fourth phase shifter **3964**, fifth phase shifter **3965**, and sixth phase shifter **3966**.

(640) In this illustrative example, nonlinear optical waveguide **3909**, has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment **3901**, second segment **3902**, third segment **3903**, fourth segment **3904**, fifth segment **3905**, sixth segment **3906**, and seventh segment **3907**.

(641) In this example, pump input coupler **3932** couples pump light **3912** introduced through pump input waveguide **3972** to first segment **3901**. Signal input coupler **3934** couples signal light **3914** introduced in signal input waveguide **3974** to first segment **3901**. Pump output coupler **3933** couples pump light **3912** from sixth segment **3906** to pump output waveguide **3973** so that pump light **3912** does not travel in seventh segment **3907**. Signal output coupler **3935** couples signal light **3914** from seventh segment **3907** to signal output waveguide **3975** prior to combiner **3980**. Combiner **3980** combines idler light **3917** from seventh segment **3907** and idler light **3918** from sixth tuning optical waveguide **3946** to form idler light **3916**. Optional idler output coupler **3939** couples idler light **3916** from seventh segment **3907** to idler output waveguide **3979**.

(642) As depicted, nonlinear optical waveguide **3909** and other components are formed on a yz plane defined by z-axis **3993** and y-axis **3992** in which x-axis **3991** is perpendicular to the plane. In an illustrative example, nonlinear optical waveguides can be fabricated from a nonlinear optical material such as x-cut lithium niobate. In this example, x-axis **3991** of the nonlinear optical material is perpendicular to the yz plane of the structure of nonlinear optical waveguide **3909** and the other components

(643) In this example, each tuning optical waveguide in optical waveguide structure **3900** can extend over more than two segments. Furthermore, each tuning optical waveguide has four folds with an electrode pair placed in the portion of the tuning optical waveguide after three of the four folds of the tuning optical waveguide. In this example, first tuning optical waveguide **3941** has first fold **3951**, second fold **3952**, third fold **3953** and fourth fold **3954**. First phase shifter **3961** comprises first electrode pair **3981** located between first fold **3951** and second fold **3952**, second electrode pair **3982** located between second fold **3952** and third fold **3953**, and third electrode pair **3983** located between third fold **3953** and fourth fold **3954**. Thus, the phase shifters in optical waveguide structure **3900** can have a larger applicable length for a given segment length in a segment in nonlinear optical waveguide **3909**. The length of the second electrode pair **3982** and the portion of first tuning optical waveguide **3941** between second fold **3952** and third fold **3953** can be increased as shown in optical waveguide structure **3900** as needed to provide a desired overall length for first phase shifter **3961**. For example, second electrode pair **3982** extends over first segment **3901**, second segment **3902**, third segment **3903**, fourth segment **3904**, and fifth segment **3905**.

(644) Likewise, third tuning optical waveguide **3943** has first fold **3956**, second fold **3957**, third fold **3958** and fourth fold **3959**. Third phase shifter **3963** comprises first electrode pair **3984** located between first fold **3956** and second fold **3957**, second electrode pair **3985** located between second fold **3957** and third fold **3958**, and third electrode pair **3986** located between third fold **3958** and fourth fold **3959**. The length of the second electrode pair **3985** of third phase shifter **3963** and the portion of third tuning optical waveguide **3943** between

second fold **3957** and third fold **3958** can be increased as needed to provide a desired overall length for third phase shifter **3963**. For example, second electrode pair **3985** extends over third segment **3903**, fourth segment **3904**, fifth segment **3905**, sixth segment **3906**, and seventh segment **3907**.

(645) In this example a first group of odd index segments comprises first segment **3901**, third segment **3903**, fifth segment **3905**, and seventh segment **3907**. A second group of even index segments comprises second segment **3902**, fourth segment **3904**, and sixth segment **3906**.

(646) Each of these groups of segments and tuning optical waveguides forms a route. As a result, 2 routes are present through which idler light **3917** and idler light **3918** travel within optical waveguide structure **3900**. For example, first route is traveled by idler light **3917** and is formed by the odd index segments and the associated tuning optical waveguides. In this example first route is comprised of first segment **3901**, third segment **3903**, fifth segment **3905**, first tuning optical waveguide **3941**, third tuning optical waveguide **3943**, and fifth tuning optical waveguide **3945**. Second route is traveled by idler light **3918** and is formed by even index segments and the associated tuning optical waveguides. In this example, second route comprises second segment **3902**, fourth segment **3904**, sixth segment **3906**, second tuning optical waveguide **3942**, fourth tuning optical waveguide **3944**, and sixth tuning optical waveguide **3946**.

(647) In this example, the tuning optical waveguides in same group can have configurations that are different from each other. For example, in the first group of the odd index segments, first tuning optical waveguide **3941** and third tuning optical waveguide **3943** each have a longer length than fifth tuning optical waveguide **3945**. In the second group of even index segments, second tuning optical waveguide **3942** and fourth tuning optical waveguide **3944** each have a longer length than sixth tuning optical waveguide **3946**. The phase shifters associated with the different tuning optical waveguides can be located at different offset distances from nonlinear optical waveguide **3909**. Each tuning optical waveguide can have a length that is selected to achieve a value that is some multiple of 2π radians for the phase walk-off of the nonlinear optical generation process that occurs between the beginning portion of the nonlinear optical waveguide segment that immediately precedes the tuning optical waveguide and the beginning portion of the subsequent segment in nonlinear optical waveguide **3909** into which that phase shifted idler light is coupled via the idler in wavelength selective coupler. For example, for third tuning optical waveguide **3943**, the immediately preceding nonlinear optical waveguide segment is third segment **3903** and the subsequent nonlinear optical waveguide segment is fifth segment **3905**.

(648) As depicted for optical waveguide structure **3900** in FIG. 39, the phase shift associated with the tuning optical waveguides is modulo 2π . For example, first tuning optical waveguide **3941** can have a phase shift of 100π ; third tuning optical waveguide **3943** can have a phase shift of 98π . As a result, latitude is present on the length of optical waveguides in optical waveguide structure **3900**.

(649) Additionally, as depicted in FIG. 39, the placement of phase shifters can be staggered. For example, first phase shifter **3961** and third phase shifter **3963** can have a “staggered” placement along z-axis **3993**. Thus, a portion of first phase shifter **3961** can overlap a portion of third phase shifter **3963** along y-axis **3992**.

(650) With reference now to FIG. 40, an illustration of an optical waveguide structure formed on a xy plane is depicted in accordance with an illustrative embodiment. As depicted, nonlinear optical waveguide **4009** is formed on a xy plane defined by x-axis **4091** and y-axis **4092**, in which a z-axis **4093** is perpendicular to the xy plane. In an illustrative example, nonlinear optical waveguides can be fabricated from a nonlinear optical material such as z-cut lithium niobate, c-axis aligned, and c-axis aligned 4H—SiC. In this example, z-axis **4093** of the nonlinear optical material such as z-cut lithium niobate, is perpendicular to the xy plane.

(651) In this example, optical waveguide structure **4000** comprises nonlinear optical waveguide **4009**, first tuning optical waveguide **4041**, second tuning optical waveguide **4042**, third tuning optical waveguide **4043**, fourth tuning optical waveguide **4044**, fifth tuning optical waveguide **4045**, sixth tuning optical waveguide **4046**, first idler out wavelength selective coupler **4021**, second idler in wave selective coupler **4022**, third idler out wavelength selective coupler **4023**, fourth idler in wavelength selective coupler **4024**, fifth idler out wavelength selective coupler **4025**, sixth idler in wavelength selective coupler **4026**, seventh idler out wavelength selective coupler **4027**, eighth idler in wavelength selective coupler **4028**, ninth idler out wavelength selective coupler **4087**, tenth idler in wavelength selective coupler **4088**, eleventh idler out wavelength selective coupler **4089**, combiner **4080**, pump input waveguide **4072**, signal input waveguide **4074**, pump output waveguide **4073**, signal output waveguide **4075**, idler output waveguide **4079**, pump input coupler **4032**, signal input coupler **4034**, pump output coupler **4033**, signal output coupler **4035**, first phase shifter **4061**, second phase shifter **4062**, third phase shifter **4063**, fourth phase shifter **4064**, fifth phase shifter **4065**, and sixth phase shifter **4066**.

(652) In this illustrative example, nonlinear optical waveguide **4009** has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment **4001**, second segment **4002**, third segment **4003**, fourth segment **4004**, fifth segment **4005**, sixth segment **4006**, and seventh segment **4007**, with seventh segment **4007** also serving as a final segment.

(653) In this example, pump input coupler **4032** couples pump light **4012** introduced through pump input waveguide **4072** to first segment **4001**. Signal input coupler **4034** couples signal light **4014** introduced in signal input waveguide **4074** to first segment **4001**. Pump output coupler **4033** couples pump light **4012** from sixth segment **4006** to pump output waveguide **4073** so that pump light **4012** is not coupled into seventh segment **4007**. Signal output coupler **4035** couples signal light **4014** from seventh segment **4007** to signal output waveguide **4075**. Pump output coupler **4033** and signal output coupler **4035** are both located prior to combiner **4080** which is located at the end of seventh segment **4007**. Combiner **4080** combines idler light **4017** from seventh segment **4007** and idler light **4018** from sixth tuning optical waveguide **4046** to form idler light **4016**. Combiner **4080** can couple idler light **4016** to idler output waveguide **4079**.

(654) In the illustrative example, materials such as z-cut lithium niobate and c-axis aligned 4H—SiC used for nonlinear optical generation involving TM polarized light in optical waveguide structure **4000** do not have the geometric constraints imposed by x-cut lithium niobate. With these types of materials, the phase shifters in optical waveguide structure **4000** can be aligned in any direction along the xy plane, which is perpendicular to the z-axis **4093**.

(655) Thus, the length of the segments in nonlinear optical waveguide **4009** do not limit the length of the phase shifters associated with the tuning optical waveguides as depicted for optical waveguide structure **4000**. Also, a phase shifter in a tuning optical waveguide does not need to have a straight path that is aligned along a particular crystallographic direction, unlike the phase shifters used in the optical waveguide structures depicted in FIG. 35, FIG. 38, and FIG. 39. Instead, a circular path as depicted in FIG. 40 for the phase shifters in optical waveguide structure **4000** can be used. The multiple phase tuning paths and phase shifters in such a nonlinear optical waveguide structure each could have a shape resembling the Greek letter capital omega, with the “feet” of the omega being the out wavelength selective coupler and in wavelength selective coupler for that tuning optical waveguide. In this example, the length of a segment is shorter than the length of a tuning optical waveguide and even is shorter than the length of a phase shifter.

(656) Thus, the configuration of optical waveguide structure **4000** can potentially provide compensation for a greater tolerance in the waveguide cross-sectional dimensions. This configuration also can be scaled readily to increasingly larger numbers of segments.

(657) Further, other shapes can be used in addition to or in place of the omega shape shown for the tuning optical waveguides in optical

waveguide structure **4000**. For example, even a meandering or irregular shape can be used for the tuning optical waveguides in addition to the omega. Additionally, the tuning optical waveguides in optical waveguide structure **4000** can have different shapes from each other.

(658) With reference next to FIGS. **41A** and **41B**, illustrations of phase shifter cross sections is depicted in accordance with an illustrative embodiment. Cross-sections **4100** can be used for shifting the phase of light traveling or propagating in optical waveguide structures using z-cut lithium niobate or c-axis aligned 4H—SiC, such as optical waveguide structure **4000** in FIG. **40**. For TM polarized light in z-cut lithium niobate or c-axis aligned 4H—SiC, one of the electrodes for an electro-optic phase shifter should be located above the core region of the optical waveguide such as shown in cross-section **4102** and cross-section **4104**.

(659) Cross-section **4102** (FIG. **41A**) is for an electro-optic (EO) phase shifter **4105** that is representative of phase shifters such as first phase shifter **4061**, second phase shifter **4062**, third phase shifter **4063**, fourth phase shifter **4064**, fifth phase shifter **4065**, and sixth phase shifter **4066**, that can be associated with tuning optical waveguides such as first tuning optical waveguide **4041**, second tuning optical waveguide **4042**, third tuning optical waveguide **4043**, fourth tuning optical waveguide **4044**, fifth tuning optical waveguide **4045**, and sixth tuning optical waveguide **4046**, respectively. As depicted, first tuning optical waveguide **4041**, second tuning optical waveguide **4042**, third tuning optical waveguide **4043**, fourth tuning optical waveguide **4044**, fifth tuning optical waveguide **4045** or sixth tuning optical waveguide **4046** comprises lithium niobate core region **4108** with silicon nitride rib structure **4110**. Electro-optic phase shifter **4105** also comprises side electrode **4112**, side electrode **4114**, and top electrode **4116**. These components are located within cladding **4118** which is located on silicon substrate **4120**.

(660) In this example, cross-section **4104** (FIG. **41B**) is for an electro-optic (EO) phase shifter **4125** that can be associated with a segment in a nonlinear optical waveguide. As depicted, the nonlinear optical waveguide comprises core region **4128** that is comprised of lithium niobate center section **4130** with silicon nitride sides **4131**. In this cross section, electro-optic phase shifter **4125** comprises side electrode **4132**, side electrode **4134**, and top electrode **4136**. These components are located within cladding **4138** which is located on silicon substrate **4140**.

(661) In this illustrative example, cross-section **4102** and cross-section **4104** have electrode arrangements that provide increased values for the electro-optic (EO) coefficient of the material to be used as compared to other arrangements of electrodes. An example of an electro-optic coefficient is the $r_{sub.33}$ coefficient of lithium niobate.

(662) With reference to FIG. **42**, an illustration of an optical waveguide structure is depicted in accordance with an illustrative embodiment. As depicted, optical waveguide structure **4200** comprises a number of different components. As depicted, optical waveguide structure **4200** comprises nonlinear optical waveguide **4209**, first tuning optical waveguide **4241**, second tuning optical waveguide **4242**, third tuning optical waveguide **4243**, fourth tuning optical waveguide **4244**, first wavelength selective coupler **4221**, second wavelength selective coupler **4222**, third wavelength selective coupler **4223**, fourth wavelength selective coupler **4224**, fifth wavelength selective coupler **4225**, sixth wavelength selective coupler **4226**, seventh wavelength selective coupler **4227**, combiner **4280**, pump input waveguide **4272**, optional signal input waveguide, optional idler input waveguide, pump output waveguide **4273**, signal output waveguide **4275**, idler output waveguide **4279**, pump input coupler **4232**, optional signal input coupler **4234**, optional idler input coupler, pump output coupler **4233**, signal output coupler **4235**, idler output coupler **4239**, first phase shifter **4261**, second phase shifter **4262**, third phase shifter **4263**, fourth phase shifter **4264**, and fifth phase shifter **4265**. For spontaneous parametric down conversion, only pump light **4212** is supplied as the source light. Both signal light **4214** and idler light **4216** are generated by the nonlinear process occurring in optical waveguide structure **4200**. For a parametric difference frequency generation or sum frequency generation process, signal light **4214** is supplied as an auxiliary source light with idler light **4216** generated by the nonlinear optical process or, alternatively, idler light **4216** is supplied as an auxiliary source light with signal light **4214** generated by the nonlinear optical process.

(663) In this illustrative example, nonlinear optical waveguide **4209** has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment **4201**, second segment **4202**, third segment **4203**, fourth segment **4204**, and fifth segment **4205**.

(664) In this example, pump input coupler **4232** couples pump light **4212** introduced through pump input waveguide **4272** to first segment **4201**. Optional signal input coupler **4234** couples signal light **4214** introduced in optional signal input waveguide to first segment **4201**. Optional idler input coupler couples idler light **4217** introduced in optional idler input waveguide to first segment **4201**. Pump output coupler **4233** couples pump light **4212** from fourth segment **4204** to pump output waveguide **4273** so that pump light **4212** is not coupled into fifth segment **4205**. Signal output coupler **4235** couples signal light **4214** from fifth segment **4205** to signal output waveguide **4275**. Idler output coupler **4239** couples idler light **4216** from fifth segment **4205** to idler output waveguide **4279**.

(665) As depicted, nonlinear optical waveguide **4209** and other components are formed on a yz plane defined by z-axis **4293** and y-axis **4292** in which an x-axis **4291** is perpendicular to the plane. In an illustrative example, nonlinear optical waveguides can be fabricated from a nonlinear optical material such as x-cut lithium niobate. In this example, x-axis **4291** of the nonlinear optical material is perpendicular to the yz plane of nonlinear optical waveguide **4209** and the other components.

(666) In optical waveguide structure **4200**, a first group of odd index segments comprises first segment **4201**, third segment **4203**, and fifth segment **4205**. A second group of even index segments comprises second segment **4202**, and fourth segment **4204**. As depicted, first tuning optical waveguide **4241**, second tuning optical waveguide **4242** are tuning optical waveguides for signal light **4215** and idler light **4217** traveling through the odd index segments. Second tuning optical waveguide **4242** and fourth tuning optical waveguide **4244** are tuning optical waveguides for signal light **4213** and idler light **4218** traveling through the even index segments.

(667) Each of these groups of segments and tuning optical waveguides forms a route. As a result, two routes are present through which signal light **4215** and **4213** and idler light **4217** and **4218** travel within optical waveguide structure **4200**. First route **4251** comprises odd index segments and the associated optical tuning waveguides. In this example, first route **4251** comprises first segment **4201**, third segment **4203**, fifth segment **4205**, first tuning optical waveguide **4241**, and third tuning optical waveguide **4243**. A combination of first tuning optical waveguide **4241** and third tuning optical waveguide **4243** can be, for example, first set of tuning optical waveguides **2941** in FIG. **29**. Second route **4252** comprises even index segments and the associated optical tuning waveguides. In this example, second route **4252** comprises second segment **4202**, fourth segment **4204**, second tuning optical waveguide **4242**, and fourth tuning optical waveguide **4244**. A combination of second tuning optical waveguide **4242** and fourth tuning optical waveguide **4244** can be, for example, second set of tuning optical waveguides **2942** in FIG. **29**.

(668) Optical waveguide structure **4200** also has tuning optical waveguides that are separate from the nonlinear optical waveguide **4209**. Idler light **4217** and signal light **4215** is routed at a first wavelength selective coupler **4221** out from nonlinear optical waveguide **4209** and then is routed back into nonlinear optical waveguide **4209** at a subsequent second wavelength selective coupler. The first segment **4201** of nonlinear optical waveguide **4209** precedes the first out-coupling point, which is first wavelength selective coupler **4221**. At the first out-coupling point, idler light **4217** and signal light **4215** generated by the nonlinear optical process in the first segment **4201** is diverted into

first tuning optical waveguide 4241 in which phases of idler light 4217 and signal light 4215 are adjusted. At the first in-coupling point, which is second wavelength selective coupler 4222, idler light 4217 and signal light 4215 from first tuning optical waveguide 4241 are routed back into nonlinear optical waveguide 4209 and into third segment 4203. The phases of the diverted idler light and signal light are adjusted by the electrically controlled tuning performed using first phase shifter 4261 to achieve at and after the first in-coupling point a constructive nonlinear optical interaction between the first generated light from the phase tuning path and the source light, which continues to propagate in the nonlinear optical waveguide 4209. As a result, additional generated idler light 4217 and signal light 4215 will continue to be produced in the portion of nonlinear optical waveguide 4209 after the first in-coupling point. This portion of nonlinear optical waveguide 4209 is the third segment 4203.

(669) Similarly, idler light 4218 and signal light 4213 generated in second segment 4202 of nonlinear optical waveguide 4209 can be diverted to second tuning optical waveguide 4242 separate from nonlinear optical waveguide 4209 at a second out-coupling point which is third wavelength selective coupler 4223 and then routed back into nonlinear optical waveguide 4209 at a subsequent second in-coupling point which is fourth wavelength selective coupler 4224. Second segment 4202 of the nonlinear optical waveguide 4209 precedes the second out-coupling point. At the second out-coupling point, idler light 4218 and signal light 4213 generated by the nonlinear optical process in second segment 4202 are routed into second tuning optical waveguide 4242 in which the phases of idler light 4218 and signal light 4213 are adjusted. At the second in-coupling point, idler light 4218 and signal light 4213 are coupled from the second tuning optical waveguide 4242 back into nonlinear optical waveguide 4209, into fourth segment 4204. The phases of idler light 4218 and signal light 4213 are adjusted by the electrically controlled tuning by second phase shifter 4262 to achieve at and after the second in-coupling point a constructive nonlinear optical interaction between the second generated light from second tuning optical waveguide 4242 and pump light 4212, which continues to propagate in the nonlinear optical waveguide 4209. As a result, idler light 4218 and signal light 4213 will continue to be produced in the portion of nonlinear optical waveguide 4209 after the second in-coupling point. This portion of nonlinear optical waveguide 4209 is fourth segment 4204.

(670) Optical waveguide structure 4200 can be used in a spontaneous parametric down conversion (SPDC) process to generate both signal light 4215 and signal light 4213 both at a signal wavelength and idler light 4217 and idler light 4218 both at an idler wavelength from source light, such as pump light 4212 at a pump wavelength. Pump light 4212 is supplied to nonlinear optical waveguide 4209 from pump input coupler 4232 and continues to propagate in nonlinear optical waveguide 4209 and is not diverted into the tuning optical waveguides. The generated signal and idler light generated prior to the first out-coupling point (i.e., generated in first segment 4201) is diverted into first tuning optical waveguide 4241. Since pump light 4212 still travels in nonlinear optical waveguide 4209, additional signal light and idler light is generated in the portion of nonlinear optical waveguide 4209 between the first out-coupling point and the first in-coupling point (i.e., in second segment 4202). The generated signal and idler light generated in the second segment 4202 is diverted into second tuning optical waveguide 4242 at the second out-coupling point. Again, since the pump light still travels in the NLO waveguide, additional signal and idler is generated in the portion of nonlinear optical waveguide 4209 between the second out-coupling point and the second in-coupling point (i.e., in third segment 4203). The generated signal and idler light generated in third segment 4203 is diverted into third tuning optical waveguide 4243 at the third out-coupling point. Since the pump light still travels in nonlinear optical waveguide 4209, additional signal and idler is generated in the portion of NLO waveguide between the third out-coupling point and the third in-coupling point (i.e., in fourth segment 4204). The generated signal light and idler light generated in the fourth segment 4204 is diverted into fourth tuning optical waveguide 4244 at the fourth out-coupling point.

(671) In this illustrative example, combiner 4280 combines signal light 4213 and signal light 4415 and also combines idler light 4218 and idler light 4417 together after the end of fifth segment 4205 to produce signal light 4214 and idler light 4216, respectively.

(672) Pump light 4212 is removed from nonlinear optical waveguide 4209 by a pump output coupler 4233 after the fourth out-coupling point. Thus, no additional signal and idler is generated in the portion of nonlinear optical waveguide 4209 between the fourth out-coupling point and the fourth in-coupling point (i.e., in fifth segment 4205). Combiner 4280 is located at the fourth in-coupling point. This optical combiner combines the signal light 4213 and idler light 4218 from fourth tuning optical waveguide 4244 with signal light 4215 and idler light 4217 from the fifth segment 4205. Signal light 4214 and idler light 4216 produced by this optical combiner is the desired output of nonlinear optical waveguide 4209. Additional at least one of the signal output coupler or idler output coupler can be used to selectively extract at least one of the generated signal light or the generated idler light.

(673) Thus, in segments with an odd index contribute constructively to produce even more generated light. The odd-indexed third tuning optical waveguides enable the generation of light in the odd-indexed segments to accumulate constructively from one odd-indexed segment to the next. Thus, the effective length of the nonlinear optical interaction can be equivalent to the total combined length of the multiple odd-indexed segments. Similarly, light generated in those segments with an even index can contribute constructively to produce even more generated light. The even-indexed phase tuning paths enable the generation of light in the even indexed segments to accumulate constructively from one even index segment to the next even index segment. Thus, the effective length of the nonlinear optical interaction can be equivalent to the total combined length of the multiple even indexed segments.

(674) When the wavelengths of the idler light 4217 and idler light 4218 and the wavelength of the signal light 4215 and 4213 are sufficiently close to each other, the same wavelength selective coupler can couple both the idler light and the signal light from a nonlinear optical waveguide segment to a tuning optical waveguide. Likewise, the same wavelength selective coupler can couple both the idler light and the signal light from a tuning optical waveguide to a nonlinear optical waveguide segment. Furthermore, the same phase shifter can shift the phase of the idler light and the phase of the signal light such that a desired phase walk-off is achieved. However, when the wavelength of the idler light and the signal light are not sufficiently close together, different wavelength selective coupler must be used to couple the signal light and to couple the idler light. Furthermore, different phase shifters must be used to adjust the phases of the signal light and of the idler light.

(675) With reference now to FIG. 43, an illustration of a nonlinear optical waveguide structure with separate tuning optical waveguides for signal light and for idler light is depicted in accordance with an illustrative embodiment. In this example, signal light 4314 and idler light 4316 can travel through optical waveguide structure 4300, using separate tuning optical waveguides. Optical waveguide structure 4300 may be especially useful for a nonlinear optical process such as spontaneous parametric down conversion (SPDC) in which both signal light 4314 and idler light 4316 are generated from source light comprising pump light 4312.

(676) As depicted, optical waveguide structure 4300 comprises nonlinear optical waveguide 4309, first idler tuning optical waveguide 4341A, first signal tuning optical waveguide 4341B, second idler tuning optical waveguide 4342A, second signal tuning optical waveguide 4342B, third idler tuning optical waveguide 4343A, third signal tuning optical waveguide 4343B, fourth idler tuning optical waveguide 4344A, fourth signal tuning optical waveguide 4344B, first idler out wavelength selective coupler 4321A, first signal out wavelength selective coupler 4321B, second idler in second idler wavelength selective coupler 4322A, second signal in second signal wavelength

selective coupler **4322B**, third idler out wavelength selective coupler **4323A**, third signal out wavelength selective coupler **4323B**, fourth idler in wavelength selective coupler **4324A**, fourth signal in fourth signal wavelength selective coupler **4324B**, fifth idler out wavelength selective coupler **4325A**, fifth signal out wavelength selective coupler **4325B**, sixth idler in wavelength selective coupler **4336A**, sixth signal in wavelength selective coupler **4336B**, seventh idler out wavelength selective coupler **4357A**, seventh signal out wavelength selective coupler **4357B**, idler combiner **4380A**, signal combiner **4380B**, pump input waveguide **4372**, pump output waveguide **4373**, signal output waveguide **4375**, idler output waveguide **4379**, pump input coupler **4332**, pump output coupler **4333**, signal output coupler **4335**, idler output coupler **4339**, first idler phase shifter **4361A**, first signal phase shifter **4361B**, second idler phase shifter **4362A**, second signal phase shifter **4362B**, third idler phase shifter **4363A**, third signal phase shifter **4363B**, fourth idler phase shifter **4364A**, fourth signal phase shifter **4364B**, and optional fifth phase shifter **4365**.

(677) In this illustrative example, nonlinear optical waveguide **4309**, has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment **4301**, second segment **4302**, third segment **4303**, fourth segment **4304**, fifth segment **4305**, and an optional output segment **4306**.

(678) In this example, pump input coupler **4332** couples pump light **4312** introduced through pump input waveguide **4372** to first segment **4301**. Pump output coupler **4333** couples pump light **4312** from fourth segment **4304** to pump output waveguide **4373** so that no pump light **4312** travels through fifth segment **4305**. Signal output coupler **4335** couples signal light **4314** from optional output segment **4306** to signal output waveguide **4375**. Idler output coupler **4339** couples idler light **4316** from optional output segment **4306** to idler output waveguide **4379**.

(679) As depicted, nonlinear optical waveguide **4309** and other components are formed on a yz plane defined by z-axis **4393** and y-axis **4392** in which an x-axis **4391** is perpendicular to the plane. In an illustrative example, the nonlinear optical waveguides can be fabricated from a nonlinear optical material such as x-cut lithium niobate. In this example, x-axis **4391** of the nonlinear optical material is perpendicular to the yz plane of the structure of nonlinear optical waveguide **4309** and the other components.

(680) In this illustrative example, idler light **4317** and **4318** and signal light **4315** and signal light **4313** are generated in the different segments in nonlinear optical waveguide **4309**. The generated idler light and signal light are obtained from the output end of nonlinear optical waveguide **4309** in optical waveguide structure **4300**.

(681) Also, separate sets of wavelength selective couplers extract idler light and signal light from nonlinear optical waveguide segments into separate tuning optical waveguides and reinsert the idler light and signal light in the separate tuning optical waveguides back into the nonlinear optical waveguide segments. In other words, both idler light and signal travel in routes formed by the odd index segments and the even index segments but the phase tuning is performed separately for idler light and signal light by the phase shifters associated with the tuning optical waveguides through which idler light and signal light are travelling separately.

(682) The portion of nonlinear optical waveguide **4309** between the pump input coupler **4332** and first idler out wavelength selective coupler **4321A** defines first segment **4301**. Idler light **4317** generated in first segment **4301** is diverted into first idler tuning optical waveguide **4341A** by the first idler out wavelength selective coupler **4321A**. This wavelength selective coupler diverts idler light **4317** into first idler tuning optical waveguide **4341A** and couples pump light **4312** and signal light **4314** to first signal out wavelength selective coupler **4321B**. Signal light **4315** generated in first segment **4301** is diverted into first signal tuning optical waveguide **4341B** by the first signal out wavelength selective coupler **4321B**. This wavelength selective coupler diverts signal light **4315** into first signal tuning optical waveguide **4341B** but couples pump light **4312** into second segment **4302** in nonlinear optical waveguide **4309**.

(683) Thus, idler light **4317** and signal light **4315** are coupled from first segment **4301** into separate tuning waveguides to be re-inserted into third segment **4303**. This process is repeated for each odd index segment with separate tuning optical waveguides for idler light **4317** and signal light **4315**.

(684) In an illustrative implementation of optical waveguide structure **4300**, such as depicted in signal light **4214** and idler light **4216**, idler light **4317** has a longer wavelength than signal light **4315**. Both idler light **4317** and signal light **4315** propagate as the fundamental mode in nonlinear optical waveguide **4309**. For the example, an idler out wavelength selective coupler, such as first idler out wavelength selective coupler **4321A**, is located immediately before a signal out wavelength selective coupler, such as first signal out wavelength selective coupler **4321B**. Also, a signal in wavelength selective coupler, such as second signal wavelength selective coupler **4322B**, is located immediately before an idler in wavelength selective coupler, such as second idler wavelength selective coupler **4322A**. However, other arrangements of the out wavelength selective couplers and in wavelength selective couplers for idler light **4317** and signal light **4315** can be used in other implementations.

(685) For a given tuning optical waveguide in optical waveguide structure **4300**, the phase shifters for signal light **4315** can be controlled separately from the phase shifters for idler light **4317** with the depicted configuration using separate tuning optical waveguides and phase shifters for idler light **4317** and signal light **4315**. Thus, the phase of the signal light re-inserted into the subsequent segment, such as third segment **4303**, in nonlinear optical waveguide **4309** can be adjusted to have that signal light interact constructively to produce additional signal light newly generated in that segment. The phase of the idler light re-inserted into the subsequent segment, such as third segment **4303**, can be adjusted to have that idler light interact constructively to produce additional idler light newly generated in that segment.

(686) As further depicted in FIG. **43**, the portion of nonlinear optical waveguide **4309** between first signal out wavelength selective coupler **4321B** and third idler out wavelength selective coupler **4323A** defines second segment **4302**. Idler light **4318** generated in second segment **4302** is diverted into second idler tuning optical waveguide **4342A** by the third idler out wavelength selective coupler **4323A**. This wavelength selective coupler diverts idler light **4318** into second idler tuning optical waveguide **4342A** and couples pump light **4312** and signal light **4314** to third signal out wavelength selective coupler **4323B**. Signal light **4313** generated in second segment **4302** is diverted into second signal tuning optical waveguide **4342B** by the third signal out wavelength selective coupler **4323B**. This wavelength selective coupler diverts signal light **4313** into second signal tuning optical waveguide **4342B** but couples pump light **4312** into third segment **4303** in nonlinear optical waveguide **4309**.

(687) Thus, idler light **4318** and signal light **4313** are coupled from second segment **4302** into separate tuning waveguides and to be re-inserted into fourth segment **4304**. This process is repeated for each even index segment with separate tuning optical waveguides for idler light **4318** and signal light **4313**.

(688) As depicted in this illustrative implementation, idler light **4318** has a longer wavelength than signal light **4313**. Both idler light **4318** and signal light **4313** propagate as the fundamental mode in nonlinear optical waveguide **4309**. For the example, an idler out wavelength selective coupler, such as third idler out wavelength selective coupler **4323A**, is located immediately before a signal out wavelength selective coupler, such as third signal wavelength selective coupler **4323B**. Also, a signal in wavelength selective coupler, such as fourth signal wavelength selective coupler **4324B**, is located immediately before an idler in wavelength selective coupler, such as fourth idler wavelength selective coupler **4324A**. However, other arrangements of the out wavelength selective couplers and in wavelength selective

couplers for idler light **4318** and signal light **4313** can be used in other implementations.

(689) For a given tuning optical waveguide in optical waveguide structure **4300**, the phase shifters for signal light **4313** can be controlled separately from the phase shifters for idler light **4318** with the depicted configuration using separate tuning optical waveguides and phase shifters for idler light **4318** and signal light **4313**. Thus, the phase of the signal light re-inserted into the subsequent segment, such as fourth segment **4304**, in nonlinear optical waveguide **4309** can be adjusted to have that signal light interact constructively to produce additional signal light newly generated in that segment. The phase of the idler light re-inserted into the subsequent segment, such as fourth segment **4304**, can be adjusted to have that idler light interact constructively to produce additional idler light newly generated in that segment.

(690) In optical waveguide structure **4300**, the adjustment of the idler phase for idler light **4317** or idler light **4318** can be controlled separately from the adjustment of the signal phase for signal light **4315** or signal light **4313**. The tuning optical waveguides in the output portion of optical waveguide structure **4300**, such as third idler tuning optical waveguides **4343A** and third signal tuning optical waveguide **4343B** as well as fourth idler tuning optical waveguides **4344A** and fourth signal tuning optical waveguide **4344B** have separate phase shifters for signal light **4315** and for idler light **4317** traveling separately through those tuning optical waveguides, **4343A** and **4343B**, and also have separate phase shifters for signal light **4313** and for idler light **4318** traveling separately through fourth idler tuning optical waveguides **4344A** and fourth signal tuning optical waveguide **4344B**. Thus, the phases of signal light **4315** and signal light **4313** in the two routes can be adjusted to achieve constructive interference at signal combiner **4380B**. Also, the phases of idler light **4317** and **4318** in the two routes can be adjusted separately to achieve constructive interference at idler combiner **4380A**.

(691) The material used to fabricate optical waveguide structure **4300** can result in constraints on the lengths of the segments and the phase shifters associated with tuning optical waveguides. For example, materials such as x-cut lithium niobate for which the electro optical effect and the nonlinear optical coefficient is much stronger for TE polarized light traveling in a particular direction with respect to the crystal axes of the material result in constraints such as the phase shifters and nonlinear optical waveguide **4309** needing to be aligned parallel with each other and with the nonlinear optical waveguide **4309** aligned parallel to the material y-axis **4392**. With phase shifters used in the examples such as the optical waveguide structures depicted in FIG. 35, FIG. 38, FIG. 39, FIG. 42 and FIG. 43, the phase shifters and the nonlinear optical waveguides are aligned parallel to the Y-axis of the exemplary lithium niobate nonlinear optical material. For these examples, the tuning optical waveguides for one group of segments (e.g., odd index) are located on one side of the nonlinear optical waveguide, such as nonlinear optical waveguide **4309**, and the tuning optical waveguides of the other group segments (e.g., even index segments) are located on the other side of nonlinear optical waveguide. Such an arrangement provides more space for longer phase shifters that use materials in which both the electro optical (EO) effect and the nonlinear optical (NLO) coefficient are much stronger for TE polarized light traveling in a particular direction with respect to the crystal axes of the material, such that the phase shifters are preferably oriented parallel to the nonlinear optical waveguide.

(692) Thus, the illustrative examples described in FIGS. 34 through 43 describe optical waveguide structures that have tuning optical waveguides that can be used to adjust the phase of light traveling through those tuning optical waveguides. In these examples, phase shifters are used to adjust the phase of light traveling through the tuning optical waveguides to obtain desired generation of light within the different optical waveguide structures.

(693) Optical waveguide structure **4300** can be compared with optical waveguide structure **3400**. Optical waveguide structure **3400** has one group of nonlinear optical waveguide segments and has separate tuning optical waveguides for the idler light and the signal light generated in each segment of the nonlinear optical waveguide. For the idler light and separately for the signal light in optical waveguide structure **3400**, the same wavelength selective coupler, rather than two separate couplers, performs the extraction from a nonlinear optical waveguide segment into a tuning optical waveguide and the insertion from that tuning optical waveguide back into another nonlinear optical waveguide segment. Since only one group of segments is present in the example of FIG. 34, optical waveguide structure **3400** does not need to have a combiner located near the output of that structure, which can be used to combine the generated light from an odd index group and an even index group of segments. In contrast, optical waveguide structure **4300** has two groups of nonlinear optical waveguide segments that are arranged into an odd index group and an even index group. Furthermore, for the idler light in each group, and separately for the signal light in each group, a wavelength selective coupler is used to out couple the idler light (or the signal light) from a nonlinear optical waveguide segment to a tuning optical waveguide for the idler light (or for the signal light); and a different wavelength selective coupler is used to in couple the idler light (or the signal light) back from the tuning optical waveguide into another nonlinear optical waveguide segment. The presence of two different groups of nonlinear optical waveguide segments that generate different components of idler light (or signal light) is addressed by the combiner that combines those two components of idler light (or signal light). The maximum overall interaction distance for nonlinear optical generation is given by the total length of the segments in a group. Thus, if both optical waveguide structure **3400** and optical waveguide structure **4300** have 7 nonlinear optical waveguide segments each of 1 mm length, for example, and if both structures achieve the desired optimal phase walk-off at the start of each segment, optical waveguide structure **3400** could have higher nonlinear optical generation efficiency than optical waveguide structure **4300**.

(694) With reference to FIG. 44, an illustration of a nonlinear optical waveguide with two groups of segments in which only one group of segments is associated with tuning optical waveguides is depicted in accordance with an illustrative embodiment. In this illustrative example, optical waveguide structure **4400** is an open-ended nonlinear optical waveguide structure.

(695) In this example, optical waveguide structure **4400** comprises a number of different components. As depicted, optical waveguide structure **4400** comprises nonlinear optical waveguide **4409**, first idler tuning optical waveguide **4441A**, first signal tuning optical waveguide **4441B**, third idler tuning optical waveguide **4443A**, third signal tuning optical waveguide **4443B**, first idler out wavelength selective coupler **4421A**, second idler in wavelength selective coupler **4422A**, first signal out wavelength selective coupler **4421B**, second signal in wavelength selective coupler **4422B**, fifth idler out wavelength selective coupler **4425A**, sixth idler in wavelength selective coupler **4426A**, fifth signal out wavelength selective coupler **4425B**, sixth signal in wavelength selective coupler **4426B**, pump input waveguide **4472**, pump output waveguide **4473**, signal output waveguide **4475**, idler output waveguide **4479**, pump input coupler **4432**, pump output coupler **4433**, signal output coupler **4435**, idler output coupler **4439**, first idler phase shifter **4461A**, first signal phase shifter **4461B**, third idler phase shifter **4463A**, and third signal phase shifter **4463B**.

(696) In this illustrative example, nonlinear optical waveguide **4409**, has nonlinear optical waveguide segments. As depicted, these nonlinear optical waveguide segments are first segment **4401**, second segment **4402**, third segment **4403**, fourth segment **4404**, and fifth segment **4405**.

(697) As depicted, nonlinear optical waveguide **4409** and other components are formed on a yz plane defined by z-axis **4493** and y-axis **4492** in which an x-axis **4491** perpendicular to the plane. In an illustrative example, nonlinear optical waveguides can be fabricated from a nonlinear optical material such as x-cut lithium niobate. In this example, the x-axis of the nonlinear optical material is perpendicular to the yz plane of the structure of nonlinear optical waveguide **4409** and the other components.

(698) In this example, pump input coupler **4432** couples pump light **4412** introduced through pump input waveguide **4472** to first segment **4401**. Pump output coupler **4433** couples pump light **4412** from fifth segment **4405** to pump output waveguide **4473**. Signal input coupler couples signal light **4415** introduced through signal input waveguide to first segment **4401**. Signal output coupler **4435** couples signal light **4415** from fifth segment **4405** to signal output waveguide **4475**. Idler output coupler **4439** couples idler light **4417** from fifth segment **4405** to idler output waveguide **4479**.

(699) In this illustrative example, first idler out wavelength selective coupler **4421A** extracts idler light **4417** from first segment **4401** into first idler tuning optical waveguide **4441A**. Second idler in wavelength selective coupler **4422A** reinserts idler light **4417** into third segment **4403** after activations have been applied using first idler phase shifter **4461A** associated with first idler tuning optical waveguide **4441A**. Each phase shifter is comprised of three pairs of electrodes in this example. First signal out wavelength selective coupler **4421B** extracts signal light **4415** from first segment **4401** into first signal tuning optical waveguide **4441B** and second signal in wavelength selective coupler **4422B** reinserts signal light **4415** into third segment **4403** after activations have been applied using first signal phase shifter **4461B** associated with first signal tuning optical waveguide **4441B**.

(700) In this illustrative example, third idler out wavelength selective coupler **4425A** extracts idler light **4417** from third segment **4403** into third idler tuning optical waveguide **4443A**. Sixth idler in wavelength selective coupler **4426A** reinserts idler light **4417** into fifth segment **4405** after activations have been applied using third idler phase shifter **4463A** associated with third idler tuning optical waveguide **4443A**. Fifth signal out wavelength selective coupler **4425B** extracts signal light **4415** from third segment **4403** into third signal tuning optical waveguide **4443B**, and sixth signal in wavelength selective coupler **4426B** reinserts signal light **4415** into fifth segment **4405** after activations have been applied using third signal phase shifter **4463B** associated with third signal tuning optical waveguide **4443B**.

(701) In this example, even index segments are second segment **4402** and fourth segment **4404**. Odd index segments are first segment **4401**, third segment **4403**, and fifth segment **4405**. The even index segments are not associated with a tuning optical waveguide. In this example, odd index segments are associated with tuning optical waveguides.

(702) Further, separate tuning optical waveguides are present for idler light **4417** and for signal light **4415**. In other words, each tuning optical waveguide is used for tuning either idler light **4417** or signal light **4415** in this example.

(703) Each of these groups of segments and tuning optical waveguides forms a route. As a result, idler route **4456** is present through which idler light **4417** travels within optical waveguide structure **4400**. Idler route **4456** comprises odd index segments, first segment **4401**, third segment **4403** and fifth segment **4405**, and the associated idler tuning optical waveguides, first idler tuning optical waveguide **4441A** and third idler tuning optical waveguide **4443A** for idler light **4417**. Signal route **4454** comprises odd index segments, first segment **4401**, third segment **4403** and fifth segment **4405**, and the associated signal tuning optical waveguides, first signal tuning optical waveguide **4441B** and third signal tuning optical waveguide **4443B**, for signal light **4415**.

(704) In optical waveguide structure **4400**, each tuning optical waveguide for idler light **4417** begins at an idler out wavelength selective coupler located at the end of the associated odd index segment. This wavelength selective coupler is designed to selectively couple idler light **4417** out of the odd index segment into a tuning optical waveguide. Each tuning optical waveguide for idler light **4417** ends at an idler in wavelength selective coupler located at the end of the even index segment that follows immediately after the associated odd index segment. This wavelength selective coupler is designed to selectively couple idler light **4417** out of the tuning optical waveguide and reinsert this light into the next odd index segment.

(705) Each tuning optical waveguide for signal light **4415** begins at a signal out wavelength selective coupler located at the end of the associated odd index segment. Each tuning optical waveguide for signal light **4415** ends at a signal in wavelength selective coupler located at the end of the even index segment that follows immediately after the associated odd index segment. For this example, the idler out wavelength selective coupler is located immediately before the signal out wavelength selective coupler. Also, the idler in wavelength selective coupler is located immediately after the signal in wavelength selective coupler.

(706) In the example of optical waveguide structure **4400**, idler light **4417** is generated from nonlinear optical interaction occurring in odd index segments such as first segment **4401** and third segment **4403**. Signal light **4415** is generated from nonlinear optical interaction occurring in odd index segments such as first segment **4401** and third segment **4403**. Besides the light generated in the odd index segments, idler light **4418** can be generated in even index segments such as second segment **4402** and fourth segment **4404**. Also, signal light **4413** can be generated from nonlinear optical interaction occurring in odd index segments such as first segment **4401** and third segment **4403**.

(707) Optical waveguide structure **4400** uses the idler in wavelength selective coupler at the end of each tuning optical waveguide to also couple out and remove idler light **4418** that is generated in the immediately preceding even index segment. This removal of idler light **4418** generated in the even index segments prevents idler light **4418** generated in the even index segments from interfering destructively with idler light **4417** coupled into next odd index segment and also from interacting destructively to reduce or reverse the generation of idler light **4417** in the next odd index segment.

(708) Similarly, the signal in wavelength selective coupler at the end of each tuning optical waveguide also couples out and removes signal light **4413** that is generated in the even index segment. This removal of signal light **4413** generated in the even index segments prevents those generated signal light from interfering destructively with signal light **4415** coupled into the next odd index waveguide segment and also from interacting destructively to reduce or reverse the generation of signal light **4415** in the next odd index segment.

(709) In optical waveguide structure **4400**, the length of each segment prior to a tuning optical waveguide can be selected to be sufficiently small that the magnitude of the phase walk-off resulting from anticipated fabrication and operational tolerances of optical waveguide structure **4400** is no greater than π radians. The phase shifters in a tuning optical waveguide can be configured to have an applicable length sufficiently large to achieve an electrically controlled phase shift of $\pm\pi$ radians. The overall length of a tuning optical waveguide for idler light **4417** is selected to achieve a relative phase shift that is 0 or a multiple of 2π radians between idler light **4417** coupled out of the tuning optical waveguide path and the newly generated idler light in the subsequent segment. Similarly, the overall length of a tuning optical waveguide for signal light **4415** can be selected to achieve a relative phase shift that is 0 or a multiple of 2π radians between signal light **4415** coupled out of the tuning optical waveguide and the newly generated signal light in the subsequent segment.

(710) The optical waveguide structure **4400** can be used to implement a parametric down conversion process, such as difference frequency generation or even spontaneous parametric down conversion, or to implement a parametric up conversion process, such as sum frequency generation and second harmonic generation. Consider, for example, a second order nonlinear optical parametric down conversion process that generates idler light from pump light and signal light supplied to the optical waveguide structure. The idler-wave amplitude $M_{\text{sub.i}}$ of the idler light **4417** in third segment **4403** can be described by

$$(711) \quad M_i(y = E_3) \approx \frac{2id_{33}}{n_i c} M_p M_s \int_{S_1}^{E_3} \exp(i(-s_1 + ky)) dy \approx M_i(S_3) + \frac{2id_{33}}{n_i c} M_p M_s \left(\frac{-i \times \exp(i(-s_3 + i \frac{kL_{S3E3}}{k}) + i \times \exp(i(-s_3))}{k} \right) \quad (1)$$

These approximate expressions assume there is negligible change in the amplitudes of the source pump light $M_{\text{sub.p}}$ and source signal light

M.sub.s, and that only the amplitude of the idler light changes as a result of the nonlinear optical interaction. The generation of the idler light over the length of third segment **4403** considers the nonlinear interactions that occur from the starting point S1 of first segment **4401** to the ending point E3 of the third segment **4403**. Since for this example, we assume the source pump light and source signal light are supplied to the nonlinear optical waveguide **4409** at the starting point S1 of the first segment. Thus, we can set the phase walk-off $\phi_{\text{sub.s1}}$ of the nonlinear optical process at starting point S1 to zero. In many cases, the phase matching in a nonlinear optical waveguide segment, such as third segment **4403**, is not perfect. Thus, there is a non-zero wave-vector mismatch $\Delta k = k_{\text{sub.p}} \pm k_{\text{sub.s}} - k_{\text{sub.i}}$ between the wave vectors for the pump light $k_{\text{sub.p}}$ and signal light $k_{\text{sub.s}}$ and the wave vector for the generated idler light $k_{\text{sub.i}}$ in a nonlinear optical waveguide segment. The expression for the wave-vector mismatch has a plus sign for an up-conversion, sum-frequency generation process and has a minus sign for a down-conversion, difference-frequency generation process. In an illustrative example, the length of a nonlinear optical waveguide segment, such as from a starting point S3 to an ending point E3 of third segment **4403** with length $L_{\text{sub.S3E3}}$, is chosen such that the phase walk-off for that segment, $\Delta k L_{\text{sub.S3E3}}$, has a value between zero and π radians.

(712) In an illustrative example, the generation of additional idler light in third segment **4403** can build upon the generation of idler light in first segment **4401**. This dependence on the idler light from first segment **4401** is shown in Expression 1 above. The term $M_{\text{sub.i}} (S_{\text{sub.3}})$ represents the amplitude of the idler wave at the starting point of third segment **4403**. To fully benefit from the idler light generated in a preceding segment of nonlinear optical waveguide and thus to have the generation of additional idler light in third segment **4403** be constructive with the generation of idler light in first segment **4401**, it is desirable to have the phase walk-off $\phi_{\text{sub.S3}}$ at the starting point of third segment **4403** equal zero or an even multiple of n radians (or a multiple of 2π radians). The phase walk-off at the starting point of third segment **4403** is given by:

$$\Phi_{\text{sub.S3}} \approx (k_{\text{sub.p}} - k_{\text{sub.i}} - k_{\text{sub.s}}) L_{\text{sub.S1E1}} + k_{\text{sub.p}} 2 L_{\text{sub.S2E2}} + \Delta \phi_{\text{sub.p2}} - k_{\text{sub.s}} T L_{\text{sub.T1s}} - \Delta \phi_{\text{sub.sT1}} - k_{\text{sub.i}} T L_{\text{sub.T1i}} - \Delta \phi_{\text{sub.iT1}} \quad (2)$$

Expression 2 is approximate and neglects the phase shifts for the pump light, signal light and idler light that may result from the wavelength selective couplers. Expression 2 illustrates that the phase walk-off at the starting point of third segment **4403** depends on the phase walk-off obtained at the ending point of first segment **4401**, as described by $\Phi_{\text{sub.E1}} = (k_{\text{sub.p}} - k_{\text{sub.i}} - k_{\text{sub.s}}) L_{\text{sub.S1E1}}$, assuming a down-conversion process. The pump light **4412** travels with wave vector $k_{\text{sub.p2}}$ through length $L_{\text{sub.S2E2}}$ of second segment **4402**. The signal light **4415** travels with wave vector $k_{\text{sub.sT}}$ through length $L_{\text{sub.T1s}}$ of first signal tuning optical waveguide **4441B**. The idler light **4417** from first segment **4401** travels with wave vector $k_{\text{sub.iT}}$ through length $L_{\text{sub.T1i}}$ of first idler tuning optical waveguide **4441A**.

(713) Phase shifter **4462** can apply an activation that produces a phase shift of $\Delta \phi_{\text{sub.p1}}$ to the pump light; phase shifter **4464** can apply an activation that produces a phase shift of $\Delta \phi_{\text{sub.p2}}$ to the pump light; first signal phase shifter **4461B** can apply an activation that produces a phase shift of $\Delta \phi_{\text{sub.sT1}}$ to the signal light; and first idler phase shifter **4461A** can apply an activation that produces a phase shift of $\Delta \phi_{\text{sub.iT1}}$ to the idler light. For optical waveguide structure **4400**, the pump light, signal light, and idler light are affected by different phase shifters. Thus, it is possible to apply electro-optic activations to those phase shifters such that the phase shift of the pump light has an opposite sign from the phase shift for the idler light. This form of push-pull control would not be possible if the same phase shifter were to apply an activation that affects both the pump light and the idler light, for example.

(714) In this example, a pump-inserting coupler, pump input coupler **4432**, is located at the start of first segment **4401** and the location of this coupler defines the start of the nonlinear optical interaction. A pump-extracting coupler, signal output coupler **4435**, is located at the end of the final segment, fifth segment **4405**, and the location of this coupler defines the end of the nonlinear optical interaction in this open-ended nonlinear optical waveguide structure. An idler output coupler can be located subsequent to the pump-extracting coupler to extract the generated idler light from the NLO waveguide. Similarly, a signal output coupler can be located subsequent to the pump-extracting coupler to extract the generated signal light from the nonlinear optical waveguide.

(715) In the different illustrative examples in FIGS. **34** through **44**, the phase shifters are implemented as electro optical phase shifters in which the electrodes for a phase shifter are constructed using electro-optical material. The use of electro optical phase shifters is not meant to limit the manner in which other illustrative examples can be implemented. For example, in another illustrative example the phase shifters can be implemented using thermal phase shifters in which the electrodes apply activation in the form of heat.

(716) The selection of the type electrodes and the configuration of the electrodes for phase shifters can be based on the type of material used for the substrate in which the optical waveguide structure is fabricated. For example, when the substrate is an x-cut material and the electrodes for the optical electrical phase shifters, at least one of folds or overlaps can be used such that the electrodes are aligned to a y-axis on a plane formed by the Y axis and z-axis. This alignment increases the effectiveness of activations applied by the electrodes to the tuning optical waveguides when electro-optical materials used to form the electrodes. Electro optical phase shifters at a faster response and can more quickly control the phase in a tuning electrode as compared to thermal phase shifters. In other words, electro optical phase shifters can provide a faster response time for controlling the phase of light as compared to using other types of phase shifters such as thermal phase shifters.

(717) In another example, when thermal phase shifters are used, this type of alignment is unnecessary. As result, folds and overlaps may not be needed. The use of thermal phase shifters can allow for less constraints with respect to the positioning and design of phase shifters and tuning optical waveguides. However, the use of temperature as an activation applied by thermal phase shifters has a slower response time for controlling the phase of light traveling through the tuning optical waveguide.

(718) Some features of the illustrative examples are described in the following clauses. These clauses are examples of features and are not intended to limit other illustrative examples.

(719) Clause 1

(720) An optical waveguide structure comprising: a nonlinear optical waveguide; a set of tuning optical waveguides; a set of wavelength selective couplers that couples light between the nonlinear optical waveguide and a tuning optical waveguide based on a wavelength of light; and a set of phase shifters located along one or more tuning optical waveguides in the set of tuning optical waveguides.

Clause 2

(721) The optical waveguide structure according to clause 1, wherein the light comprises a first wavelength light and a second wavelength light produced from the first wavelength light through a nonlinear optical interaction occurring within the nonlinear optical waveguide; the optical waveguide structure further comprising: a first segment in the nonlinear optical waveguide; a second segment in the nonlinear optical waveguide; a wavelength selective coupler in the set of wavelength selective couplers that couples the first wavelength light from the first segment into the second segment, wherein the wavelength selective coupler couples the second wavelength light from the first segment into the tuning optical waveguide; and a phase shifter in the set of phase shifters that applies an activation to the tuning optical waveguide to change a phase shift for the second wavelength light in the tuning optical waveguide. Clause 3

(722) The optical waveguide structure according to clause 2, wherein the second wavelength light in the tuning optical waveguide is

coupled from the tuning optical waveguide to the second segment by the wavelength selective coupler.

(723) Clause 4

(724) The optical waveguide structure according to clause 3, wherein the phase shifter in the set of phase shifters applies the activation to the tuning optical waveguide to change the phase shift for the second wavelength light in the tuning optical waveguide such that a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from a starting point where the first wavelength light is supplied to the first segment to a junction between the wavelength selective coupler and the second segment is zero or an even multiple of π radians.

(725) Clause 5

(726) The optical waveguide structure according to one of clauses 2, 3, or 4, wherein the wavelength selective coupler is a first wavelength selective coupler and the optical waveguide structure further comprises: a third segment in the nonlinear optical waveguide; and a second wavelength selective coupler in the set of wavelength selective couplers, wherein the second wavelength selective coupler in the set of wavelength selective couplers couples first wavelength light from the second segment into the third segment.

Clause 6

(727) The optical waveguide structure according to clause 5, wherein the second wavelength light in the tuning optical waveguide is coupled from the tuning optical waveguide to the third segment by the second wavelength selective coupler.

(728) Clause 7

(729) The optical waveguide structure according to clause 6, wherein the phase shifter in the set of phase shifters applies the activation to the tuning optical waveguide to change the phase shift for the second wavelength light in the tuning optical waveguide such that a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from a starting point where the first wavelength light is supplied to the first segment to a junction between the second wavelength selective coupler and the third segment is zero or an even multiple of π radians.

(730) Clause 8

(731) The optical waveguide structure according to one of clauses 5, 6, or 7 wherein the tuning optical waveguide is a first tuning optical waveguide and the phase shifter in the set of phase shifters is a first phase shifter; the optical waveguide structure further comprising: a second tuning optical waveguide; a second phase shifter in the set of phase shifters located along the second tuning optical waveguide; and a third wavelength selective coupler in the set of wavelength selective couplers that couples first wavelength light from the second segment into the third segment and couples second wavelength light from the second segment into the second tuning optical waveguide.

Clause 9

(732) The optical waveguide structure according to clause 8 further comprising: a fourth segment in the nonlinear optical waveguide; and a fourth wavelength selective coupler in the set of wavelength selective couplers that couples second wavelength light from the second tuning optical waveguide into the fourth segment, wherein the fourth wavelength selective coupler couples the first wavelength light from the third segment into the fourth segment.

Clause 10

(733) The optical waveguide structure according to clause 9, wherein the second phase shifter applies an activation to the second tuning optical waveguide to change a phase shift for the second wavelength light in the second tuning optical waveguide such that a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from a first junction between the first wavelength selective coupler and the second segment to a second junction between the fourth wavelength selective coupler and the fourth segment is zero or an even multiple of π radians.

(734) Clause 11

(735) The optical waveguide structure according to clause 9 further comprising: a third tuning optical waveguide; a fifth wavelength selective coupler in the set of wavelength selective couplers that couples second wavelength light from the third segment into the third tuning optical waveguide and couples first wavelength light from the third segment into the fourth segment; and a sixth wavelength selective coupler in the set of wavelength selective couplers that couples second wavelength light from the third tuning optical waveguide into a fifth segment and couples first wavelength light from the fourth segment into the fifth segment.

Clause 12

(736) The optical waveguide structure according to clause 11, further comprising: a third phase shifter in the set of phase shifters located along the third tuning optical waveguide, wherein the third phase shifter applies an activation to the third tuning optical waveguide to change a phase shift for the second wavelength light in the third tuning optical waveguide such that a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from a third junction between the second wavelength selective coupler and the third segment to a fourth junction between the sixth wavelength selective coupler and the fifth segment is zero or an even multiple of π radians.

Clause 13

(737) The optical waveguide structure according to one of clauses 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 further comprising: a phase shifter located along a segment in the nonlinear optical waveguide, wherein the phase shifter located along the segment in the nonlinear optical waveguide applies an activation to the segment to change a phase shift for the first wavelength light in the segment, and wherein the phase shift for the first wavelength light in the segment resulting from the activation applied to the segment has an opposite sign from the phase shift for the second wavelength light resulting from the activation applied to the tuning optical waveguide.

Clause 14

(738) The optical waveguide structure according to one of clauses 9, 10, 11, 12, or 13 further comprising: a combiner that combines the second wavelength light from the first segment and the second wavelength light from the second segment to form a combined wavelength light.

Clause 15

(739) The optical waveguide structure according to clause 14, wherein a third phase shifter in the set of phase shifters located along a third tuning optical waveguide and a fourth phase shifter in the set of phase shifters located along the fourth tuning optical waveguide apply activations that produce a difference between a phase of the second wavelength light from the first segment and a phase of the second wavelength light from the second segment that equals zero or an even multiple of 2π radians at the combiner.

(740) Clause 16

(741) The optical waveguide structure according to one of clauses 9, 10, 11, 12, 13, 14, or 15, further comprising: a source output coupler that couples the first wavelength light out of a final segment in the nonlinear optical waveguide.

Clause 17

(742) The optical waveguide structure according to clause 16 further comprising: a final phase shifter in the set of phase shifters located

along the final segment.

Clause 18

(743) The optical waveguide structure according to one of clauses 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, or 17, wherein the first wavelength light is a pump light and the second wavelength light is one of a signal light and an idler light.

(744) Clause 19

(745) The optical waveguide structure according to one of clauses 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, or 18, wherein the set of phase shifters is associated with the tuning optical waveguide being at least one of adjacent to part of the tuning optical waveguide, connected to part of the tuning optical waveguide, or integrated as part of the tuning optical waveguide.

(746) Clause 20

(747) The optical waveguide structure according to one of clauses 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, or 19, wherein the set of phase shifters is selected from at least one of a tuning electrode, a thermal element, shape memory alloy element, or piezo electric element.

(748) Clause 21

(749) The optical waveguide structure according to one of clauses 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20, wherein the tuning optical waveguide has a set of folds, wherein a pair of electrodes for a phase shifter in the set of phase shifters is present along a length of the tuning optical waveguide occurring after a fold in the set of folds such that the phase shifter and wherein the length is parallel to a crystal axis of a nonlinear optical material in which the optical waveguide structure and the set of tuning waveguides is formed.

(750) Clause 22

(751) The optical waveguide structure according to clause 21, wherein the nonlinear optical material is an x-cut lithium niobate and wherein the optical waveguide structure is formed on a yz plane and an x-axis of the nonlinear optical material is perpendicular to the yz plane of the optical waveguide structure and the set of tuning optical waveguides.

(752) Clause 23

(753) An optical waveguide structure comprising: a nonlinear optical waveguide; a tuning optical waveguide; a set of wavelength selective couplers that couples light between the nonlinear optical waveguide and the tuning optical waveguide based on a wavelength of light; and a set of phase shifters located along the set of tuning optical waveguide.

Clause 24

(754) The optical waveguide structure according to clause 23, wherein the nonlinear optical waveguide comprises segments alternating between odd index segments and even index segments, wherein a first wavelength light travels through the odd index segments and the even index segments and further comprising: tuning optical waveguides including the tuning optical waveguide; a first route through odd index segments and a first set of wavelength selective couplers that couples an odd index second wavelength light generated in the odd index segments from the odd index segments into a first set of tuning optical waveguides in the tuning optical waveguides and from the first set of tuning optical waveguides back into the odd index segments; a second route through even index segments and a second set of wavelength selective couplers that couples an even index second wavelength light generated in the even index segments from the even index segments into a second set of tuning optical waveguides in the tuning optical waveguides and from the second set of tuning optical waveguides back into the even index segments; phase shifters including the set of phase shifters, wherein a first set of phase shifters is associated with the first set of tuning optical waveguides and applies first activations to adjust a first phase of the odd index second wavelength light and a second set of phase shifters is associated with the second set of tuning optical waveguides and applies second activations to adjust a second phase of the even index second wavelength light; and a combiner, wherein the combiner receives the odd index second wavelength light from a first end of the first route, receives the even index second wavelength light from a second end of the second route and combines the odd index second wavelength light and the even index second wavelength light to form a second wavelength light.

Clause 25

(755) The optical waveguide structure according to one of clauses 23 or 24, wherein the light comprises a first wavelength light and a second wavelength light produced from the first wavelength light through a nonlinear optical interaction occurring within the nonlinear optical waveguide; the optical waveguide structure further comprising: a first wavelength selective coupler in the set of wavelength selective couplers that couples the second wavelength light from an exit location in the nonlinear optical waveguide to a starting point in the tuning optical waveguide; a second wavelength selective coupler in the set of wavelength selective couplers that couples the second wavelength light from an ending point in the tuning optical waveguide an entry location in the nonlinear optical waveguide; and a phase shifter in the set of phase shifters located between the starting point and the ending point in the tuning optical waveguide, wherein the phase shifter applies an activation to the second wavelength light in the tuning optical waveguide to change a phase shift for the second wavelength light in the tuning optical waveguide.

Clause 26

(756) A method for a nonlinear optical interaction, the method comprising: coupling, by a wavelength selective coupler, a first wavelength light from a first segment in a nonlinear optical waveguide into a second segment in the nonlinear optical waveguide; coupling, by the wavelength selective coupler, a second wavelength light from the first segment in the nonlinear optical waveguide into a tuning optical waveguide; and applying, by a phase shifter, an activation to the tuning optical waveguide to change a phase shift for the second wavelength light in the tuning optical waveguide.

Clause 27

(757) The method according to clause 26, wherein the wavelength selective coupler is a first wavelength selective coupler and further comprising: coupling by a second wavelength selective coupler, the first wavelength light from the second segment in the nonlinear optical waveguide into a third segment in the nonlinear optical waveguide; and coupling, by the second wavelength selective coupler, the second wavelength light from the tuning optical waveguide into the third segment in the nonlinear optical waveguide.

Clause 28

(758) The method according to clause 27, wherein the change in the phase shift for the second wavelength light in the tuning optical waveguide modifies a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from an entry location where the first wavelength light is supplied to the first segment to an entry location in the third segment to have a value that is zero or an even multiple of π radians.

(759) Clause 29

(760) The method according to one of clauses 26, 27, or 28 further comprising: coupling, by the wavelength selective coupler, the second

wavelength light from the tuning optical waveguide into the nonlinear optical waveguide, wherein the change in the phase shift for the second wavelength light in the tuning optical waveguide modifies a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from an entry location where the first wavelength light is supplied to the first segment to an entry location into the second segment to have a value that is zero or an even multiple of π radians.

Clause 30

(761) The method according to one of clauses 26, 27, 28, or 29, wherein the tuning optical waveguide is a first tuning optical waveguide and the phase shifter is a first phase shifter and further comprising: coupling, by a third wavelength selective coupler, the first wavelength light from the second segment in the nonlinear optical waveguide into a second tuning optical waveguide; applying, by a second phase shifter, a second activation to the second tuning optical waveguide to change the phase shift for the second wavelength light in the second tuning optical waveguide; coupling, by a fourth wavelength selective coupler, the first wavelength light from a third segment in the nonlinear optical waveguide into a fourth segment in the nonlinear optical waveguide; and coupling, by the fourth wavelength selective coupler, the second wavelength light from the second tuning optical waveguide into the fourth segment in the nonlinear optical waveguide, wherein the phase shift for the second wavelength light is changed by the second activation applied to the second tuning optical waveguide, wherein the change in the phase shift for the second wavelength light in the second tuning optical waveguide modifies a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from an entry location where the first wavelength light is supplied to the second segment to an entry location into the fourth segment to have a value that is zero or an even multiple of π radians.

Clause 31

(762) The method according to one of clauses 27, 28, 29, or 30 further comprising: coupling, by a combiner, the second wavelength light from a first tuning optical waveguide into an output segment in the nonlinear optical waveguide; and coupling, by the combiner, the second wavelength light from a second tuning optical waveguide into the output segment in the nonlinear optical waveguide.

Clause 32

(763) The method according to one of clauses 30 or 31 further comprising: coupling, by a combiner, the second wavelength light from the third segment in the nonlinear optical waveguide into an output segment in the nonlinear optical waveguide; and coupling, by the combiner, the second wavelength light from the fourth segment into the output segment in the nonlinear optical waveguide.

Clause 33

(764) The method according to clause 32 further comprising: applying, by a third phase shifter, a third activation to the second wavelength light in the third segment; and applying, by a fourth phase shifter, a fourth activation the second wavelength light in the fourth segment, wherein the third activation adjusts a phase of the second wavelength light in the third segment, wherein the fourth activation adjusts the phase of the second wavelength light in the fourth segment such that a difference between the phase of the second wavelength light in the third segment and a phase of the second wavelength light in the fourth segment is an even multiple of π radians where the second wavelength light in the third segment and second wavelength light in the third segment and the second wavelength light in the fourth segment are coupled by the combiner into the output segment. The description of the different illustrative embodiments has been presented for purposes of illustration and description and is not intended to be exhaustive or limited to the embodiments in the form disclosed. The different illustrative examples describe components that perform actions or operations. In an illustrative embodiment, a component can be configured to perform the action or operation described. For example, the component can have a configuration or design for a structure that provides the component an ability to perform the action or operation that is described in the illustrative examples as being performed by the component. Further, To the extent that terms “includes”, “including”, “has”, “contains”, and variants thereof are used herein, such terms are intended to be inclusive in a manner similar to the term “comprises” as an open transition word without precluding any additional or other elements.

(765) Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other desirable embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. An optical waveguide structure comprising: a nonlinear optical waveguide; a set of tuning optical waveguides; a set of wavelength selective couplers that couples light between the nonlinear optical waveguide and a tuning optical waveguide based on a wavelength of light; and a set of phase shifters located along one or more tuning optical waveguides in the set of tuning optical waveguides; wherein the light comprises a first wavelength light and a second wavelength light produced from the first wavelength light through a nonlinear optical interaction occurring within the nonlinear optical waveguide; the optical waveguide structure further comprising: a first segment in the nonlinear optical waveguide; a second segment in the nonlinear optical waveguide; a wavelength selective coupler in the set of wavelength selective couplers that couples the first wavelength light from the first segment into the second segment, wherein the wavelength selective coupler couples the second wavelength light from the first segment into the tuning optical waveguide; and a phase shifter in the set of phase shifters that applies an activation to the tuning optical waveguide to change a phase shift for the second wavelength light in the tuning optical waveguide.
2. The optical waveguide structure of claim 1, wherein the second wavelength light in the tuning optical waveguide is coupled from the tuning optical waveguide to the second segment by the wavelength selective coupler.
3. The optical waveguide structure of claim 2, wherein the phase shifter in the set of phase shifters applies the activation to the tuning optical waveguide to change the phase shift for the second wavelength light in the tuning optical waveguide such that a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from a starting point where the first wavelength light is supplied to the first segment to a junction between the wavelength selective coupler and the second segment is zero or an even multiple of π radians.
4. The optical waveguide structure of claim 1, wherein the wavelength selective coupler is a first wavelength selective coupler and the optical waveguide structure further comprises: a third segment in the nonlinear optical waveguide; and a second wavelength selective coupler in the set of wavelength selective couplers, wherein the second wavelength selective coupler in the set of wavelength selective couplers couples the first wavelength light from the second segment into the third segment.
5. The optical waveguide structure of claim 4, wherein the second wavelength light in the tuning optical waveguide is coupled from the tuning optical waveguide to the third segment by the second wavelength selective coupler.
6. The optical waveguide structure of claim 5, wherein the phase shifter in the set of phase shifters applies the activation to the tuning optical waveguide to change the phase shift for the second wavelength light in the tuning optical waveguide such that a phase walk-off for

the nonlinear optical interaction in the nonlinear optical waveguide from a starting point where the first wavelength light is supplied to the first segment to a junction between the second wavelength selective coupler and the third segment is zero or an even multiple of π radians.

7. The optical waveguide structure of claim 4, wherein the tuning optical waveguide is a first tuning optical waveguide and the phase shifter in the set of phase shifters is a first phase shifter; the optical waveguide structure further comprising: a second tuning optical waveguide; a second phase shifter in the set of phase shifters located along the second tuning optical waveguide; and a third wavelength selective coupler in the set of wavelength selective couplers that couples the first wavelength light from the second segment into the third segment and couples the second wavelength light from the second segment into the second tuning optical waveguide.

8. The optical waveguide structure of claim 7 further comprising: a fourth segment in the nonlinear optical waveguide; and a fourth wavelength selective coupler in the set of wavelength selective couplers that couples the second wavelength light from the second tuning optical waveguide into the fourth segment, wherein the fourth wavelength selective coupler couples the first wavelength light from the third segment into the fourth segment.

9. The optical waveguide structure of claim 8, wherein the second phase shifter applies an activation to the second tuning optical waveguide to change a phase shift for the second wavelength light in the second tuning optical waveguide such that a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from a first junction between the first wavelength selective coupler and the second segment to a second junction between the fourth wavelength selective coupler and the fourth segment is zero or an even multiple of π radians.

10. The optical waveguide structure of claim 8 further comprising: a third tuning optical waveguide; a fifth wavelength selective coupler in the set of wavelength selective couplers that couples the second wavelength light from the third segment into the third tuning optical waveguide and couples the first wavelength light from the third segment into the fourth segment; and a sixth wavelength selective coupler in the set of wavelength selective couplers that couples the second wavelength light from the third tuning optical waveguide into a fifth segment and couples the first wavelength light from the fourth segment into the fifth segment.

11. The optical waveguide structure of claim 10, further comprising: a third phase shifter in the set of phase shifters located along the third tuning optical waveguide, wherein the third phase shifter applies an activation to the third tuning optical waveguide to change a phase shift for the second wavelength light in the third tuning optical waveguide such that a phase walk-off for the nonlinear optical interaction in the nonlinear optical waveguide from a third junction between the second wavelength selective coupler and the third segment to a fourth junction between the sixth wavelength selective coupler and the fifth segment is zero or an even multiple of π radians.

12. The optical waveguide structure of claim 8 further comprising: a combiner that combines the second wavelength light from the first segment and the second wavelength light from the second segment to form a combined second wavelength light.

13. The optical waveguide structure of claim 12, wherein a third phase shifter in the set of phase shifters located along a third tuning optical waveguide and a fourth phase shifter in the set of phase shifters located along a fourth tuning optical waveguide apply activations that produce a difference between a phase of the second wavelength light from the first segment and a phase of the second wavelength light from the second segment that equals zero or an even multiple of 2π radians at the combiner.

14. The optical waveguide structure of claim 8, further comprising: a source output coupler that couples the first wavelength light out of a final segment in the nonlinear optical waveguide.

15. The optical waveguide structure of claim 14 further comprising: a final phase shifter in the set of phase shifters located along the final segment.

16. The optical waveguide structure of claim 1 further comprising: a phase shifter located along a segment in the nonlinear optical waveguide, wherein the phase shifter located along the segment in the nonlinear optical waveguide applies an activation to the segment to change a phase shift for the first wavelength light in the segment, and wherein the phase shift for the first wavelength light in the segment resulting from the activation applied to the segment has an opposite sign from the phase shift for the second wavelength light resulting from the activation applied to the tuning optical waveguide.

17. The optical waveguide structure of claim 1, wherein the first wavelength light is a pump light and the second wavelength light is one of a signal light and an idler light.

18. The optical waveguide structure of claim 1, wherein the set of phase shifters is associated with the tuning optical waveguide being at least one of adjacent to part of the tuning optical waveguide, connected to part of the tuning optical waveguide, or integrated as part of the tuning optical waveguide.

19. The optical waveguide structure of claim 1, wherein the set of phase shifters is selected from at least one of a tuning electrode, a thermal element, shape memory alloy element, or piezo electric element.

20. The optical waveguide structure of claim 1, wherein the nonlinear optical material is an x-cut lithium niobate and wherein the optical waveguide structure is formed on a yz plane and an x-axis of the nonlinear optical material is perpendicular to the yz plane of the optical waveguide structure and the set of tuning optical waveguides.

21. An optical waveguide structure comprising: a nonlinear optical waveguide; a tuning optical waveguide; a set of wavelength selective couplers that couples light between the nonlinear optical waveguide and the tuning optical waveguide based on a wavelength of light; and a set of phase shifters located along the tuning optical waveguide; wherein the nonlinear optical waveguide comprises segments alternating between odd index segments and even index segments, wherein a first wavelength light travels through the odd index segments and the even index segments and further comprising: tuning optical waveguides including the tuning optical waveguide; a first route through odd index segments and a first set of wavelength selective couplers that couples an odd index second wavelength light generated in the odd index segments from the odd index segments into a first set of tuning optical waveguides in the tuning optical waveguides and from the first set of tuning optical waveguides back into the odd index segments; a second route through even index segments and a second set of wavelength selective couplers that couples an even index second wavelength light generated in the even index segments from the even index segments into a second set of tuning optical waveguides in the tuning optical waveguides and from the second set of tuning optical waveguides back into the even index segments; phase shifters including the set of phase shifters, wherein a first set of phase shifters is associated with the first set of tuning optical waveguides and applies first activations to adjust a first phase of the odd index second wavelength light and a second set of phase shifters is associated with the second set of tuning optical waveguides and applies second activations to adjust a second phase of the even index second wavelength light; and a combiner, wherein the combiner receives the odd index second wavelength light from a first end of the first route, receives the even index second wavelength light from a second end of the second route and combines the odd index second wavelength light and the even index second wavelength light to form a combined second wavelength light.

22. The optical waveguide structure of claim 21, wherein the light comprises a first wavelength light and a second wavelength light produced from the first wavelength light through a nonlinear optical interaction occurring within the nonlinear optical waveguide; the

optical waveguide structure further comprising: a first wavelength selective coupler in the set of wavelength selective couplers that couples the second wavelength light from an exit location in the nonlinear optical waveguide to a starting point in the tuning optical waveguide; a second wavelength selective coupler in the set of wavelength selective couplers that couples the second wavelength light from an ending point in the tuning optical waveguide an entry location in the nonlinear optical waveguide; and a phase shifter in the set of phase shifters located between the starting point and the ending point in the tuning optical waveguide, wherein the phase shifter applies an activation to the second wavelength light in the tuning optical waveguide to change a phase shift for the second wavelength light in the tuning optical waveguide.
